In [1]:
```
import glob
import os
import pandas as pd
import numpy as np
from sklearn.metrics import accuracy_score, precision_score, recall_score, c
pd.set_option('display.max_columns', None)

events = pd.read_parquet("../data/gdelt/events/6_final/events_dataset_v2.par
events = events[events['period'] <= '202406']
```

In [2]:
```
print(len(events.index))
print(events.columns)
events.tail(3)
```

```
71293
Index(['ADMIN0', 'ADMIN1', 'ADMIN2', 'CS_score', 'period', 'SQLDATE',
       'EventCode', 'SOURCEURL', 'NumMentions', 'NumSources', 'NumArticles',
       'valid_SOURCEURL', 'clean_text_list', 'NER_admin0_list',
       'NER_admin1_list', 'NER_admin2_list', 'compound_score_list',
       'neg_score_list', 'neu_score_list', 'pos_score_list',
       'fatalities_freq_list', 'displaced_freq_list', 'detained_freq_list',
       'injured_freq_list', 'sexual_violence_freq_list', 'torture_freq_lis
t',
       'economic_shocks_freq_list', 'agriculture_freq_list',
       'weather_freq_list', 'food_insecurity_freq_list',
       'fatalities_count_list', 'displaced_count_list', 'detained_count_lis
t',
       'injured_count_list', 'sexual_violence_count_list',
       'torture_count_list', 'sentiment.compound_list', 'sentiment.neg_lis
t',
       'sentiment.neu_list', 'sentiment.pos_list', 'pred_impact_type_list',
       'pred_urgency_list', 'pred_resource_food_list',
       'pred_resource_water_list', 'pred_resource_cash_aid_list',
       'pred_resource_healthcare_list', 'pred_resource_shelter_list',
       'pred_resource_livelihoods_list', 'pred_resource_education_list',
       'pred_resource_infrastructure_list', 'pred_resource_none_list'],
      dtype='object')
```

Out[2]:

	ADMIN0	ADMIN1	ADMIN2	CS_score	period	SQLDATE	EventCode	
271887	nigeria	zamfara	talata mafara	NaN	201907	[]	[20.0, 20.0, 43.0, 73.0, 46.0, 100.0, 43.0, 20...	[
271898	nigeria	zamfara	tsafe	NaN	201601	[]	[100.0, 42.0, 36.0, 100.0, 36.0, 20.0, 20.0, 1...	[http://
272100	niger	zinder	tanout	NaN	202308	[]	[42.0, 46.0, 42.0, 10.0, 10.0, 42.0, 172.0, 20...	[h

In [3]:
```
# Forecasting Configuration
# ============================
# Set the forecast horizon: predict CS_score N periods ahead
# Options: 1 (nowcasting), 2, 3, or 4 (forecasting)
# Literature shows GDELT features are more valuable for forecasting (2-4 per
# because baseline (previous_CS) becomes less powerful with longer horizons

FORECAST_HORIZON = 3  # Predict 3 periods ahead (can be changed to 2 or 4)

print(f"=== Forecasting Configuration ===")
print(f"Forecast Horizon: {FORECAST_HORIZON} period(s) ahead")
if FORECAST_HORIZON == 1:
    print("Mode: NOWCASTING (predicting next period)")
else:
    print(f"Mode: FORECASTING (predicting {FORECAST_HORIZON} periods ahead)"
print(f"\nWhy forecasting helps GDELT features:")
print(f"  - Baseline 'previous_CS' becomes less predictive ({FORECAST_HORIZO
print(f"  - GDELT temporal patterns can capture early warning signals")
print(f"  - More realistic for early warning systems\n")
```

```
=== Forecasting Configuration ===
Forecast Horizon: 3 period(s) ahead
Mode: FORECASTING (predicting 3 periods ahead)

Why forecasting helps GDELT features:
  - Baseline 'previous_CS' becomes less predictive (3 periods old)
  - GDELT temporal patterns can capture early warning signals
  - More realistic for early warning systems
```

In [4]:
```
df = events.copy()
print(f"Total rows: {len(df.index)}")
print(f"Period range: {min(df['period'])} to {max(df['period'])}")
print(f"Columns: {len(df.columns)}")

# Drop rows where ADMIN0 is missing
mask_missing = df['ADMIN0'].isna() | (df['ADMIN0'] == 'None') | (df['ADMIN0'
```

```
if mask_missing.sum() > 0:
    print(f"Dropping {mask_missing.sum()} rows with ADMIN0=None")
    df = df[~mask_missing].copy()

print(f"Final rows: {len(df)}")
df.tail()
```

```
Total rows: 71293
Period range: 201601 to 202406
Columns: 51
Dropping 429 rows with ADMIN0=None
Final rows: 70864
```

Out[4]:

	ADMIN0	ADMIN1	ADMIN2	CS_score	period	SQLDATE	EventCode	
271876	nigeria	zamfara	talata mafara	NaN	201804	[]	[10.0, 42.0, 71.0, 43.0, 43.0, 20.0, 43.0, 43....	[I
271883	nigeria	zamfara	talata mafara	NaN	201901	[]	[42.0, 10.0, 10.0, 20.0, 10.0, 10.0, 10.0, 100...	
271887	nigeria	zamfara	talata mafara	NaN	201907	[]	[20.0, 20.0, 43.0, 73.0, 46.0, 100.0, 43.0, 20...	[
271898	nigeria	zamfara	tsafe	NaN	201601	[]	[100.0, 42.0, 36.0, 100.0, 36.0, 20.0, 20.0, 1...	[http://
272100	niger	zinder	tanout	NaN	202308	[]	[42.0, 46.0, 42.0, 10.0, 10.0, 42.0, 172.0, 20...	[h

In [5]:

```
key_cols = ["ADMIN0", "ADMIN1", "ADMIN2", "period"]

def check_duplicates(df, name):
    dupes = (
        df
        .groupby(key_cols)
        .size()
        .reset_index(name="n")
        .query("n > 1")
    )
    print(f"{name}: {len(dupes)} duplicated keys")
    return dupes

dupes = check_duplicates(df, "events")
```

```
events: 19 duplicated keys
```

In [6]:

```python
# BETWEEN-ASSESSMENT FORECASTING SETUP
# ====================================
# CRITICAL: This implements the correct "between-assessment forecasting" set
#
# Key principle: Observed IPC defines targets; unobserved months are predict
#
# Target: y_next = next observed IPC assessment (forward-lagged label)
# Feature: IPC_last = last observed IPC before this row (forward-filled, cor
#
# Rules:
# - Only use IPC_last as IPC feature (no rolling, no multiple lags, no trend
# - CS_score is NOT the direct target (y_next is)
# - Keep ALL rows (including last periods without future assessments)
# - Filter to rows with valid y_next only when training models

print("="*70)
print("BETWEEN-ASSESSMENT FORECASTING SETUP")
print("="*70)
print("\nKey principle: Observed IPC defines targets; unobserved months are

# Create region identifiers
df['region'] = df['ADMIN0'] + '-' + df['ADMIN1']
df['district'] = df['ADMIN0'] + '-' + df['ADMIN1'] + '-' + df['ADMIN2']

# Sort by district (ADMIN2) and period to ensure deterministic operations
df = df.sort_values(['ADMIN0', 'ADMIN1', 'ADMIN2', 'period']).reset_index(dr

# CS_score is already in the events dataset - just ensure numeric
df['CS_score'] = pd.to_numeric(df['CS_score'], errors='coerce')

# Create IPC_last: last observed IPC before this row (forward-filled)
# This is the ONLY IPC feature allowed - it's stale and constant between ass
print("1. Creating IPC_last (last observed IPC, forward-filled)...")
df['IPC_last'] = df.groupby(['ADMIN0', 'ADMIN1', 'ADMIN2'], sort=False)['CS_
    lambda x: x.ffill()
)

# Create y_next: next observed IPC assessment (forward-lagged label)
# This is the target variable - we predict the next assessment outcome
print("2. Creating y_next (next observed IPC assessment - THE TARGET)...")

def get_next_assessment(group):
    """Get the next observed IPC assessment for each row"""
    cs_values = group['CS_score'].values
    result = []

    for i in range(len(cs_values)):
        # Look forward to find next non-null value
        found = False
        for j in range(i + 1, len(cs_values)):
            if pd.notna(cs_values[j]) and 1 <= cs_values[j] <= 5:
                result.append(cs_values[j])
                found = True
                break
```

```
        if not found:
            result.append(np.nan)

    return pd.Series(result, index=group.index)

df['y_next'] = df.groupby(['ADMIN0', 'ADMIN1', 'ADMIN2'], sort=False, group_
    get_next_assessment
)

# Keep ALL rows - don't drop periods without future assessments
# Rows without y_next (like 202402) can still be used for feature engineerir
# Filter to rows with valid y_next only when training models
rows_with_y_next = df['y_next'].notna().sum()
rows_without_y_next = df['y_next'].isna().sum()
df = df[df['IPC_last'].isin([1, 2, 3, 4, 5])]
print(f"   Rows with y_next (can be used for training): {rows_with_y_next}")
print(f"   Rows without y_next (last periods, feature engineering only): {ro
print(f"   Total rows kept: {len(df)} (no periods dropped)")

# Clean y_next where it exists: ensure valid range (1-5) and convert to int
# Only clean where y_next is not NaN
mask_valid_y_next = df['y_next'].notna()
df.loc[mask_valid_y_next, 'y_next'] = df.loc[mask_valid_y_next, 'y_next'].ro
df.loc[mask_valid_y_next, 'y_next'] = df.loc[mask_valid_y_next, 'y_next'].cl
df.loc[mask_valid_y_next, 'y_next'] = df.loc[mask_valid_y_next, 'y_next'].as

# Summary statistics
print(f"\n=== Summary ===")
print(f"IPC_last: last observed IPC (forward-filled, constant between assess
print(f"y_next: next observed IPC assessment (THE TARGET)")
print(f"\nIPC_last distribution:")
print(df['IPC_last'].value_counts().sort_index())
print(f"\ny_next distribution (TARGET):")
print(df['y_next'].value_counts().sort_index())
print(f"\nRows with valid y_next: {df['y_next'].notna().sum()}")
print(f"Rows without y_next (last periods): {df['y_next'].isna().sum()}")
print(f"Current DataFrame shape: {df.shape}")
print(f"Period range: {df['period'].min()} to {df['period'].max()}")

# Show example for verification
print(f"\n=== Example (first ADMIN2) ===")
if len(df) > 0:
    example_admin2 = df['ADMIN2'].iloc[0]
    example = df[df['ADMIN2'] == example_admin2].head(12)[['ADMIN2', 'perioc
    print(example.to_string(index=False))
    print(f"\nNote: IPC_last is constant between assessments, y_next is the

print("\n" + "="*70)
```

```
======================================================================
BETWEEN-ASSESSMENT FORECASTING SETUP
======================================================================

Key principle: Observed IPC defines targets; unobserved months are predictio
n times.

1. Creating IPC_last (last observed IPC, forward-filled)...
2. Creating y_next (next observed IPC assessment - THE TARGET)...
   Rows with y_next (can be used for training): 66446
   Rows without y_next (last periods, feature engineering only): 4418
   Total rows kept: 69645 (no periods dropped)

=== Summary ===
IPC_last: last observed IPC (forward-filled, constant between assessments)
y_next: next observed IPC assessment (THE TARGET)

IPC_last distribution:
IPC_last
1.0    34059
2.0    18841
3.0    14936
4.0     1805
5.0        4
Name: count, dtype: int64

y_next distribution (TARGET):
y_next
1.0    30949
2.0    18339
3.0    14603
4.0     1754
5.0        4
Name: count, dtype: int64

Rows with valid y_next: 65649
Rows without y_next (last periods): 3996
Current DataFrame shape: (69645, 55)
Period range: 201602 to 202406

=== Example (first ADMIN2) ===
ADMIN2 period  CS_score  IPC_last  y_next
  bale 201602       1.0       1.0     1.0
  bale 201606       1.0       1.0     1.0
  bale 201610       1.0       1.0     1.0
  bale 201702       1.0       1.0     1.0
  bale 201706       1.0       1.0     1.0
  bale 201710       1.0       1.0     1.0
  bale 202012       NaN       1.0     1.0
  bale 202106       1.0       1.0     1.0
  bale 202110       1.0       1.0     1.0
  bale 202202       1.0       1.0     1.0
  bale 202306       1.0       1.0     1.0
  bale 202310       1.0       1.0     1.0

Note: IPC_last is constant between assessments, y_next is the next assessmen
```

```
t value

================================================================================
```

```
/var/folders/3y/cdj5wjs107z_64y5f9ws4cj40000gn/T/ipykernel_55538/3370721709.
py:60: FutureWarning: DataFrameGroupBy.apply operated on the grouping column
s. This behavior is deprecated, and in a future version of pandas the groupi
ng columns will be excluded from the operation. Either pass `include_groups=
False` to exclude the groupings or explicitly select the grouping columns af
ter groupby to silence this warning.
  df['y_next'] = df.groupby(['ADMIN0', 'ADMIN1', 'ADMIN2'], sort=False, grou
p_keys=False).apply(
```

In [7]:
```
# Article coverage by IPC score and ADMIN0
# ========================================

df_ipc = df[df['IPC_last'].notna()].copy()
df_ipc['has_articles'] = df_ipc['valid_SOURCEURL'].apply(
    lambda x: x.size > 0 if isinstance(x, np.ndarray) else bool(x) if x is r
)

total = len(df_ipc)
print(f"Total rows (with IPC_last): {total:,}")
print(f"Rows with articles: {df_ipc['has_articles'].sum():,} ({df_ipc['has_a

# Breakdown by IPC_last
print("\nBreakdown by IPC_last:")
breakdown = df_ipc.groupby('IPC_last').agg(
    total=('has_articles', 'size'),
    with_articles=('has_articles', 'sum'),
)
breakdown['pct_articles'] = (breakdown['with_articles'] / breakdown['total']
print(breakdown)

# Breakdown by ADMIN0
print("\nBreakdown by ADMIN0:")
breakdown_admin0 = df_ipc.groupby('ADMIN0').agg(
    total=('has_articles', 'size'),
    with_articles=('has_articles', 'sum'),
    admin2_total=('ADMIN2', 'nunique'),
)
admin2_cov = df_ipc[df_ipc['has_articles']].groupby('ADMIN0')['ADMIN2'].nuni
breakdown_admin0 = breakdown_admin0.join(admin2_cov, how='left').fillna({'ad
breakdown_admin0['admin2_with_coverage'] = breakdown_admin0['admin2_with_cov
breakdown_admin0['pct_admin2_covered'] = (breakdown_admin0['admin2_with_cove
breakdown_admin0['pct_articles'] = (breakdown_admin0['with_articles'] / brea
breakdown_admin0 = breakdown_admin0.sort_values('total', ascending=False)
print(breakdown_admin0)

# Admin1 propagation gain
admin1_coverage = df_ipc.groupby(['ADMIN0', 'ADMIN1', 'period'])['has_articl
o, p = df_ipc['has_articles'].sum(), admin1_coverage.sum()
print(f"\nAdmin1 propagation: {o:,} ({o/total*100:.1f}%) -> {p:,} ({p/total*

# Breakdown by ADMIN0 after admin1 propagation
print("\nBreakdown by ADMIN0 (after admin1 propagation):")
df_ipc['has_articles_admin1'] = admin1_coverage.values
```

```
breakdown_prop = df_ipc.groupby('ADMIN0').agg(
    total=('has_articles', 'size'),
    admin2_total=('ADMIN2', 'nunique'),
    pct_admin2=('has_articles', 'mean'),
    pct_admin1=('has_articles_admin1', 'mean'),
)
breakdown_prop['pct_admin2'] = (breakdown_prop['pct_admin2'] * 100).round(1)
breakdown_prop['pct_admin1'] = (breakdown_prop['pct_admin1'] * 100).round(1)
breakdown_prop['gain_pp'] = (breakdown_prop['pct_admin1'] - breakdown_prop['
breakdown_prop = breakdown_prop.sort_values('total', ascending=False)
print(breakdown_prop)
```

Total rows (with IPC_last): 69,645
Rows with articles: 14,939 (21.5%)

Breakdown by IPC_last:

IPC_last	total	with_articles	pct_articles
1.0	34059	6964	20.4
2.0	18841	4243	22.5
3.0	14936	3274	21.9
4.0	1805	454	25.2
5.0	4	4	100.0

Breakdown by ADMIN0:

ADMIN0	total	with_articles	admin2_total	admin2_with_coverage	pct_admin2_covered
nigeria	19989	3133	729	364	49.9
kenya	7183	1467	269	171	63.6
yemen	6586	980	300	130	43.3
sudan	4522	942	167	51	30.5
mozambique	3861	502	154	58	37.7
uganda	3142	870	112	76	67.9
democratic republic of the congo	2772	242	181	41	22.7
ethiopia	2428	862	76	53	69.7
zimbabwe	2414	1211	74	64	86.5
somalia	2339	799	74	51	68.9
niger	1917	303	72	24	33.3
south sudan	1879	477	79	28	35.4
chad	1831	290	68	21	30.9
burundi	1498	92	111		
mali	1464	459	50		
malawi	854	430	27		
cameroon	838	176	58		
zambia	837	355	89		
burkina faso	794	260	45		
liberia	723	216	131		
madagascar	671	271	22		
mauritania	505	165	45		
guinea	333	212	34		
sierra leone	265	225	13		

```
burundi                                          22               19.8
mali                                             40               80.0
malawi                                           25               92.6
cameroon                                         36               62.1
zambia                                           53               59.6
burkina faso                                     28               62.2
liberia                                          18               13.7
madagascar                                       20               90.9
mauritania                                       12               26.7
guinea                                           12               35.3
sierra leone                                     10               76.9

                                 pct_articles
ADMIN0
nigeria                                  15.7
kenya                                    20.4
yemen                                    14.9
sudan                                    20.8
mozambique                               13.0
uganda                                   27.7
democratic republic of the congo          8.7
ethiopia                                 35.5
zimbabwe                                 50.2
somalia                                  34.2
niger                                    15.8
south sudan                              25.4
chad                                     15.8
burundi                                   6.1
mali                                     31.4
malawi                                   50.4
cameroon                                 21.0
zambia                                   42.4
burkina faso                             32.7
liberia                                  29.9
madagascar                               40.4
mauritania                               32.7
guinea                                   63.7
sierra leone                             84.9

Admin1 propagation: 14,939 (21.5%) -> 46,010 (66.1%), gain +31,071 (44.6%)

Breakdown by ADMIN0 (after admin1 propagation):
                                 total  admin2_total  pct_admin2  pct_admin
1  \
ADMIN0
nigeria                          19989           729        15.7        76.
8
kenya                             7183           269        20.4        54.
3
yemen                             6586           300        14.9        67.
7
sudan                             4522           167        20.8        58.
4
mozambique                        3861           154        13.0        54.
6
uganda                            3142           112        27.7        99.
```

democratic republic of the congo	2772	181	8.7	31.6
ethiopia	2428	76	35.5	82.5
zimbabwe	2414	74	50.2	94.2
somalia	2339	74	34.2	64.0
niger	1917	72	15.8	49.0
south sudan	1879	79	25.4	75.6
chad	1831	68	15.8	21.9
burundi	1498	111	6.1	23.6
mali	1464	50	31.4	68.2
malawi	854	27	50.4	98.2
cameroon	838	58	21.0	59.3
zambia	837	89	42.4	71.3
burkina faso	794	45	32.7	44.1
liberia	723	131	29.9	40.8
madagascar	671	22	40.4	64.4
mauritania	505	45	32.7	38.8
guinea	333	34	63.7	69.4
sierra leone	265	13	84.9	89.8

ADMIN0	gain_pp
nigeria	61.1
kenya	33.9
yemen	52.8
sudan	37.6
mozambique	41.6
uganda	71.3
democratic republic of the congo	22.9
ethiopia	47.0
zimbabwe	44.0
somalia	29.8
niger	33.2
south sudan	50.2
chad	6.1
burundi	17.5
mali	36.8
malawi	47.8

```
cameroon                    38.3
zambia                      28.9
burkina faso                11.4
liberia                     10.9
madagascar                  24.0
mauritania                   6.1
guinea                       5.7
sierra leone                 4.9
```

In [8]:
```
print(min(df['period']))
print(max(df['period']))
df = df.sort_values(['ADMIN0', 'ADMIN1', 'ADMIN2', 'period'])
df.tail()
```

```
201602
202406
```

Out[8]:

	ADMIN0	ADMIN1	ADMIN2	CS_score	period	SQLDATE	EventCode
70859	zimbabwe	midlands	zvishavane urban	2.0	202206	[]	[70.0, 70.0, 42.0]
70860	zimbabwe	midlands	zvishavane urban	2.0	202210	[]	[10.0, 173.0, 173.0, 10.0, 42.0]
70861	zimbabwe	midlands	zvishavane urban	2.0	202302	[]	[173.0, 112.0]
70862	zimbabwe	midlands	zvishavane urban	2.0	202306	[20230604.0]	[173.0, 71.0, 10.0, 30.0, 43.0, 71.0]
70863	zimbabwe	midlands	zvishavane urban	2.0	202402	[]	[173.0, 42.0, 10.0]

Feature engineering

In [9]:
```
# COMPREHENSIVE FEATURE ENGINEERING
# ==================================
# This cell performs ALL feature engineering for IPC and GDELT Events featur
# At this point: lists are NOT aggregated yet, y_next and IPC_last exist

print("="*70)
print("COMPREHENSIVE FEATURE ENGINEERING")
print("="*70)
print(f"\nStarting with {len(df)} rows")
print(f"Columns before feature engineering: {len(df.columns)}\n")
```

```python
# ============================================================================
# 1. AGGREGATE GDELT LIST FEATURES
# ============================================================================
print("1. Aggregating GDELT list features...")

# Helper functions (already defined in previous cell, but ensure they exist)
def safe_list_agg(lst, func):
    """Safely aggregate a list, handling None, empty lists, and non-numeric
    if lst is None:
        return np.nan
    if isinstance(lst, np.ndarray):
        if lst.size == 0:
            return np.nan
        lst = lst.tolist()
    try:
        if pd.isna(lst):
            return np.nan
    except (ValueError, TypeError):
        pass
    if isinstance(lst, (int, float)):
        return float(lst)
    if isinstance(lst, str):
        try:
            if lst.startswith('[') or lst.startswith('('):
                lst = eval(lst)
            else:
                return float(lst)
        except:
            return np.nan
    if not isinstance(lst, (list, tuple)):
        return np.nan
    if len(lst) == 0:
        return np.nan
    try:
        numeric_lst = []
        for x in lst:
            try:
                if pd.isna(x) or x is None:
                    continue
            except (ValueError, TypeError):
                pass
            try:
                val = float(x)
                if not np.isinf(val) and not np.isnan(val):
                    numeric_lst.append(val)
            except (ValueError, TypeError):
                continue
        if len(numeric_lst) == 0:
            return np.nan
        result = func(numeric_lst)
        return float(result) if not np.isnan(result) and not np.isinf(result
    except Exception as e:
        return np.nan

def safe_list_count(x):
```

```
    """Safely count elements in a list/array"""
    if x is None:
        return 0
    if isinstance(x, np.ndarray):
        return x.size if x.size > 0 else 0
    if isinstance(x, (list, tuple)):
        return len(x)
    try:
        if pd.isna(x):
            return 0
        return 1
    except (ValueError, TypeError):
        return 1

def aggregate_list_features(df, list_cols, prefix=""):
    """Aggregate list columns into meaningful statistical features for NLP-d

    For noisy NLP features, we keep only:
    - mean: central tendency
    - count: volume indicator
    - sum: total (especially useful for frequencies/counts)
    - max: peak intensity (only for crisis indicators)

    We skip: min (usually 0), std (variance not meaningful for noisy signals
    """
    new_cols = {}
    cols_to_drop = []

    for col in list_cols:
        if col not in df.columns:
            continue

        base_name = col.replace('_list', '')
        if prefix:
            base_name = f"{prefix}_{base_name}"

        # For text columns (NER, clean_text), only create count
        if 'NER' in col or 'clean_text' in col:
            new_cols[f"{base_name}_count"] = df[col].apply(safe_list_count)
        else:
            # For numeric columns, create meaningful aggregations only
            # Mean: central tendency (most important for noisy signals)
            new_cols[f"{base_name}_mean"] = df[col].apply(lambda x: safe_lis

            # Count: volume indicator (how many articles/events)
            new_cols[f"{base_name}_count"] = df[col].apply(safe_list_count)

            # Sum: total (especially useful for frequencies and counts)
            new_cols[f"{base_name}_sum"] = df[col].apply(lambda x: safe_list

            # Max: peak intensity (only for crisis indicators where peaks ma
            # Check if this is a crisis-related feature
            is_crisis_feature = any(x in col.lower() for x in ['fatalities',
                                                               'violence',
            if is_crisis_feature:
                new_cols[f"{base_name}_max"] = df[col].apply(lambda x: safe_
```

```
            # Skip: min (usually 0), std (variance not meaningful for noisy

        cols_to_drop.append(col)

    if new_cols:
        new_df = pd.DataFrame(new_cols, index=df.index)
        df = pd.concat([df, new_df], axis=1)

    if cols_to_drop:
        df = df.drop(columns=cols_to_drop)

    return df

# Find and aggregate list columns (events only, no merge suffix)
list_cols = [c for c in df.columns if '_list' in c]
print(f"   Found {len(list_cols)} list columns")

if list_cols:
    print("   Aggregating events features...")
    df = aggregate_list_features(df, list_cols, prefix="evt")

# ================================================================================
# 2. CLEAN AND PREPARE IPC FEATURES
# ================================================================================
print("\n2. Cleaning IPC features...")

# Clean IPC_last: round to nearest integer and ensure valid range
df['IPC_last'] = df['IPC_last'].round()
# Keep IPC_last as float for now (can filter invalid values later if needed)
# Don't drop rows here - keep all periods

print(f"   IPC_last range: {df['IPC_last'].min():.1f} to {df['IPC_last'].max
print(f"   Valid IPC_last (1-5): {(df['IPC_last'].between(1, 5, inclusive='b

# ================================================================================
# 3. CREATE TEMPORAL FEATURES FOR GDELT
# ================================================================================
print("\n3. Creating temporal features for GDELT...")

# Focus on most reliable features for temporal patterns
# Prioritize: sentiment, crisis indicators, food security
key_gdelt_features = [
    # Sentiment (most reliable NLP signals)
    'evt_compound_score_mean', 'evt_sentiment.compound_mean',

    # Crisis indicators (high signal-to-noise)
    'evt_fatalities_freq_mean', 'evt_displaced_freq_mean',

    # Food security (directly relevant to IPC)
    'evt_food_insecurity_freq_mean',

    # Economic/agricultural (relevant but less reliable)
    'evt_economic_shocks_freq_mean', 'evt_agriculture_freq_mean',
]
```

```
# Filter to features that actually exist
key_gdelt_features = [f for f in key_gdelt_features if f in df.columns]

print(f"   Creating temporal features for {len(key_gdelt_features)} key GDEL

# Ensure sorted for lag operations
df = df.sort_values(['ADMIN0', 'ADMIN1', 'ADMIN2', 'period']).reset_index(dr

# Create rolling windows (smoothing noise - most useful for NLP features)
# Use 3 and 6 periods to capture short and medium-term trends
for window in [3, 6]:
    for feat in key_gdelt_features:
        df[f"{feat}_rolling_{window}"] = df.groupby(['ADMIN0', 'ADMIN1', 'AD
            lambda x: x.rolling(window=window, min_periods=1).mean()
        )

# Create lags (1 and 2 periods only - reduce noise)
# Lags can capture trends but too many lags add noise
for lag in [1, 2]:
    for feat in key_gdelt_features:
        df[f"{feat}_lag{lag}"] = df.groupby(['ADMIN0', 'ADMIN1', 'ADMIN2'],

# Create escalation indicators (binary - less noisy than continuous change)
# Use rolling mean comparison instead of raw change to reduce noise
for feat in key_gdelt_features:
    rolling_3 = df.groupby(['ADMIN0', 'ADMIN1', 'ADMIN2'], sort=False)[feat]
        lambda x: x.rolling(window=3, min_periods=1).mean()
    )
    rolling_6 = df.groupby(['ADMIN0', 'ADMIN1', 'ADMIN2'], sort=False)[feat]
        lambda x: x.rolling(window=6, min_periods=2).mean()
    )
    # Escalation: current 3-period average > 6-period average
    df[f"{feat}_escalation"] = (rolling_3 > rolling_6).astype(int)

# Skip: change, pct_change, anomaly (too noisy for NLP-derived features)

print(f"   Created temporal features: rolling windows (3, 6 periods), lags (

# ============================================================================
# 4. CREATE FEATURE INDICATORS AND INTERACTIONS
# ============================================================================
print("\n4. Creating feature indicators and interactions...")

# GDELT feature availability indicator
gdelt_cols = [c for c in df.columns if c.startswith('evt_')]
if gdelt_cols:
    has_gdelt = df[gdelt_cols].notna().any(axis=1)
    df['has_gdelt'] = has_gdelt.astype(int)
else:
    df['has_gdelt'] = 0

# Sentiment features from events

# Combine crisis indicators
crisis_features = [c for c in df.columns if any(x in c for x in ['fatalities
                   and c.endswith('_mean') and c.startswith('evt_')]
```

```python
if crisis_features:
    df['crisis_intensity'] = df[crisis_features].fillna(0).sum(axis=1)

# Food security combined
food_features = [c for c in df.columns if 'food_insecurity' in c and c.endsw
                 and c.startswith('evt_')]
if food_features:
    df['food_security_combined'] = df[food_features].fillna(0).mean(axis=1)

print(f"   Created interaction and combined features")

# ============================================================================
# 5. CREATE GEOGRAPHIC DUMMY VARIABLES
# ============================================================================
print("\n5. Creating geographic dummy variables...")

# Only create dummies for base geographic identifiers (ADMIN0, ADMIN1, ADMIN
# Note: 'region' and 'district' are redundant (derived from ADMIN0+ADMIN1 ar
geographic_cols = []
for col in ['ADMIN0', 'ADMIN1', 'ADMIN2']:
    if col in df.columns:
        # Create dummy variables (one-hot encoding)
        dummies = pd.get_dummies(df[col], prefix=col, drop_first=False)
        df = pd.concat([df, dummies], axis=1)
        geographic_cols.extend(dummies.columns.tolist())
        print(f"   Created {len(dummies.columns)} dummy variables for {col}"

geographic_features = geographic_cols
print(f"   Total geographic features: {len(geographic_features)}")
print(f"   Note: 'region' and 'district' are kept as categorical identifiers

# ============================================================================
# 6. SUMMARY
# ============================================================================
print("\n" + "="*70)
print("FEATURE ENGINEERING COMPLETE")
print("="*70)

print(f"\nFinal dataset: {len(df)} rows")
print(f"Total columns: {len(df.columns)}")

# Feature counts
ipc_features = ['IPC_last']

# GDELT base features: evt_ prefixed features
gdelt_base = [c for c in df.columns if c.startswith('evt_')
              and not any(x in c for x in ['_lag', '_rolling', '_escalation'

# GDELT temporal features: evt_ prefixed features with temporal patterns
gdelt_temporal = [c for c in df.columns if c.startswith('evt_')
                  and any(x in c for x in ['_lag', '_rolling', '_escalatior

# Geographic features were created as dummies in section 5 above
# geographic_features is already set to geographic_cols

# Geographic features: only ADMIN0, ADMIN1, ADMIN2 are converted to dummies
```

```
# 'region' and 'district' are kept as categorical identifiers for grouping b
# Exclude original categorical columns from feature lists (but keep them in
geographic_categorical_cols = ['ADMIN0', 'ADMIN1', 'ADMIN2', 'region', 'dist

# Exclude all non-feature columns (geographic categoricals, metadata, scores
excluded_cols = (ipc_features + gdelt_base + gdelt_temporal + geographic_fea
                 ['y_next', 'has_gdelt', 'has_articles',
                  'CS_score',
                  'period', 'SQLDATE', 'EventCode', 'SOURCEURL', 'NumMentior
                  'NumArticles', 'valid_SOURCEURL'] +
                 geographic_categorical_cols +
                 # Also exclude any remaining list columns that weren't aggr
                 [c for c in df.columns if '_list' in c])
other_features = [c for c in df.columns if c not in excluded_cols]

print(f"\nFeature breakdown:")
print(f"  - IPC features: {len(ipc_features)}")
print(f"  - GDELT base features: {len(gdelt_base)} (mean, count, sum, max fo
print(f"  - GDELT temporal features: {len(gdelt_temporal)} (rolling windows,
print(f"  - Geographic features: {len(geographic_features)} (dummy variables
print(f"  - Other features: {len(other_features)}")
print(f"  - Total features: {len(ipc_features) + len(gdelt_base) + len(gdelt
print(f"\nNote: Simplified feature set for noisy NLP-derived signals:")
print(f"  - Base: mean, count, sum (all features) + max (crisis features onl
print(f"  - Temporal: rolling windows (3, 6 periods), lags (1, 2 periods), e
print(f"  - Removed: min, std, change, pct_change, anomaly (too noisy for NL

print(f"\nRows with y_next: {df['y_next'].notna().sum()} ({df['y_next'].notr
print(f"Rows with IPC_last: {df['IPC_last'].notna().sum()} ({df['IPC_last'].
print(f"Rows with GDELT features: {df['has_gdelt'].sum()} ({df['has_gdelt'].

# ============================================================================
# 6. DROP PROCESSED/METADATA COLUMNS
# ============================================================================
print("\n6. Dropping processed metadata columns...")

cols_to_drop = ['SQLDATE', 'EventCode', 'SOURCEURL', 'NumMentions',
                'NumSources', 'NumArticles', 'valid_SOURCEURL']
# Only drop columns that exist
cols_to_drop = [c for c in cols_to_drop if c in df.columns]
if cols_to_drop:
    df.drop(cols_to_drop, axis=1, inplace=True)
    print(f"   Dropped {len(cols_to_drop)} metadata columns")

print("\n" + "="*70)
print("READY FOR MODELING")
print("="*70)
print("\nFeature sets available:")
print(f"  - IPC only: {len(ipc_features)} IPC features + {len(geographic_fea
print(f"  - IPC + GDELT base: {len(ipc_features)} IPC + {len(gdelt_base)} GD
print(f"  - IPC + GDELT + temporal: {len(ipc_features)} IPC + {len(gdelt_bas
print("="*70)
```

```
================================================================================
COMPREHENSIVE FEATURE ENGINEERING
================================================================================

Starting with 69645 rows
Columns before feature engineering: 55

1. Aggregating GDELT list features...
   Found 39 list columns
   Aggregating events features...

2. Cleaning IPC features...
   IPC_last range: 1.0 to 5.0
   Valid IPC_last (1-5): 69645 rows

3. Creating temporal features for GDELT...
   Creating temporal features for 7 key GDELT features
   Created temporal features: rolling windows (3, 6 periods), lags (1, 2 per
iods), escalation indicators

4. Creating feature indicators and interactions...
   Created interaction and combined features

5. Creating geographic dummy variables...
   Created 24 dummy variables for ADMIN0
   Created 338 dummy variables for ADMIN1
   Created 2969 dummy variables for ADMIN2
   Total geographic features: 3331
   Note: 'region' and 'district' are kept as categorical identifiers but not
converted to dummies (redundant with ADMIN0+ADMIN1+ADMIN2)

================================================================================
FEATURE ENGINEERING COMPLETE
================================================================================

Final dataset: 69645 rows
Total columns: 3504

Feature breakdown:
  - IPC features: 1
  - GDELT base features: 119 (mean, count, sum, max for crisis)
  - GDELT temporal features: 35 (rolling windows, lags, escalation)
  - Geographic features: 3331 (dummy variables)
  - Other features: 2
  - Total features: 3488

Note: Simplified feature set for noisy NLP-derived signals:
  - Base: mean, count, sum (all features) + max (crisis features only)
  - Temporal: rolling windows (3, 6 periods), lags (1, 2 periods), escalatio
n indicators
  - Removed: min, std, change, pct_change, anomaly (too noisy for NLP featur
es)

Rows with y_next: 65649 (94.3%)
Rows with IPC_last: 69645 (100.0%)
Rows with GDELT features: 69645 (100.0%)
```

```
6. Dropping processed metadata columns...
   Dropped 7 metadata columns

======================================================================
READY FOR MODELING
======================================================================

Feature sets available:
  - IPC only: 1 IPC features + 3331 geographic features
  - IPC + GDELT base: 1 IPC + 119 GDELT base + 3331 geographic
  - IPC + GDELT + temporal: 1 IPC + 119 GDELT base + 35 GDELT temporal + 333
1 geographic
======================================================================
```

In [10]:
```
print(min(df['period']))
print(max(df['period']))
df = df.sort_values(['ADMIN0', 'ADMIN1', 'ADMIN2', 'period'])
df.tail()
```

```
201602
202406
```

Out[10]:

	ADMIN0	ADMIN1	ADMIN2	CS_score	period	region	district	IPC
69640	zimbabwe	midlands	zvishavane urban	2.0	202206	zimbabwe-midlands	zimbabwe-midlands-zvishavane urban	
69641	zimbabwe	midlands	zvishavane urban	2.0	202210	zimbabwe-midlands	zimbabwe-midlands-zvishavane urban	
69642	zimbabwe	midlands	zvishavane urban	2.0	202302	zimbabwe-midlands	zimbabwe-midlands-zvishavane urban	
69643	zimbabwe	midlands	zvishavane urban	2.0	202306	zimbabwe-midlands	zimbabwe-midlands-zvishavane urban	
69644	zimbabwe	midlands	zvishavane urban	2.0	202402	zimbabwe-midlands	zimbabwe-midlands-zvishavane urban	

Configure train and test, and loops

In [11]:
```
# ======================================================================
# NAIVE BASELINE MODELS
# ======================================================================
```

```python
# PPS     — Previous Period Score: predict y_next = IPC_last (no-change)
# SPLY    — Same Period Last Year: CS_score from ~12 months ago
# Max-2PP — Max of 2 Previous Periods: max CS_score of last 2 assessments
#
# Evaluated on every assessment period that has y_next data, then averaged.
# ============================================================================

from sklearn.metrics import accuracy_score, precision_score, recall_score, f

admin_cols = ['ADMIN0', 'ADMIN1', 'ADMIN2']

# All assessment periods with valid CS_score data
eval_periods = sorted(
    df.loc[df['CS_score'].between(1, 5), 'period'].unique()
)

print(f"Evaluating baselines on {len(eval_periods)} assessment periods: {eva

# ── Evaluate each baseline across all periods ──────────────────────────
baseline_names = ['PPS', 'SPLY', 'Max-2PP']
period_results = {name: [] for name in baseline_names}

for test_period in eval_periods:
    test = df[(df['period'] == test_period) & df['CS_score'].between(1, 5)].
    train = df[(df['period'] < test_period) & df['CS_score'].between(1, 5)].
    train_periods = sorted(train['period'].unique())

    if len(test) == 0 or len(train_periods) < 2:
        continue

    y_test = test['CS_score'].astype(int)

    # PPS: previous assessment's CS_score (last training period, NOT current
    pps_lookup = (train[train['period'] == train_periods[-1]][admin_cols + [
                  .drop_duplicates(admin_cols))
    pps_merged = test.merge(pps_lookup.rename(columns={'CS_score': 'pred'}),
                            on=admin_cols, how='left')
    pps_pred = pps_merged['pred'].fillna(0).astype(int)

    # SPLY: CS_score from closest assessment ~12 months before test
    target_sply = str(int(test_period) - 100)
    sply_period = min(train_periods, key=lambda p: abs(int(p) - int(target_s
    sply_lookup = (train[train['period'] == sply_period][admin_cols + ['CS_s
                   .drop_duplicates(admin_cols))
    sply_merged = test.merge(sply_lookup.rename(columns={'CS_score': 'pred'}
                             on=admin_cols, how='left')
    sply_pred = sply_merged['pred'].fillna(0).astype(int)

    # Max-2PP: max CS_score from last 2 assessment periods
    last2 = train_periods[-2:]
    max2pp_lookup = (train[train['period'].isin(last2)]
                     .groupby(admin_cols)['CS_score'].max().reset_index())
    max2pp_merged = test.merge(max2pp_lookup.rename(columns={'CS_score': 'pr
                               on=admin_cols, how='left')
    max2pp_pred = max2pp_merged['pred'].fillna(0).astype(int)
```

```
    preds = {'PPS': pps_pred, 'SPLY': sply_pred, 'Max-2PP': max2pp_pred}

    for name in baseline_names:
        period_results[name].append({
            'period': test_period,
            'n_test': len(test),
            'accuracy': accuracy_score(y_test, preds[name]),
            'precision': precision_score(y_test, preds[name], average='weigh
            'recall': recall_score(y_test, preds[name], average='weighted',
            'f1': f1_score(y_test, preds[name], average='weighted', zero_div
        })

# ── Summary ──────────────────────────────────────────────────────────────
rows = []
for name in baseline_names:
    res = period_results[name]
    rows.append({
        'Model': name,
        'Accuracy': np.mean([r['accuracy'] for r in res]),
        'Precision': np.mean([r['precision'] for r in res]),
        'Recall': np.mean([r['recall'] for r in res]),
        'F1': np.mean([r['f1'] for r in res]),
        'N Periods': len(res),
        'Total Test': sum(r['n_test'] for r in res),
    })

baseline_summary = pd.DataFrame(rows)
print(baseline_summary.to_string(index=False))
```

```
Evaluating baselines on 26 assessment periods: 201602 → 202406

  Model  Accuracy  Precision   Recall       F1  N Periods  Total Test
    PPS  0.707565   0.780899 0.707565 0.727106         24       55423
   SPLY  0.676253   0.741235 0.676253 0.695393         24       55423
Max-2PP  0.730698   0.768227 0.730698 0.738371         24       55423
```

ML from here

In [12]:
```
n_training_periods = 20     # months of training data before each test asses
n_eval_periods = 26          # number of IPC assessment periods to evaluate

# Optional: set to None to auto-detect latest assessment, or specify e.g. 20
start_period = None
```

In [13]:
```
# ============================================================================
# ROLLING WINDOW EVALUATION — TEST ON IPC ASSESSMENT PERIODS
# ============================================================================
# Strategy:
#   - Identify IPC assessment periods (months where FEWS NET published score
#     for >= min_obs_per_assessment admin2 areas).
#   - Test on each assessment period: rows with valid y_next → predict next
#   - Train on all prior periods (up to n_training_periods months before).
#   - No leakage: IPC_last tells the model the *current* assessment;
#     y_next is the *future* assessment — different values.
```

```python
# ================================================================================

from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.preprocessing import StandardScaler
from sklearn.base import clone
from sklearn.metrics import accuracy_score, precision_score, recall_score, f
from catboost import CatBoostClassifier

all_periods = sorted(df['period'].unique())

# ── 1. Identify IPC assessment periods ──────────────────────────────────────
ipc_obs_per_period = df[df['CS_score'].between(1, 5)].groupby('period').size
assessment_periods = sorted(ipc_obs_per_period.index.tolist())

print("="*70)
print("IPC ASSESSMENT PERIODS")
print("="*70)
print(f"\nTotal periods in data: {len(all_periods)}  ({all_periods[0]} → {al
print(f"Assessment periods (with CS_score data): {len(assessment_periods)}\r
for p in assessment_periods:
    print(f"  {p}:  {ipc_obs_per_period[p]:>5,} scored")

# ── 2. Select evaluation assessment periods (backwards from latest) ─────────
if start_period is not None:
    candidates = [p for p in assessment_periods if p <= start_period]
else:
    candidates = list(assessment_periods)

eval_assessments = candidates[-n_eval_periods:]  # latest N

print(f"\n{'='*70}")
print("ROLLING WINDOW EVALUATION SETUP")
print(f"{'='*70}")
print(f"\nConfiguration:")
print(f"  - Training window:       {n_training_periods} periods before test
print(f"  - Assessments to test:   {len(eval_assessments)} (of {len(assessme
print(f"  - Assessment periods:    {len(assessment_periods)}")

# ── 3. Create evaluation splits ─────────────────────────────────────────────
evaluation_splits = []

for i, test_period in enumerate(reversed(eval_assessments)):
    test_idx = all_periods.index(test_period)
    train_end_idx = test_idx - 1  # everything BEFORE test period
    train_start_idx = max(0, train_end_idx - n_training_periods + 1)

    if train_end_idx < 0:
        print(f"\n  ⚠️  Assessment {test_period}: no training data available
        continue

    train_periods = all_periods[train_start_idx:train_end_idx + 1]

    evaluation_splits.append({
        'eval_id': i + 1,
        'test_period': test_period,
```

```
        'n_scored': int(ipc_obs_per_period.get(test_period, 0)),
        'train_periods': train_periods,
        'train_start': train_periods[0],
        'train_end': train_periods[-1],
        'n_train_periods': len(train_periods),
    })

print(f"\nEvaluation schedule (latest → earliest):\n")
for s in evaluation_splits:
    print(f"  Eval {s['eval_id']}: test {s['test_period']}  "
          f"({s['n_scored']:,} scored)  |  "
          f"train {s['train_start']} → {s['train_end']}  "
          f"({s['n_train_periods']} periods)")

print(f"\n{'='*70}")
print(f"Created {len(evaluation_splits)} evaluation splits")
print("="*70)

# ============================================================================
# DEFINE FEATURE SETS
# ============================================================================
if 'ipc_features' in globals():
    ipc_feat = [c for c in ipc_features if c in df.columns]
else:
    ipc_feat = ['IPC_last']

_temporal_suffixes = ('_rolling_', '_lag', '_escalation')

if 'gdelt_features' in globals():
    gdelt_base_feat = [c for c in gdelt_features if c in df.columns]
else:
    gdelt_base_feat = [c for c in df.columns
                       if c.startswith('evt_') and not any(s in c for s in _

if 'temporal_features' in globals():
    gdelt_temporal_feat = [c for c in temporal_features if c in df.columns]
else:
    gdelt_temporal_feat = [c for c in df.columns
                           if c.startswith('evt_') and any(s in c for s in _

if 'geographic_features' in globals():
    geographic_feat = [c for c in geographic_features if c in df.columns]
else:
    geographic_feat = [c for c in df.columns if c.startswith('geo_')]

feature_sets = {
    # With geographic dummies (location fixed effects)
    'IPC only':               ipc_feat + geographic_feat,
    'IPC + GDELT base':       ipc_feat + gdelt_base_feat + geographic_feat,
    'IPC + GDELT + temporal': ipc_feat + gdelt_base_feat + gdelt_temporal_fe
    'GDELT only':             gdelt_base_feat + gdelt_temporal_feat + geogra
    # Without geographic dummies (pure feature signal)
    'IPC only (no geo)':               ipc_feat,
    'IPC + GDELT base (no geo)':       ipc_feat + gdelt_base_feat,
    'IPC + GDELT + temporal (no geo)': ipc_feat + gdelt_base_feat + gdelt_te
    'GDELT only (no geo)':             gdelt_base_feat + gdelt_temporal_feat
```

```
}

print("\nFeature sets available:")
for name, features in feature_sets.items():
    n_ipc = len([c for c in features if c in ipc_feat])
    n_gdelt_base = len([c for c in features if c in gdelt_base_feat])
    n_gdelt_temp = len([c for c in features if c in gdelt_temporal_feat])
    n_geo = len([c for c in features if c in geographic_feat])
    print(f"  - {name}: {n_ipc} IPC + {n_gdelt_base} GDELT base + {n_gdelt_t

# ============================================================================
# DEFINE MODELS
# ============================================================================
models = {
    'Logistic Regression': {
        'model': LogisticRegression(max_iter=1000, random_state=42, solver='
        'needs_scaling': True,
    },
    'Random Forest': {
        'model': RandomForestClassifier(
            n_estimators=150, max_depth=12, min_samples_split=15,
            min_samples_leaf=8, max_features='sqrt',
            class_weight='balanced', random_state=42, n_jobs=-1
        ),
        'needs_scaling': False,
    },
    'CatBoost': {
        'model': CatBoostClassifier(
            iterations=300, depth=4, learning_rate=0.05,
            loss_function='MultiClass', auto_class_weights='Balanced',
            l2_leaf_reg=5, colsample_bylevel=0.7,
            random_seed=42, verbose=0
        ),
        'needs_scaling': False,
    },
}

print(f"\nModels to evaluate: {', '.join(models.keys())}")

# ============================================================================
# TRAIN AND EVALUATE
# ============================================================================
print("\n" + "="*70)
print("TRAINING AND EVALUATION")
print("="*70)

all_results = {}  # keyed by (model_name, feature_set_name)
rf_importances = {}  # keyed by feature_set_name → list of (feature_cols, im

for model_name, model_cfg in models.items():
    for feature_set_name, feature_cols in feature_sets.items():
        result_key = (model_name, feature_set_name)
        print(f"\n{'='*70}")
        print(f"MODEL: {model_name}  |  FEATURE SET: {feature_set_name}")
        print(f"{'='*70}")
        print(f"Using {len(feature_cols)} features")
```

```
    feature_results = []

    for split in evaluation_splits:
        print(f"\n  ── Eval {split['eval_id']}: Assessment period {split

        # ── Training set: all prior periods with valid y_next ────────
        #    (includes inter-assessment rows with GDELT features)
        train_df = df[
            (df['period'].isin(split['train_periods'])) &
            (df['y_next'].notna())
        ].copy()

        # ── Test set: rows from assessment period with valid CS_score -
        test_df = df[
            (df['period'] == split['test_period']) &
            (df['CS_score'].between(1, 5))
        ].copy()

        print(f"  Training samples:   {len(train_df):,}")
        print(f"  Test samples:       {len(test_df):,}  (admin2 areas ir

        if len(train_df) == 0:
            print(f"  ⚠  Skipping — no training data")
            continue
        if len(test_df) == 0:
            print(f"  ⚠  Skipping — no test data")
            continue

        # ── Prepare features and target ──────────────────────────────
        X_train = train_df[feature_cols].fillna(0)
        y_train = train_df['y_next'].astype(int)

        X_test = test_df[feature_cols].fillna(0)
        y_test = test_df['CS_score'].astype(int)

        # ── Fix IPC_last leakage: at assessment periods IPC_last == CS_
        #    so replace it with the previous assessment's score ────────
        if 'IPC_last' in feature_cols:
            ipc_col_idx = feature_cols.index('IPC_last')
            # Find previous assessment period in training window
            train_assessments = sorted(
                p for p in assessment_periods if p in split['train_perio
            )
            if train_assessments:
                prev_period = train_assessments[-1]
                prev_ipc = (
                    df.loc[(df['period'] == prev_period) & df['CS_score'
                           ['ADMIN0', 'ADMIN1', 'ADMIN2', 'CS_score']]
                    .drop_duplicates(['ADMIN0', 'ADMIN1', 'ADMIN2'])
                    .rename(columns={'CS_score': '_prev_ipc'})
                )
                test_with_prev = test_df.merge(prev_ipc, on=['ADMIN0', '
                if isinstance(X_test, np.ndarray):
                    X_test[:, ipc_col_idx] = test_with_prev['_prev_ipc']
                else:
```

```
                        X_test['IPC_last'] = test_with_prev['_prev_ipc'].fil

            # ── Scale (only for models that need it) ──────────────────
            if model_cfg['needs_scaling']:
                scaler = StandardScaler()
                X_train_fit = scaler.fit_transform(X_train)
                X_test_fit = scaler.transform(X_test)
            else:
                X_train_fit = X_train
                X_test_fit = X_test

            # ── Train ─────────────────────────────────────────────────
            print(f"  Training {model_name}...")
            model = clone(model_cfg['model'])
            model.fit(X_train_fit, y_train)

            y_pred = model.predict(X_test_fit)

            # ── Capture Random Forest feature importances ─────────────
            if model_name == 'Random Forest':
                if feature_set_name not in rf_importances:
                    rf_importances[feature_set_name] = []
                rf_importances[feature_set_name].append(
                    pd.Series(model.feature_importances_, index=feature_cols
                )

            # ── Metrics ───────────────────────────────────────────────
            accuracy  = accuracy_score(y_test, y_pred)
            precision = precision_score(y_test, y_pred, average='weighted',
            recall    = recall_score(y_test, y_pred, average='weighted', zer
            f1        = f1_score(y_test, y_pred, average='weighted', zero_di
            cm        = confusion_matrix(y_test, y_pred)

            feature_results.append({
                'eval_id': split['eval_id'],
                'test_period': split['test_period'],
                'n_train': len(train_df),
                'n_test': len(test_df),
                'accuracy': accuracy,
                'precision': precision,
                'recall': recall,
                'f1': f1,
                'confusion_matrix': cm
            })

            print(f"\n  Results  (n_test={len(test_df):,}):")
            print(f"    Accuracy:  {accuracy:.4f}")
            print(f"    Precision: {precision:.4f}")
            print(f"    Recall:    {recall:.4f}")
            print(f"    F1 Score:  {f1:.4f}")
            print(f"\n  Confusion Matrix:\n    {cm}")

        all_results[result_key] = feature_results

# ============================================================================
# SUMMARY
```

```python
# ============================================================================
if any(len(v) > 0 for v in all_results.values()):
    print("\n" + "="*70)
    print("SUMMARY: AVERAGE RESULTS BY MODEL & FEATURE SET")
    print("="*70)

    summary_data = []
    for (model_name, feature_set_name), results in all_results.items():
        if len(results) == 0:
            continue

        avg_acc  = np.mean([r['accuracy']  for r in results])
        avg_prec = np.mean([r['precision'] for r in results])
        avg_rec  = np.mean([r['recall']    for r in results])
        avg_f1   = np.mean([r['f1']        for r in results])
        total_test = sum(r['n_test'] for r in results)

        print(f"\n{model_name} — {feature_set_name}:")
        print("-" * 70)
        print(f"  Average Metrics (across {len(results)} assessments, {total
        print(f"    Accuracy:  {avg_acc:.4f}")
        print(f"    Precision: {avg_prec:.4f}")
        print(f"    Recall:    {avg_rec:.4f}")
        print(f"    F1 Score:  {avg_f1:.4f}")

        # Per-evaluation breakdown
        print(f"\n  Per assessment:")
        for r in results:
            print(f"    {r['test_period']}:  n={r['n_test']:>5,}  acc={r['ac

        n_features = len(feature_sets[feature_set_name])
        n_ipc = len([c for c in feature_sets[feature_set_name] if c in ipc_f
        n_gdelt_base = len([c for c in feature_sets[feature_set_name] if c i
        n_gdelt_temp = len([c for c in feature_sets[feature_set_name] if c i
        n_geo = len([c for c in feature_sets[feature_set_name] if c in geogr

        summary_data.append({
            'Model': model_name,
            'Feature Set': feature_set_name,
            'N Features': n_features,
            'IPC': n_ipc,
            'GDELT Base': n_gdelt_base,
            'GDELT Temporal': n_gdelt_temp,
            'Geographic': n_geo,
            'Accuracy': avg_acc,
            'Precision': avg_prec,
            'Recall': avg_rec,
            'F1 Score': avg_f1,
            'Total Test': total_test,
            'N Evals': len(results),
        })

    summary_df = pd.DataFrame(summary_data)
    summary_df = summary_df.sort_values('F1 Score', ascending=False)

    print(f"\n{'='*70}")
```

```
    print("SUMMARY TABLE")
    print("="*70)
    print(summary_df.to_string(index=False))

else:
    print("\n⚠️  No evaluations completed — check data availability")

# ============================================================================
# RANDOM FOREST — AVERAGE FEATURE IMPORTANCES
# ============================================================================
if rf_importances:
    print("\n" + "="*70)
    print("RANDOM FOREST: AVERAGE FEATURE IMPORTANCES")
    print("="*70)

    for fs_name, imp_list in rf_importances.items():
        avg_imp = pd.concat(imp_list, axis=1).mean(axis=1).sort_values(ascer
        print(f"\n── {fs_name} (averaged over {len(imp_list)} splits) ──")
        print(f"{'Feature':<55} {'Importance':>10}")
        print("-" * 67)
        for feat, imp in avg_imp.head(25).items():
            print(f"  {feat:<53} {imp:>10.4f}")
        if len(avg_imp) > 25:
            print(f"  ... ({len(avg_imp) - 25} more features omitted)")
```

```
================================================================================
IPC ASSESSMENT PERIODS
================================================================================

Total periods in data: 101  (201602 → 202406)
Assessment periods (with CS_score data): 26

  201602:  2,639 scored
  201606:  2,635 scored
  201610:  2,748 scored
  201702:  2,449 scored
  201706:  2,573 scored
  201710:  2,577 scored
  201802:  2,441 scored
  201806:  2,440 scored
  201810:  1,081 scored
  201902:  2,400 scored
  201906:  2,400 scored
  201910:  2,400 scored
  202002:  1,588 scored
  202006:  2,570 scored
  202010:  2,491 scored
  202102:  2,489 scored
  202106:  2,536 scored
  202110:  2,535 scored
  202202:  2,535 scored
  202206:  2,568 scored
  202210:  2,399 scored
  202302:  1,376 scored
  202306:  2,613 scored
  202310:  1,919 scored
  202402:  2,376 scored
  202406:  1,919 scored

================================================================================
ROLLING WINDOW EVALUATION SETUP
================================================================================

Configuration:
  - Training window:       20 periods before test assessment
  - Assessments to test:   26 (of 26 total)
  - Assessment periods:    26

  ⚠️  Assessment 201602: no training data available — skipping

Evaluation schedule (latest → earliest):

  Eval 1: test 202406  (1,919 scored)  |  train 202210 → 202405  (20 period
s)
  Eval 2: test 202402  (2,376 scored)  |  train 202206 → 202401  (20 period
s)
  Eval 3: test 202310  (1,919 scored)  |  train 202202 → 202309  (20 period
s)
  Eval 4: test 202306  (2,613 scored)  |  train 202110 → 202305  (20 period
s)
  Eval 5: test 202302  (1,376 scored)  |  train 202106 → 202301  (20 period
```

```
s)
  Eval 6: test 202210  (2,399 scored)  |  train 202102 → 202209  (20 period
s)
  Eval 7: test 202206  (2,568 scored)  |  train 202010 → 202205  (20 period
s)
  Eval 8: test 202202  (2,535 scored)  |  train 202006 → 202201  (20 period
s)
  Eval 9: test 202110  (2,535 scored)  |  train 202002 → 202109  (20 period
s)
  Eval 10: test 202106  (2,536 scored)  |  train 201910 → 202105  (20 period
s)
  Eval 11: test 202102  (2,489 scored)  |  train 201906 → 202101  (20 period
s)
  Eval 12: test 202010  (2,491 scored)  |  train 201902 → 202009  (20 period
s)
  Eval 13: test 202006  (2,570 scored)  |  train 201810 → 202005  (20 period
s)
  Eval 14: test 202002  (1,588 scored)  |  train 201806 → 202001  (20 period
s)
  Eval 15: test 201910  (2,400 scored)  |  train 201802 → 201909  (20 period
s)
  Eval 16: test 201906  (2,400 scored)  |  train 201710 → 201905  (20 period
s)
  Eval 17: test 201902  (2,400 scored)  |  train 201706 → 201901  (20 period
s)
  Eval 18: test 201810  (1,081 scored)  |  train 201702 → 201809  (20 period
s)
  Eval 19: test 201806  (2,440 scored)  |  train 201610 → 201805  (20 period
s)
  Eval 20: test 201802  (2,441 scored)  |  train 201606 → 201801  (20 period
s)
  Eval 21: test 201710  (2,577 scored)  |  train 201602 → 201709  (20 period
s)
  Eval 22: test 201706  (2,573 scored)  |  train 201602 → 201705  (16 period
s)
  Eval 23: test 201702  (2,449 scored)  |  train 201602 → 201701  (12 period
s)
  Eval 24: test 201610  (2,748 scored)  |  train 201602 → 201609  (8 period
s)
  Eval 25: test 201606  (2,635 scored)  |  train 201602 → 201605  (4 period
s)

======================================================================
Created 25 evaluation splits
======================================================================

Feature sets available:
  - IPC only: 1 IPC + 0 GDELT base + 0 GDELT temporal + 3331 geographic = 33
32 total
  - IPC + GDELT base: 1 IPC + 119 GDELT base + 0 GDELT temporal + 3331 geogr
aphic = 3451 total
  - IPC + GDELT + temporal: 1 IPC + 119 GDELT base + 35 GDELT temporal + 333
1 geographic = 3486 total
  - GDELT only: 0 IPC + 119 GDELT base + 35 GDELT temporal + 3331 geographic
= 3485 total
  - IPC only (no geo): 1 IPC + 0 GDELT base + 0 GDELT temporal + 0 geographi
```

```
c = 1 total
  - IPC + GDELT base (no geo): 1 IPC + 119 GDELT base + 0 GDELT temporal + 0
geographic = 120 total
  - IPC + GDELT + temporal (no geo): 1 IPC + 119 GDELT base + 35 GDELT tempo
ral + 0 geographic = 155 total
  - GDELT only (no geo): 0 IPC + 119 GDELT base + 35 GDELT temporal + 0 geog
raphic = 154 total

Models to evaluate: Logistic Regression, Random Forest, CatBoost

======================================================================
TRAINING AND EVALUATION
======================================================================

======================================================================
MODEL: Logistic Regression  |  FEATURE SET: IPC only
======================================================================
Using 3332 features

  —— Eval 1: Assessment period 202406 ——
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Training Logistic Regression...

  Results  (n_test=1,919):
    Accuracy:  0.7092
    Precision: 0.7283
    Recall:    0.7092
    F1 Score:  0.7087

  Confusion Matrix:
    [[423  57   0   0]
 [270 526  34   0]
 [  2 135 319  12]
 [  0   2  46  93]]

  —— Eval 2: Assessment period 202402 ——
  Training samples:   12,107
  Test samples:       2,376  (admin2 areas in 202402)
  Training Logistic Regression...

  Results  (n_test=2,376):
    Accuracy:  0.7908
    Precision: 0.8051
    Recall:    0.7908
    F1 Score:  0.7920

  Confusion Matrix:
    [[711 241   5   0]
 [ 60 690  84   0]
 [  4  36 411  25]
 [  1   1  40  67]]

  —— Eval 3: Assessment period 202310 ——
  Training samples:   12,645
  Test samples:       1,919  (admin2 areas in 202310)
```

```
 Training Logistic Regression...

 Results  (n_test=1,919):
   Accuracy:  0.7598
   Precision: 0.7846
   Recall:    0.7598
   F1 Score:  0.7679

 Confusion Matrix:
   [[761 207   2   0]
[ 73 312  71   6]
[  4  39 346  40]
[  1   4  14  39]]

 —— Eval 4: Assessment period 202306 ——
 Training samples:   12,765
 Test samples:       2,613  (admin2 areas in 202306)
 Training Logistic Regression...

 Results  (n_test=2,613):
   Accuracy:  0.7784
   Precision: 0.8059
   Recall:    0.7784
   F1 Score:  0.7805

 Confusion Matrix:
   [[582  53   1   0]
[266 728  54   9]
[ 10  99 620  84]
[  0   0   3 104]]

 —— Eval 5: Assessment period 202302 ——
 Training samples:   13,772
 Test samples:       1,376  (admin2 areas in 202302)
 Training Logistic Regression...

 Results  (n_test=1,376):
   Accuracy:  0.8183
   Precision: 0.8248
   Recall:    0.8183
   F1 Score:  0.8195

 Confusion Matrix:
   [[448 132   0   0]
[ 75 525  10   0]
[ 11  19 153   3]
[  0   0   0   0]]

 —— Eval 6: Assessment period 202210 ——
 Training samples:   13,866
 Test samples:       2,399  (admin2 areas in 202210)
 Training Logistic Regression...

 Results  (n_test=2,399):
   Accuracy:  0.7824
   Precision: 0.7863
```

```
  Recall:    0.7824
  F1 Score:  0.7803

 Confusion Matrix:
   [[750  92   0   0]
[244 532  42   2]
[ 12  82 529  26]
[  0   0  22  66]]

 —— Eval 7: Assessment period 202206 ——
 Training samples:   13,795
 Test samples:       2,568  (admin2 areas in 202206)
 Training Logistic Regression...

 Results  (n_test=2,568):
  Accuracy:  0.8318
  Precision: 0.8331
  Recall:    0.8318
  F1 Score:  0.8244

 Confusion Matrix:
   [[795  10   1   0]
[167 725  26   0]
[  8 131 577   8]
[  0  19  62  39]]

 —— Eval 8: Assessment period 202202 ——
 Training samples:   13,862
 Test samples:       2,535  (admin2 areas in 202202)
 Training Logistic Regression...

 Results  (n_test=2,535):
  Accuracy:  0.8213
  Precision: 0.8283
  Recall:    0.8213
  F1 Score:  0.8176

 Confusion Matrix:
   [[876  38   3   0]
[216 676  16   0]
[ 10 144 521   1]
[  0   5  20   9]]

 —— Eval 9: Assessment period 202110 ——
 Training samples:   12,950
 Test samples:       2,535  (admin2 areas in 202110)
 Training Logistic Regression...

 Results  (n_test=2,535):
  Accuracy:  0.8659
  Precision: 0.8654
  Recall:    0.8659
  F1 Score:  0.8634

 Confusion Matrix:
   [[1061   54    3    0]
```

```
[ 161  648   58    0]
[   2   56  478    1]
[   0    0    5    8]]

 ── Eval 10: Assessment period 202106 ──
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Training Logistic Regression...

 Results  (n_test=2,536):
   Accuracy:  0.7973
   Precision: 0.8148
   Recall:    0.7973
   F1 Score:  0.7965

 Confusion Matrix:
   [[832  34  14   0]
[295 741  42   0]
[  7  90 442  25]
[  1   0   6   7]]

 ── Eval 11: Assessment period 202102 ──
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Training Logistic Regression...

 Results  (n_test=2,489):
   Accuracy:  0.8630
   Precision: 0.8610
   Recall:    0.8630
   F1 Score:  0.8612

 Confusion Matrix:
   [[1183   49    0    0]
[  95  559   50    0]
[   4  105  376   15]
[   0    1   22   30]]

 ── Eval 12: Assessment period 202010 ──
 Training samples:   12,858
 Test samples:       2,491  (admin2 areas in 202010)
 Training Logistic Regression...

 Results  (n_test=2,491):
   Accuracy:  0.8230
   Precision: 0.8346
   Recall:    0.8230
   F1 Score:  0.8261

 Confusion Matrix:
   [[1054   76    6    0]
[ 172  592   50    0]
[   6   88  400   42]
[   0    0    1    4]]

 ── Eval 13: Assessment period 202006 ──
```

```
 Training samples:   11,506
 Test samples:       2,570  (admin2 areas in 202006)
 Training Logistic Regression...

 Results  (n_test=2,570):
   Accuracy:  0.8054
   Precision: 0.8072
   Recall:    0.8054
   F1 Score:  0.8018

 Confusion Matrix:
   [[1049   71    8    0]
[ 244  529   33    0]
[  27   98  433    2]
[   0    0   17   59]]

 —— Eval 14: Assessment period 202002 ——
 Training samples:   12,388
 Test samples:       1,588  (admin2 areas in 202002)
 Training Logistic Regression...

 Results  (n_test=1,588):
   Accuracy:  0.8489
   Precision: 0.8563
   Recall:    0.8489
   F1 Score:  0.8462

 Confusion Matrix:
   [[627  36   5   0]
[ 39 305 117   0]
[ 19  15 368   0]
[  0   0   9  48]]

 —— Eval 15: Assessment period 201910 ——
 Training samples:   12,533
 Test samples:       2,400  (admin2 areas in 201910)
 Training Logistic Regression...

 Results  (n_test=2,400):
   Accuracy:  0.8450
   Precision: 0.8383
   Recall:    0.8450
   F1 Score:  0.8372

 Confusion Matrix:
   [[1185    9    1    0]
[ 147  327   84    1]
[  17   88  467   14]
[   0    8    3   49]]

 —— Eval 16: Assessment period 201906 ——
 Training samples:   12,605
 Test samples:       2,400  (admin2 areas in 201906)
 Training Logistic Regression...

 Results  (n_test=2,400):
```

```
  Accuracy:  0.8400
  Precision: 0.8385
  Recall:    0.8400
  F1 Score:  0.8346

 Confusion Matrix:
   [[1157   22    0    0]
[ 180  341   29    0]
[  18  115  459    2]
[   0    0   18   59]]

 —— Eval 17: Assessment period 201902 ——
 Training samples:   12,785
 Test samples:       2,400  (admin2 areas in 201902)
 Training Logistic Regression...

 Results  (n_test=2,400):
  Accuracy:  0.8633
  Precision: 0.8734
  Recall:    0.8633
  F1 Score:  0.8609

 Confusion Matrix:
   [[1313  112    3    0]
[  56  347    9    0]
[  22   69  399    3]
[   0    0   54   13]]

 —— Eval 18: Assessment period 201810 ——
 Training samples:   13,957
 Test samples:       1,081  (admin2 areas in 201810)
 Training Logistic Regression...

 Results  (n_test=1,081):
  Accuracy:  0.8871
  Precision: 0.8895
  Recall:    0.8871
  F1 Score:  0.8853

 Confusion Matrix:
   [[624  20   0   0]
[ 45 202  32   4]
[  1  10 119   7]
[  0   0   3  14]]

 —— Eval 19: Assessment period 201806 ——
 Training samples:   14,374
 Test samples:       2,440  (admin2 areas in 201806)
 Training Logistic Regression...

 Results  (n_test=2,440):
  Accuracy:  0.8160
  Precision: 0.8061
  Recall:    0.8160
  F1 Score:  0.8042
```

```
 Confusion Matrix:
   [[1261   64    6    0]
[ 214  242   59    8]
[  30   42  477   10]
[   0    0   16   11]]

 ── Eval 20: Assessment period 201802 ──
 Training samples:   14,614
 Test samples:       2,441  (admin2 areas in 201802)
 Training Logistic Regression...

 Results  (n_test=2,441):
   Accuracy:  0.9062
   Precision: 0.9090
   Recall:    0.9062
   F1 Score:  0.9070

 Confusion Matrix:
   [[1468   40    3    0]
[  67  262   28    0]
[  11   42  474   26]
[   0    2   10    8]]

 ── Eval 21: Assessment period 201710 ──
 Training samples:   14,902
 Test samples:       2,577  (admin2 areas in 201710)
 Training Logistic Regression...

 Results  (n_test=2,577):
   Accuracy:  0.8840
   Precision: 0.8854
   Recall:    0.8840
   F1 Score:  0.8829

 Confusion Matrix:
   [[1567   49   42    0]
[  60  232   75    1]
[   2   27  456   20]
[   0    4   19   23]]

 ── Eval 22: Assessment period 201706 ──
 Training samples:   11,953
 Test samples:       2,573  (admin2 areas in 201706)
 Training Logistic Regression...

 Results  (n_test=2,573):
   Accuracy:  0.7773
   Precision: 0.7646
   Recall:    0.7773
   F1 Score:  0.7673

 Confusion Matrix:
   [[1347   45   74    0]
[ 254  233   13    0]
[   4  115  388   17]
[   5    2   44   32]]
```

```
  ── Eval 23: Assessment period 201702 ──
  Training samples:   9,303
  Test samples:       2,449  (admin2 areas in 201702)
  Training Logistic Regression...

  Results  (n_test=2,449):
    Accuracy:  0.9000
    Precision: 0.8948
    Recall:    0.9000
    F1 Score:  0.8938

  Confusion Matrix:
    [[1713   11    0    0    0]
 [ 116  278   30    0    0]
 [  18   48  194    0    0]
 [   1    8   12   19    0]
 [   0    0    0    1    0]]

  ── Eval 24: Assessment period 201610 ──
  Training samples:   6,140
  Test samples:       2,748  (admin2 areas in 201610)
  Training Logistic Regression...

  Results  (n_test=2,748):
    Accuracy:  0.9269
    Precision: 0.9521
    Recall:    0.9269
    F1 Score:  0.9350

  Confusion Matrix:
    [[1782   18    0   27]
 [  30  335   14   86]
 [   4   14  391    0]
 [   0    1    7   39]]

  ── Eval 25: Assessment period 201606 ──
  Training samples:   3,099
  Test samples:       2,635  (admin2 areas in 201606)
  Training Logistic Regression...

  Results  (n_test=2,635):
    Accuracy:  1.0000
    Precision: 1.0000
    Recall:    1.0000
    F1 Score:  1.0000

  Confusion Matrix:
    [[1696    0    0    0]
 [   0  365    0    0]
 [   0    0  520    0]
 [   0    0    0   54]]

======================================================================
MODEL: Logistic Regression  |  FEATURE SET: IPC + GDELT base
======================================================================
```

```
Using 3451 features

  ── Eval 1: Assessment period 202406 ──
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Training Logistic Regression...

  Results  (n_test=1,919):
    Accuracy:  0.7113
    Precision: 0.7313
    Recall:    0.7113
    F1 Score:  0.7111

  Confusion Matrix:
    [[422  57   1   0]
 [272 525  33   0]
 [  2 133 323  10]
 [  0   2  44  95]]

  ── Eval 2: Assessment period 202402 ──
  Training samples:   12,107
  Test samples:       2,376  (admin2 areas in 202402)
  Training Logistic Regression...

  Results  (n_test=2,376):
    Accuracy:  0.7912
    Precision: 0.8053
    Recall:    0.7912
    F1 Score:  0.7925

  Confusion Matrix:
    [[713 239   5   0]
 [ 61 689  82   2]
 [  4  39 410  23]
 [  1   1  39  68]]

  ── Eval 3: Assessment period 202310 ──
  Training samples:   12,645
  Test samples:       1,919  (admin2 areas in 202310)
  Training Logistic Regression...

  Results  (n_test=1,919):
    Accuracy:  0.7603
    Precision: 0.7864
    Recall:    0.7603
    F1 Score:  0.7688

  Confusion Matrix:
    [[762 205   2   1]
 [ 67 316  73   6]
 [  5  40 343  41]
 [  1   4  15  38]]

  ── Eval 4: Assessment period 202306 ──
  Training samples:   12,765
  Test samples:       2,613  (admin2 areas in 202306)
```

```
 Training Logistic Regression...

 Results  (n_test=2,613):
   Accuracy:  0.7819
   Precision: 0.8065
   Recall:    0.7819
   F1 Score:  0.7844

 Confusion Matrix:
   [[568  67   1   0]
[253 744  51   9]
[ 11  94 628  80]
[  0   2   2 103]]

 ── Eval 5: Assessment period 202302 ──
 Training samples:   13,772
 Test samples:       1,376  (admin2 areas in 202302)
 Training Logistic Regression...

 Results  (n_test=1,376):
   Accuracy:  0.8140
   Precision: 0.8198
   Recall:    0.8140
   F1 Score:  0.8151

 Confusion Matrix:
   [[450 130   0   0]
[ 75 520  15   0]
[ 11  22 150   3]
[  0   0   0   0]]

 ── Eval 6: Assessment period 202210 ──
 Training samples:   13,866
 Test samples:       2,399  (admin2 areas in 202210)
 Training Logistic Regression...

 Results  (n_test=2,399):
   Accuracy:  0.7795
   Precision: 0.7828
   Recall:    0.7795
   F1 Score:  0.7771

 Confusion Matrix:
   [[747  95   0   0]
[245 528  44   3]
[ 13  77 533  26]
[  0   0  26  62]]

 ── Eval 7: Assessment period 202206 ──
 Training samples:   13,795
 Test samples:       2,568  (admin2 areas in 202206)
 Training Logistic Regression...

 Results  (n_test=2,568):
   Accuracy:  0.8306
   Precision: 0.8345
```

```
  Recall:    0.8306
  F1 Score:  0.8237

 Confusion Matrix:
   [[795  11   0   0]
[177 715  26   0]
[  9 129 581   5]
[  0  18  60  42]]

 ── Eval 8: Assessment period 202202 ──
 Training samples:   13,862
 Test samples:       2,535  (admin2 areas in 202202)
 Training Logistic Regression...

 Results  (n_test=2,535):
   Accuracy:  0.8036
   Precision: 0.8091
   Recall:    0.8036
   F1 Score:  0.8005

 Confusion Matrix:
   [[860  54   3   0]
[244 641  22   1]
[ 10 136 525   5]
[  0   5  18  11]]

 ── Eval 9: Assessment period 202110 ──
 Training samples:   12,950
 Test samples:       2,535  (admin2 areas in 202110)
 Training Logistic Regression...

 Results  (n_test=2,535):
   Accuracy:  0.8639
   Precision: 0.8630
   Recall:    0.8639
   F1 Score:  0.8615

 Confusion Matrix:
   [[1064   52    2    0]
[ 156  647   62    2]
[   3   61  471    2]
[   0    0    5    8]]

 ── Eval 10: Assessment period 202106 ──
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Training Logistic Regression...

 Results  (n_test=2,536):
   Accuracy:  0.7953
   Precision: 0.8138
   Recall:    0.7953
   F1 Score:  0.7953

 Confusion Matrix:
   [[828  35  15   2]
```

```
[289 740  49   0]
[  7  85 443  29]
[  2   1   5   6]]

 ── Eval 11: Assessment period 202102 ──
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Training Logistic Regression...

 Results  (n_test=2,489):
   Accuracy:  0.8562
   Precision: 0.8543
   Recall:    0.8562
   F1 Score:  0.8547

 Confusion Matrix:
   [[1173   59    0    0]
[  94  553   56    1]
[   5  104  376   15]
[   0    1   23   29]]

 ── Eval 12: Assessment period 202010 ──
 Training samples:   12,858
 Test samples:       2,491  (admin2 areas in 202010)
 Training Logistic Regression...

 Results  (n_test=2,491):
   Accuracy:  0.8202
   Precision: 0.8322
   Recall:    0.8202
   F1 Score:  0.8236

 Confusion Matrix:
   [[1048   79    6    3]
[ 175  589   49    1]
[   5   90  402   39]
[   0    0    1    4]]

 ── Eval 13: Assessment period 202006 ──
 Training samples:   11,506
 Test samples:       2,570  (admin2 areas in 202006)
 Training Logistic Regression...

 Results  (n_test=2,570):
   Accuracy:  0.8008
   Precision: 0.8016
   Recall:    0.8008
   F1 Score:  0.7970

 Confusion Matrix:
   [[1044   71    9    4]
[ 240  531   32    3]
[  30   98  427    5]
[   0    0   20   56]]

 ── Eval 14: Assessment period 202002 ──
```

```
 Training samples:   12,388
 Test samples:       1,588  (admin2 areas in 202002)
 Training Logistic Regression...

 Results  (n_test=1,588):
   Accuracy:  0.8438
   Precision: 0.8490
   Recall:    0.8438
   F1 Score:  0.8409

 Confusion Matrix:
   [[623  41   4   0]
[ 41 303 114   3]
[ 18  16 366   2]
[  0   0   9  48]]

 —— Eval 15: Assessment period 201910 ——
 Training samples:   12,533
 Test samples:       2,400  (admin2 areas in 201910)
 Training Logistic Regression...

 Results  (n_test=2,400):
   Accuracy:  0.8442
   Precision: 0.8379
   Recall:    0.8442
   F1 Score:  0.8370

 Confusion Matrix:
   [[1181   12    1    1]
[ 146  333   79    1]
[  18   89  464   15]
[   0    8    4   48]]

 —— Eval 16: Assessment period 201906 ——
 Training samples:   12,605
 Test samples:       2,400  (admin2 areas in 201906)
 Training Logistic Regression...

 Results  (n_test=2,400):
   Accuracy:  0.8329
   Precision: 0.8320
   Recall:    0.8329
   F1 Score:  0.8270

 Confusion Matrix:
   [[1155   22    2    0]
[ 188  337   25    0]
[  28  113  449    4]
[   0    0   19   58]]

 —— Eval 17: Assessment period 201902 ——
 Training samples:   12,785
 Test samples:       2,400  (admin2 areas in 201902)
 Training Logistic Regression...

 Results  (n_test=2,400):
```

```
  Accuracy:  0.8625
  Precision: 0.8709
  Recall:    0.8625
  F1 Score:  0.8598

 Confusion Matrix:
   [[1314  111    3    0]
[  56  344   12    0]
[  20   69  400    4]
[   0    0   55   12]]

 ── Eval 18: Assessment period 201810 ──
 Training samples:   13,957
 Test samples:       1,081  (admin2 areas in 201810)
 Training Logistic Regression...

 Results  (n_test=1,081):
  Accuracy:  0.8844
  Precision: 0.8865
  Recall:    0.8844
  F1 Score:  0.8822

 Confusion Matrix:
   [[625  19   0   0]
[ 46 201  34   2]
[  2   9 119   7]
[  0   0   6  11]]

 ── Eval 19: Assessment period 201806 ──
 Training samples:   14,374
 Test samples:       2,440  (admin2 areas in 201806)
 Training Logistic Regression...

 Results  (n_test=2,440):
  Accuracy:  0.8180
  Precision: 0.8088
  Recall:    0.8180
  F1 Score:  0.8062

 Confusion Matrix:
   [[1262   63    6    0]
[ 214  244   58    7]
[  32   38  479   10]
[   0    0   16   11]]

 ── Eval 20: Assessment period 201802 ──
 Training samples:   14,614
 Test samples:       2,441  (admin2 areas in 201802)
 Training Logistic Regression...

 Results  (n_test=2,441):
  Accuracy:  0.8968
  Precision: 0.9004
  Recall:    0.8968
  F1 Score:  0.8980
```

```
 Confusion Matrix:
   [[1460   47    4    0]
[  73  252   32    0]
[  13   41  469   30]
[   0    2   10    8]]

 —— Eval 21: Assessment period 201710 ——
 Training samples:   14,902
 Test samples:       2,577  (admin2 areas in 201710)
 Training Logistic Regression...

 Results  (n_test=2,577):
   Accuracy:  0.8820
   Precision: 0.8832
   Recall:    0.8820
   F1 Score:  0.8815

 Confusion Matrix:
   [[1567   51   40    0    0]
[  63  238   66    1    0]
[   2   31  448   24    0]
[   0    6   19   20    1]
[   0    0    0    0    0]]

 —— Eval 22: Assessment period 201706 ——
 Training samples:   11,953
 Test samples:       2,573  (admin2 areas in 201706)
 Training Logistic Regression...

 Results  (n_test=2,573):
   Accuracy:  0.7804
   Precision: 0.7719
   Recall:    0.7804
   F1 Score:  0.7721

 Confusion Matrix:
   [[1331   60   75    0]
[ 250  237   13    0]
[   7  108  403    6]
[   2    3   41   37]]

 —— Eval 23: Assessment period 201702 ——
 Training samples:   9,303
 Test samples:       2,449  (admin2 areas in 201702)
 Training Logistic Regression...

 Results  (n_test=2,449):
   Accuracy:  0.9008
   Precision: 0.8954
   Recall:    0.9008
   F1 Score:  0.8958

 Confusion Matrix:
   [[1707   16    1    0    0]
[ 108  284   30    1    1]
[  16   49  193    2    0]
```

```
 [   1    6   11   22    0]
 [   0    0    0    1    0]]

  ── Eval 24: Assessment period 201610 ──
  Training samples:   6,140
  Test samples:       2,748  (admin2 areas in 201610)
  Training Logistic Regression...

  Results  (n_test=2,748):
    Accuracy:  0.9203
    Precision: 0.9439
    Recall:    0.9203
    F1 Score:  0.9277

  Confusion Matrix:
    [[1771   22    7   27]
 [  27  330   28   80]
 [   4   15  389    1]
 [   2    1    5   39]]

  ── Eval 25: Assessment period 201606 ──
  Training samples:   3,099
  Test samples:       2,635  (admin2 areas in 201606)
  Training Logistic Regression...

  Results  (n_test=2,635):
    Accuracy:  1.0000
    Precision: 1.0000
    Recall:    1.0000
    F1 Score:  1.0000

  Confusion Matrix:
    [[1696    0    0    0]
 [   0  365    0    0]
 [   0    0  520    0]
 [   0    0    0   54]]

======================================================================
MODEL: Logistic Regression  |  FEATURE SET: IPC + GDELT + temporal
======================================================================
Using 3486 features

  ── Eval 1: Assessment period 202406 ──
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Training Logistic Regression...

  Results  (n_test=1,919):
    Accuracy:  0.7118
    Precision: 0.7306
    Recall:    0.7118
    F1 Score:  0.7113

  Confusion Matrix:
    [[423  56   1   0]
 [270 526  33   1]
```

```
[  3 125 324  16]
[  0   3  45  93]]

 ── Eval 2: Assessment period 202402 ──
 Training samples:   12,107
 Test samples:       2,376  (admin2 areas in 202402)
 Training Logistic Regression...

 Results  (n_test=2,376):
   Accuracy:  0.7875
   Precision: 0.8013
   Recall:    0.7875
   F1 Score:  0.7891

 Confusion Matrix:
   [[712 240   5   0]
[ 64 687  81   2]
[  4  42 401  29]
[  1   1  36  71]]

 ── Eval 3: Assessment period 202310 ──
 Training samples:   12,645
 Test samples:       1,919  (admin2 areas in 202310)
 Training Logistic Regression...

 Results  (n_test=1,919):
   Accuracy:  0.7650
   Precision: 0.7898
   Recall:    0.7650
   F1 Score:  0.7731

 Confusion Matrix:
   [[769 198   2   1]
[ 69 317  70   6]
[  6  42 340  41]
[  1   4  11  42]]

 ── Eval 4: Assessment period 202306 ──
 Training samples:   12,765
 Test samples:       2,613  (admin2 areas in 202306)
 Training Logistic Regression...

 Results  (n_test=2,613):
   Accuracy:  0.7857
   Precision: 0.8096
   Recall:    0.7857
   F1 Score:  0.7885

 Confusion Matrix:
   [[569  66   1   0]
[246 758  45   8]
[  9  99 625  80]
[  0   3   3 101]]

 ── Eval 5: Assessment period 202302 ──
 Training samples:   13,772
```

```
 Test samples:       1,376  (admin2 areas in 202302)
 Training Logistic Regression...

 Results  (n_test=1,376):
   Accuracy:  0.8176
   Precision: 0.8250
   Recall:    0.8176
   F1 Score:  0.8187

 Confusion Matrix:
   [[449 131   0   0]
[ 68 528  14   0]
[  9  26 148   3]
[  0   0   0   0]]

 —— Eval 6: Assessment period 202210 ——
 Training samples:   13,866
 Test samples:       2,399  (admin2 areas in 202210)
 Training Logistic Regression...

 Results  (n_test=2,399):
   Accuracy:  0.7853
   Precision: 0.7893
   Recall:    0.7853
   F1 Score:  0.7837

 Confusion Matrix:
   [[740 101   0   1]
[237 539  41   3]
[ 14  67 539  29]
[  0   0  22  66]]

 —— Eval 7: Assessment period 202206 ——
 Training samples:   13,795
 Test samples:       2,568  (admin2 areas in 202206)
 Training Logistic Regression...

 Results  (n_test=2,568):
   Accuracy:  0.8069
   Precision: 0.8106
   Recall:    0.8069
   F1 Score:  0.8011

 Confusion Matrix:
   [[788  17   1   0]
[232 661  25   0]
[  6 127 574  17]
[  0  15  56  49]]

 —— Eval 8: Assessment period 202202 ——
 Training samples:   13,862
 Test samples:       2,535  (admin2 areas in 202202)
 Training Logistic Regression...

 Results  (n_test=2,535):
   Accuracy:  0.8059
```

```
  Precision: 0.8105
  Recall:    0.8059
  F1 Score:  0.8032

 Confusion Matrix:
   [[859  53   4   1]
[236 647  21   4]
[  8 134 527   7]
[  1   7  16  10]]

 —— Eval 9: Assessment period 202110 ——
 Training samples:   12,950
 Test samples:       2,535  (admin2 areas in 202110)
 Training Logistic Regression...

 Results  (n_test=2,535):
  Accuracy:  0.8659
  Precision: 0.8652
  Recall:    0.8659
  F1 Score:  0.8637

 Confusion Matrix:
   [[1062   53    2    1]
[ 151  651   63    2]
[   2   59  473    3]
[   0    0    4    9]]

 —— Eval 10: Assessment period 202106 ——
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Training Logistic Regression...

 Results  (n_test=2,536):
  Accuracy:  0.7989
  Precision: 0.8152
  Recall:    0.7989
  F1 Score:  0.7983

 Confusion Matrix:
   [[831  32  16   1]
[284 745  48   1]
[  4  90 445  25]
[  2   0   7   5]]

 —— Eval 11: Assessment period 202102 ——
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Training Logistic Regression...

 Results  (n_test=2,489):
  Accuracy:  0.8546
  Precision: 0.8533
  Recall:    0.8546
  F1 Score:  0.8536

 Confusion Matrix:
```

```
   [[1163   67    1    1]
[  96  551   53    4]
[   6   97  383   14]
[   0    0   23   30]]

 ── Eval 12: Assessment period 202010 ──
 Training samples:   12,858
 Test samples:       2,491  (admin2 areas in 202010)
 Training Logistic Regression...

 Results  (n_test=2,491):
   Accuracy:  0.8177
   Precision: 0.8303
   Recall:    0.8177
   F1 Score:  0.8213

 Confusion Matrix:
   [[1049   76    7    4]
[ 181  579   52    2]
[   6   85  405   40]
[   0    0    1    4]]

 ── Eval 13: Assessment period 202006 ──
 Training samples:   11,506
 Test samples:       2,570  (admin2 areas in 202006)
 Training Logistic Regression...

 Results  (n_test=2,570):
   Accuracy:  0.8074
   Precision: 0.8081
   Recall:    0.8074
   F1 Score:  0.8034

 Confusion Matrix:
   [[1050   66    9    3]
[ 237  528   38    3]
[  29   89  437    5]
[   0    0   16   60]]

 ── Eval 14: Assessment period 202002 ──
 Training samples:   12,388
 Test samples:       1,588  (admin2 areas in 202002)
 Training Logistic Regression...

 Results  (n_test=1,588):
   Accuracy:  0.8375
   Precision: 0.8423
   Recall:    0.8375
   F1 Score:  0.8343

 Confusion Matrix:
   [[624  38   6   0]
[ 44 298 117   2]
[ 20  20 358   4]
[  0   0   7  50]]
```

```
 ── Eval 15: Assessment period 201910 ──
 Training samples:   12,533
 Test samples:       2,400  (admin2 areas in 201910)
 Training Logistic Regression...

 Results  (n_test=2,400):
   Accuracy:  0.8479
   Precision: 0.8424
   Recall:    0.8479
   F1 Score:  0.8408

 Confusion Matrix:
   [[1182   11    1    1]
[ 145  336   76    2]
[  19   82  470   15]
[   0    6    7   47]]

 ── Eval 16: Assessment period 201906 ──
 Training samples:   12,605
 Test samples:       2,400  (admin2 areas in 201906)
 Training Logistic Regression...

 Results  (n_test=2,400):
   Accuracy:  0.8358
   Precision: 0.8345
   Recall:    0.8358
   F1 Score:  0.8299

 Confusion Matrix:
   [[1154   23    2    0]
[ 189  335   26    0]
[  25  105  461    3]
[   0    1   20   56]]

 ── Eval 17: Assessment period 201902 ──
 Training samples:   12,785
 Test samples:       2,400  (admin2 areas in 201902)
 Training Logistic Regression...

 Results  (n_test=2,400):
   Accuracy:  0.8633
   Precision: 0.8707
   Recall:    0.8633
   F1 Score:  0.8616

 Confusion Matrix:
   [[1311  114    3    0]
[  56  343   13    0]
[  16   66  404    7]
[   0    0   53   14]]

 ── Eval 18: Assessment period 201810 ──
 Training samples:   13,957
 Test samples:       1,081  (admin2 areas in 201810)
 Training Logistic Regression...
```

```
 Results  (n_test=1,081):
   Accuracy:  0.8760
   Precision: 0.8812
   Recall:    0.8760
   F1 Score:  0.8753

 Confusion Matrix:
   [[623  20   0   1]
[ 43 199  35   6]
[  3  10 115   9]
[  0   0   7  10]]

 —— Eval 19: Assessment period 201806 ——
 Training samples:   14,374
 Test samples:       2,440  (admin2 areas in 201806)
 Training Logistic Regression...

 Results  (n_test=2,440):
   Accuracy:  0.8139
   Precision: 0.8047
   Recall:    0.8139
   F1 Score:  0.8027

 Confusion Matrix:
   [[1258   67    5    1]
[ 214  245   57    7]
[  35   40  473   11]
[   1    0   16   10]]

 —— Eval 20: Assessment period 201802 ——
 Training samples:   14,614
 Test samples:       2,441  (admin2 areas in 201802)
 Training Logistic Regression...

 Results  (n_test=2,441):
   Accuracy:  0.8964
   Precision: 0.8992
   Recall:    0.8964
   F1 Score:  0.8973

 Confusion Matrix:
   [[1458   49    4    0]
[  74  250   33    0]
[  13   40  472   28]
[   0    2   10    8]]

 —— Eval 21: Assessment period 201710 ——
 Training samples:   14,902
 Test samples:       2,577  (admin2 areas in 201710)
 Training Logistic Regression...

 Results  (n_test=2,577):
   Accuracy:  0.8766
   Precision: 0.8792
   Recall:    0.8766
   F1 Score:  0.8772
```

```
 Confusion Matrix:
   [[1566   53   39    0]
[  62  239   66    1]
[   1   40  432   32]
[   0    6   18   22]]

 —— Eval 22: Assessment period 201706 ——
 Training samples:   11,953
 Test samples:       2,573  (admin2 areas in 201706)
 Training Logistic Regression...

 Results  (n_test=2,573):
   Accuracy:  0.7843
   Precision: 0.7747
   Recall:    0.7843
   F1 Score:  0.7776

 Confusion Matrix:
   [[1342   53   68    3]
[ 240  247   11    2]
[   7  115  382   20]
[   3    5   28   47]]

 —— Eval 23: Assessment period 201702 ——
 Training samples:   9,303
 Test samples:       2,449  (admin2 areas in 201702)
 Training Logistic Regression...

 Results  (n_test=2,449):
   Accuracy:  0.9024
   Precision: 0.8976
   Recall:    0.9024
   F1 Score:  0.8985

 Confusion Matrix:
   [[1699   20    2    3    0]
[  96  298   27    3    0]
[  16   54  185    5    0]
[   2    4    6   28    0]
[   0    0    0    1    0]]

 —— Eval 24: Assessment period 201610 ——
 Training samples:   6,140
 Test samples:       2,748  (admin2 areas in 201610)
 Training Logistic Regression...

 Results  (n_test=2,748):
   Accuracy:  0.9152
   Precision: 0.9207
   Recall:    0.9152
   F1 Score:  0.9151

 Confusion Matrix:
   [[1763   34   29    1]
[  30  332   97    6]
```

```
 [  13   14  381    1]
 [   1    2    5   39]]

  ── Eval 25: Assessment period 201606 ──
  Training samples:   3,099
  Test samples:       2,635  (admin2 areas in 201606)
  Training Logistic Regression...

  Results  (n_test=2,635):
    Accuracy:  1.0000
    Precision: 1.0000
    Recall:    1.0000
    F1 Score:  1.0000

  Confusion Matrix:
    [[1696    0    0    0]
 [   0  365    0    0]
 [   0    0  520    0]
 [   0    0    0   54]]

================================================================================
MODEL: Logistic Regression  |  FEATURE SET: GDELT only
================================================================================
Using 3485 features

  ── Eval 1: Assessment period 202406 ──
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Training Logistic Regression...

  Results  (n_test=1,919):
    Accuracy:  0.6884
    Precision: 0.7126
    Recall:    0.6884
    F1 Score:  0.6868

  Confusion Matrix:
    [[425  55   0   0]
 [315 472  42   1]
 [  1 124 326  17]
 [  0   3  40  98]]

  ── Eval 2: Assessment period 202402 ──
  Training samples:   12,107
  Test samples:       2,376  (admin2 areas in 202402)
  Training Logistic Regression...

  Results  (n_test=2,376):
    Accuracy:  0.8304
    Precision: 0.8342
    Recall:    0.8304
    F1 Score:  0.8308

  Confusion Matrix:
    [[812 143   2   0]
 [ 80 693  61   0]
```

```
[  0  59 400  17]
[  0   1  40  68]]

 ── Eval 3: Assessment period 202310 ──
 Training samples:   12,645
 Test samples:       1,919  (admin2 areas in 202310)
 Training Logistic Regression...

 Results  (n_test=1,919):
   Accuracy:  0.7759
   Precision: 0.8147
   Recall:    0.7759
   F1 Score:  0.7856

 Confusion Matrix:
   [[741 226   3   0]
[ 29 354  78   1]
[  1  41 352  35]
[  0   1  15  42]]

 ── Eval 4: Assessment period 202306 ──
 Training samples:   12,765
 Test samples:       2,613  (admin2 areas in 202306)
 Training Logistic Regression...

 Results  (n_test=2,613):
   Accuracy:  0.8086
   Precision: 0.8187
   Recall:    0.8086
   F1 Score:  0.8083

 Confusion Matrix:
   [[587  48   1   0]
[236 766  51   4]
[  8  94 691  20]
[  0   8  30  69]]

 ── Eval 5: Assessment period 202302 ──
 Training samples:   13,772
 Test samples:       1,376  (admin2 areas in 202302)
 Training Logistic Regression...

 Results  (n_test=1,376):
   Accuracy:  0.8488
   Precision: 0.8532
   Recall:    0.8488
   F1 Score:  0.8498

 Confusion Matrix:
   [[490  90   0   0]
[ 55 535  20   0]
[  7  33 143   3]
[  0   0   0   0]]

 ── Eval 6: Assessment period 202210 ──
 Training samples:   13,866
```

```
 Test samples:       2,399  (admin2 areas in 202210)
 Training Logistic Regression...

 Results  (n_test=2,399):
   Accuracy:  0.8062
   Precision: 0.8065
   Recall:    0.8062
   F1 Score:  0.8055

 Confusion Matrix:
   [[725 116   0   1]
[171 604  43   2]
[ 11  77 549  12]
[  0   0  32  56]]

 —— Eval 7: Assessment period 202206 ——
 Training samples:   13,795
 Test samples:       2,568  (admin2 areas in 202206)
 Training Logistic Regression...

 Results  (n_test=2,568):
   Accuracy:  0.8201
   Precision: 0.8187
   Recall:    0.8201
   F1 Score:  0.8128

 Confusion Matrix:
   [[789  16   1   0]
[167 730  21   0]
[  7 152 549  16]
[  0  19  63  38]]

 —— Eval 8: Assessment period 202202 ——
 Training samples:   13,862
 Test samples:       2,535  (admin2 areas in 202202)
 Training Logistic Regression...

 Results  (n_test=2,535):
   Accuracy:  0.8174
   Precision: 0.8217
   Recall:    0.8174
   F1 Score:  0.8147

 Confusion Matrix:
   [[872  43   1   1]
[217 666  21   4]
[  5 141 523   7]
[  0   7  16  11]]

 —— Eval 9: Assessment period 202110 ——
 Training samples:   12,950
 Test samples:       2,535  (admin2 areas in 202110)
 Training Logistic Regression...

 Results  (n_test=2,535):
   Accuracy:  0.8789
```

```
  Precision: 0.8780
  Recall:    0.8789
  F1 Score:  0.8779

 Confusion Matrix:
   [[1056   59    2    1]
[ 113  691   61    2]
[   1   61  472    3]
[   0    0    4    9]]

 ── Eval 10: Assessment period 202106 ──
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Training Logistic Regression...

 Results  (n_test=2,536):
   Accuracy:  0.7914
   Precision: 0.8080
   Recall:    0.7914
   F1 Score:  0.7906

 Confusion Matrix:
   [[824  44  12   0]
[302 730  45   1]
[  3  89 447  25]
[  0   0   8   6]]

 ── Eval 11: Assessment period 202102 ──
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Training Logistic Regression...

 Results  (n_test=2,489):
   Accuracy:  0.8578
   Precision: 0.8559
   Recall:    0.8578
   F1 Score:  0.8564

 Confusion Matrix:
   [[1168   64    0    0]
[  99  548   53    4]
[   7   92  389   12]
[   0    0   23   30]]

 ── Eval 12: Assessment period 202010 ──
 Training samples:   12,858
 Test samples:       2,491  (admin2 areas in 202010)
 Training Logistic Regression...

 Results  (n_test=2,491):
   Accuracy:  0.8198
   Precision: 0.8320
   Recall:    0.8198
   F1 Score:  0.8233

 Confusion Matrix:
```

```
   [[1047   77    8    4]
[ 169  593   50    2]
[   8   92  398   38]
[   0    0    1    4]]

 ── Eval 13: Assessment period 202006 ──
 Training samples:   11,506
 Test samples:       2,570  (admin2 areas in 202006)
 Training Logistic Regression...

 Results  (n_test=2,570):
   Accuracy:  0.7907
   Precision: 0.7933
   Recall:    0.7907
   F1 Score:  0.7831

 Confusion Matrix:
   [[1067   51    8    2]
[ 292  466   45    3]
[  31   87  439    3]
[   0    0   16   60]]

 ── Eval 14: Assessment period 202002 ──
 Training samples:   12,388
 Test samples:       1,588  (admin2 areas in 202002)
 Training Logistic Regression...

 Results  (n_test=1,588):
   Accuracy:  0.8463
   Precision: 0.8518
   Recall:    0.8463
   F1 Score:  0.8451

 Confusion Matrix:
   [[614  47   7   0]
[ 30 316 113   2]
[ 16  20 364   2]
[  0   0   7  50]]

 ── Eval 15: Assessment period 201910 ──
 Training samples:   12,533
 Test samples:       2,400  (admin2 areas in 201910)
 Training Logistic Regression...

 Results  (n_test=2,400):
   Accuracy:  0.8529
   Precision: 0.8479
   Recall:    0.8529
   F1 Score:  0.8467

 Confusion Matrix:
   [[1181   12    1    1]
[ 144  350   64    1]
[  20   85  471   10]
[   0    7    8   45]]
```

```
 ── Eval 16: Assessment period 201906 ──
 Training samples:   12,605
 Test samples:       2,400  (admin2 areas in 201906)
 Training Logistic Regression...

 Results  (n_test=2,400):
   Accuracy:  0.8158
   Precision: 0.8119
   Recall:    0.8158
   F1 Score:  0.8042

 Confusion Matrix:
   [[1159   18    2    0]
[ 215  305   30    0]
[  24  101  466    3]
[   0    1   48   28]]

 ── Eval 17: Assessment period 201902 ──
 Training samples:   12,785
 Test samples:       2,400  (admin2 areas in 201902)
 Training Logistic Regression...

 Results  (n_test=2,400):
   Accuracy:  0.8692
   Precision: 0.8665
   Recall:    0.8692
   F1 Score:  0.8642

 Confusion Matrix:
   [[1347   71    9    1]
[  72  310   30    0]
[  17   54  416    6]
[   0    0   54   13]]

 ── Eval 18: Assessment period 201810 ──
 Training samples:   13,957
 Test samples:       1,081  (admin2 areas in 201810)
 Training Logistic Regression...

 Results  (n_test=1,081):
   Accuracy:  0.8612
   Precision: 0.8683
   Recall:    0.8612
   F1 Score:  0.8599

 Confusion Matrix:
   [[622  21   0   1]
[ 41 185  50   7]
[  5  12 113   7]
[  0   0   6  11]]

 ── Eval 19: Assessment period 201806 ──
 Training samples:   14,374
 Test samples:       2,440  (admin2 areas in 201806)
 Training Logistic Regression...
```

```
 Results  (n_test=2,440):
   Accuracy:  0.8004
   Precision: 0.7890
   Recall:    0.8004
   F1 Score:  0.7892

 Confusion Matrix:
   [[1258   64    8    1]
[ 215  243   60    5]
[  41   66  443    9]
[   1    1   16    9]]

 ── Eval 20: Assessment period 201802 ──
 Training samples:   14,614
 Test samples:       2,441  (admin2 areas in 201802)
 Training Logistic Regression...

 Results  (n_test=2,441):
   Accuracy:  0.8972
   Precision: 0.8996
   Recall:    0.8972
   F1 Score:  0.8975

 Confusion Matrix:
   [[1464   40    7    0]
[  85  246   26    0]
[  12   42  471   28]
[   0    2    9    9]]

 ── Eval 21: Assessment period 201710 ──
 Training samples:   14,902
 Test samples:       2,577  (admin2 areas in 201710)
 Training Logistic Regression...

 Results  (n_test=2,577):
   Accuracy:  0.8828
   Precision: 0.8847
   Recall:    0.8828
   F1 Score:  0.8836

 Confusion Matrix:
   [[1578   44   36    0]
[  61  255   51    1]
[   4   49  420   32]
[   0    8   16   22]]

 ── Eval 22: Assessment period 201706 ──
 Training samples:   11,953
 Test samples:       2,573  (admin2 areas in 201706)
 Training Logistic Regression...

 Results  (n_test=2,573):
   Accuracy:  0.7641
   Precision: 0.7492
   Recall:    0.7641
   F1 Score:  0.7533
```

```
 Confusion Matrix:
   [[1336   59   68    3]
[ 281  206   11    2]
[   7  112  385   20]
[   2    7   35   39]]

 —— Eval 23: Assessment period 201702 ——
 Training samples:   9,303
 Test samples:       2,449  (admin2 areas in 201702)
 Training Logistic Regression...

 Results  (n_test=2,449):
   Accuracy:  0.9053
   Precision: 0.9008
   Recall:    0.9053
   F1 Score:  0.9017

 Confusion Matrix:
   [[1699   20    2    3    0]
[  99  300   23    2    0]
[  11   54  190    5    0]
[   1    5    6   28    0]
[   0    0    0    1    0]]

 —— Eval 24: Assessment period 201610 ——
 Training samples:   6,140
 Test samples:       2,748  (admin2 areas in 201610)
 Training Logistic Regression...

 Results  (n_test=2,748):
   Accuracy:  0.8948
   Precision: 0.8910
   Recall:    0.8948
   F1 Score:  0.8910

 Confusion Matrix:
   [[1760   51   16    0]
[ 125  293   46    1]
[  10   21  369    9]
[   0    4    6   37]]

 —— Eval 25: Assessment period 201606 ——
 Training samples:   3,099
 Test samples:       2,635  (admin2 areas in 201606)
 Training Logistic Regression...

 Results  (n_test=2,635):
   Accuracy:  1.0000
   Precision: 1.0000
   Recall:    1.0000
   F1 Score:  1.0000

 Confusion Matrix:
   [[1696    0    0    0]
[   0  365    0    0]
```

```
 [   0    0  520    0]
 [   0    0    0   54]]

======================================================================
MODEL: Logistic Regression  |  FEATURE SET: IPC only (no geo)
======================================================================
Using 1 features

  ── Eval 1: Assessment period 202406 ──
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Training Logistic Regression...

  Results  (n_test=1,919):
    Accuracy:  0.6514
    Precision: 0.6803
    Recall:    0.6514
    F1 Score:  0.6483

  Confusion Matrix:
    [[429  51   0   0]
 [359 439  32   0]
 [  0 157 292  19]
 [  0   2  49  90]]

  ── Eval 2: Assessment period 202402 ──
  Training samples:   12,107
  Test samples:       2,376  (admin2 areas in 202402)
  Training Logistic Regression...

  Results  (n_test=2,376):
    Accuracy:  0.6772
    Precision: 0.6738
    Recall:    0.6772
    F1 Score:  0.6451

  Confusion Matrix:
    [[906  51   0   0]
 [418 370  46   0]
 [103  40 333   0]
 [  0   1 108   0]]

  ── Eval 3: Assessment period 202310 ──
  Training samples:   12,645
  Test samples:       1,919  (admin2 areas in 202310)
  Training Logistic Regression...

  Results  (n_test=1,919):
    Accuracy:  0.6295
    Precision: 0.7438
    Recall:    0.6295
    F1 Score:  0.6321

  Confusion Matrix:
    [[493 471   6   0]
 [  4 326 132   0]
```

```
[  0  40 389   0]
[  0   0  58   0]]

 ── Eval 4: Assessment period 202306 ──
 Training samples:   12,765
 Test samples:       2,613  (admin2 areas in 202306)
 Training Logistic Regression...

 Results  (n_test=2,613):
   Accuracy:  0.4646
   Precision: 0.6057
   Recall:    0.4646
   F1 Score:  0.4336

 Confusion Matrix:
   [[598  36   2   0]
[494 500  63   0]
[623  74 116   0]
[102   0   5   0]]

 ── Eval 5: Assessment period 202302 ──
 Training samples:   13,772
 Test samples:       1,376  (admin2 areas in 202302)
 Training Logistic Regression...

 Results  (n_test=1,376):
   Accuracy:  0.7369
   Precision: 0.7456
   Recall:    0.7369
   F1 Score:  0.7368

 Confusion Matrix:
   [[386 189   5]
[ 94 479  37]
[  0  37 149]]

 ── Eval 6: Assessment period 202210 ──
 Training samples:   13,866
 Test samples:       2,399  (admin2 areas in 202210)
 Training Logistic Regression...

 Results  (n_test=2,399):
   Accuracy:  0.7353
   Precision: 0.7095
   Recall:    0.7353
   F1 Score:  0.7213

 Confusion Matrix:
   [[636 205   1   0]
[166 570  84   0]
[  4  87 558   0]
[  0   0  88   0]]

 ── Eval 7: Assessment period 202206 ──
 Training samples:   13,795
 Test samples:       2,568  (admin2 areas in 202206)
```

```
 Training Logistic Regression...

 Results  (n_test=2,568):
   Accuracy:  0.7371
   Precision: 0.7045
   Recall:    0.7371
   F1 Score:  0.7188

 Confusion Matrix:
   [[687 119   0   0]
[197 656  65   0]
[ 48 126 550   0]
[ 31   2  87   0]]

 ── Eval 8: Assessment period 202202 ──
 Training samples:   13,862
 Test samples:       2,535  (admin2 areas in 202202)
 Training Logistic Regression...

 Results  (n_test=2,535):
   Accuracy:  0.7744
   Precision: 0.7710
   Recall:    0.7744
   F1 Score:  0.7673

 Confusion Matrix:
   [[846  71   0   0]
[268 618  22   0]
[  4 173 499   0]
[  0   5  29   0]]

 ── Eval 9: Assessment period 202110 ──
 Training samples:   12,950
 Test samples:       2,535  (admin2 areas in 202110)
 Training Logistic Regression...

 Results  (n_test=2,535):
   Accuracy:  0.7815
   Precision: 0.8008
   Recall:    0.7815
   F1 Score:  0.7828

 Confusion Matrix:
   [[809 307   2   0]
[ 70 703  94   0]
[  0  68 469   0]
[  0   0  13   0]]

 ── Eval 10: Assessment period 202106 ──
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Training Logistic Regression...

 Results  (n_test=2,536):
   Accuracy:  0.7169
   Precision: 0.7365
```

```
  Recall:    0.7169
  F1 Score:  0.7055

 Confusion Matrix:
   [[805  59  16   0]
[470 549  59   0]
[  4  96 464   0]
[  0   0  14   0]]

 —— Eval 11: Assessment period 202102 ——
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Training Logistic Regression...

 Results  (n_test=2,489):
   Accuracy:  0.7790
   Precision: 0.7901
   Recall:    0.7790
   F1 Score:  0.7763

 Confusion Matrix:
   [[1007  225    0    0]
[ 120  511   73    0]
[   7   75  417    1]
[   2    1   46    4]]

 —— Eval 12: Assessment period 202010 ——
 Training samples:   12,858
 Test samples:       2,491  (admin2 areas in 202010)
 Training Logistic Regression...

 Results  (n_test=2,491):
   Accuracy:  0.7354
   Precision: 0.7456
   Recall:    0.7354
   F1 Score:  0.7399

 Confusion Matrix:
   [[936 189  11   0]
[190 505 119   0]
[  2 106 387  41]
[  0   0   1   4]]

 —— Eval 13: Assessment period 202006 ——
 Training samples:   11,506
 Test samples:       2,570  (admin2 areas in 202006)
 Training Logistic Regression...

 Results  (n_test=2,570):
   Accuracy:  0.7054
   Precision: 0.7276
   Recall:    0.7054
   F1 Score:  0.6865

 Confusion Matrix:
   [[1077   48    3    0]
```

```
[ 482  314   10    0]
[  86   99  370    5]
[   5    0   19   52]]

 ── Eval 14: Assessment period 202002 ──
 Training samples:   12,388
 Test samples:       1,588  (admin2 areas in 202002)
 Training Logistic Regression...

 Results  (n_test=1,588):
   Accuracy:  0.7752
   Precision: 0.7906
   Recall:    0.7752
   F1 Score:  0.7737

 Confusion Matrix:
   [[569  78  21   0]
[ 29 257 167   8]
[ 13  24 363   2]
[  0   1  14  42]]

 ── Eval 15: Assessment period 201910 ──
 Training samples:   12,533
 Test samples:       2,400  (admin2 areas in 201910)
 Training Logistic Regression...

 Results  (n_test=2,400):
   Accuracy:  0.8017
   Precision: 0.8032
   Recall:    0.8017
   F1 Score:  0.8021

 Confusion Matrix:
   [[1086  108    1    0]
[  85  341  132    1]
[   8  101  449   28]
[   0    0   12   48]]

 ── Eval 16: Assessment period 201906 ──
 Training samples:   12,605
 Test samples:       2,400  (admin2 areas in 201906)
 Training Logistic Regression...

 Results  (n_test=2,400):
   Accuracy:  0.7704
   Precision: 0.7326
   Recall:    0.7704
   F1 Score:  0.7448

 Confusion Matrix:
   [[1158   19    2    0]
[ 256  252   42    0]
[  14  141  439    0]
[   0    0   77    0]]

 ── Eval 17: Assessment period 201902 ──
```

```
 Training samples:   12,785
 Test samples:       2,400  (admin2 areas in 201902)
 Training Logistic Regression...

 Results  (n_test=2,400):
   Accuracy:  0.6746
   Precision: 0.6610
   Recall:    0.6746
   F1 Score:  0.6250

 Confusion Matrix:
   [[1336   92    0    0]
[ 234  167   11    0]
[ 353   24  116    0]
[  40    0   27    0]]

 —— Eval 18: Assessment period 201810 ——
 Training samples:   13,957
 Test samples:       1,081  (admin2 areas in 201810)
 Training Logistic Regression...

 Results  (n_test=1,081):
   Accuracy:  0.8511
   Precision: 0.8476
   Recall:    0.8511
   F1 Score:  0.8475

 Confusion Matrix:
   [[580  64   0   0]
[ 25 220  38   0]
[  3  14 120   0]
[  2   0  15   0]]

 —— Eval 19: Assessment period 201806 ——
 Training samples:   14,374
 Test samples:       2,440  (admin2 areas in 201806)
 Training Logistic Regression...

 Results  (n_test=2,440):
   Accuracy:  0.8152
   Precision: 0.7949
   Recall:    0.8152
   F1 Score:  0.7986

 Confusion Matrix:
   [[1279   42   10    0]
[ 211  244   68    0]
[  22   71  466    0]
[   0    0   27    0]]

 —— Eval 20: Assessment period 201802 ——
 Training samples:   14,614
 Test samples:       2,441  (admin2 areas in 201802)
 Training Logistic Regression...

 Results  (n_test=2,441):
```

```
  Accuracy:  0.8992
  Precision: 0.8925
  Recall:    0.8992
  F1 Score:  0.8958

 Confusion Matrix:
   [[1449   59    3    0]
[  64  255   38    0]
[  11   51  491    0]
[   0    3   17    0]]

 —— Eval 21: Assessment period 201710 ——
 Training samples:   14,902
 Test samples:       2,577  (admin2 areas in 201710)
 Training Logistic Regression...

 Results  (n_test=2,577):
  Accuracy:  0.7800
  Precision: 0.7970
  Recall:    0.7800
  F1 Score:  0.7836

 Confusion Matrix:
   [[1352  291   15    0]
[  93  193   82    0]
[  25   15  465    0]
[   0    1   45    0]]

 —— Eval 22: Assessment period 201706 ——
 Training samples:   11,953
 Test samples:       2,573  (admin2 areas in 201706)
 Training Logistic Regression...

 Results  (n_test=2,573):
  Accuracy:  0.6246
  Precision: 0.5744
  Recall:    0.6246
  F1 Score:  0.5878

 Confusion Matrix:
   [[1275  108   83    0]
[ 283  200   17    0]
[ 276  115  132    1]
[  14    1   68    0]]

 —— Eval 23: Assessment period 201702 ——
 Training samples:   9,303
 Test samples:       2,449  (admin2 areas in 201702)
 Training Logistic Regression...

 Results  (n_test=2,449):
  Accuracy:  0.8399
  Precision: 0.8206
  Recall:    0.8399
  F1 Score:  0.8283
```

```
  Confusion Matrix:
    [[1664   60    0    0    0]
 [ 156  249   19    0    0]
 [   6  110  144    0    0]
 [   0    9   31    0    0]
 [   0    0    1    0    0]]

  ── Eval 24: Assessment period 201610 ──
  Training samples:   6,140
  Test samples:       2,748  (admin2 areas in 201610)
  Training Logistic Regression...

  Results  (n_test=2,748):
    Accuracy:  0.8130
    Precision: 0.7989
    Recall:    0.8130
    F1 Score:  0.8013

  Confusion Matrix:
    [[1653  139   35    0]
 [ 152  200  113    0]
 [   6   22  381    0]
 [   0    2   45    0]]

  ── Eval 25: Assessment period 201606 ──
  Training samples:   3,099
  Test samples:       2,635  (admin2 areas in 201606)
  Training Logistic Regression...

  Results  (n_test=2,635):
    Accuracy:  0.8516
    Precision: 0.8352
    Recall:    0.8516
    F1 Score:  0.8414

  Confusion Matrix:
    [[1659   37    0    0]
 [ 113  227   25    0]
 [  24  138  358    0]
 [   0    4   50    0]]

======================================================================
MODEL: Logistic Regression  |  FEATURE SET: IPC + GDELT base (no geo)
======================================================================
Using 120 features

  ── Eval 1: Assessment period 202406 ──
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Training Logistic Regression...

  Results  (n_test=1,919):
    Accuracy:  0.6503
    Precision: 0.6790
    Recall:    0.6503
    F1 Score:  0.6471
```

```
 Confusion Matrix:
   [[428  52   0   0]
[358 438  34   0]
[  1 154 295  18]
[  0   2  52  87]]

 ── Eval 2: Assessment period 202402 ──
 Training samples:   12,107
 Test samples:       2,376  (admin2 areas in 202402)
 Training Logistic Regression...

 Results  (n_test=2,376):
   Accuracy:  0.6785
   Precision: 0.7198
   Recall:    0.6785
   F1 Score:  0.6482

 Confusion Matrix:
   [[905  52   0   0]
[416 372  46   0]
[103  41 332   0]
[  0   1 105   3]]

 ── Eval 3: Assessment period 202310 ──
 Training samples:   12,645
 Test samples:       1,919  (admin2 areas in 202310)
 Training Logistic Regression...

 Results  (n_test=1,919):
   Accuracy:  0.6295
   Precision: 0.7537
   Recall:    0.6295
   F1 Score:  0.6339

 Confusion Matrix:
   [[493 470   7   0]
[  5 327 130   0]
[  0  39 386   4]
[  0   0  56   2]]

 ── Eval 4: Assessment period 202306 ──
 Training samples:   12,765
 Test samples:       2,613  (admin2 areas in 202306)
 Training Logistic Regression...

 Results  (n_test=2,613):
   Accuracy:  0.4657
   Precision: 0.6023
   Recall:    0.4657
   F1 Score:  0.4346

 Confusion Matrix:
   [[597  37   2   0]
[489 504  64   0]
[620  76 116   1]
```

```
[100   1   6   0]]

 ── Eval 5: Assessment period 202302 ──
 Training samples:   13,772
 Test samples:       1,376  (admin2 areas in 202302)
 Training Logistic Regression...

 Results  (n_test=1,376):
   Accuracy:  0.7362
   Precision: 0.7449
   Recall:    0.7362
   F1 Score:  0.7361

 Confusion Matrix:
   [[386 188   6]
[ 93 478  39]
[  0  37 149]]

 ── Eval 6: Assessment period 202210 ──
 Training samples:   13,866
 Test samples:       2,399  (admin2 areas in 202210)
 Training Logistic Regression...

 Results  (n_test=2,399):
   Accuracy:  0.7349
   Precision: 0.7284
   Recall:    0.7349
   F1 Score:  0.7243

 Confusion Matrix:
   [[635 206   1   0]
[166 569  84   1]
[  4  87 554   4]
[  0   0  83   5]]

 ── Eval 7: Assessment period 202206 ──
 Training samples:   13,795
 Test samples:       2,568  (admin2 areas in 202206)
 Training Logistic Regression...

 Results  (n_test=2,568):
   Accuracy:  0.7375
   Precision: 0.7513
   Recall:    0.7375
   F1 Score:  0.7208

 Confusion Matrix:
   [[686 120   0   0]
[196 655  67   0]
[ 48 126 550   0]
[ 31   3  83   3]]

 ── Eval 8: Assessment period 202202 ──
 Training samples:   13,862
 Test samples:       2,535  (admin2 areas in 202202)
 Training Logistic Regression...
```

```
 Results  (n_test=2,535):
   Accuracy:  0.7708
   Precision: 0.7739
   Recall:    0.7708
   F1 Score:  0.7643

 Confusion Matrix:
   [[842  74   1   0]
[269 613  26   0]
[  4 173 498   1]
[  0   5  28   1]]

 ── Eval 9: Assessment period 202110 ──
 Training samples:   12,950
 Test samples:       2,535  (admin2 areas in 202110)
 Training Logistic Regression...

 Results  (n_test=2,535):
   Accuracy:  0.7795
   Precision: 0.8042
   Recall:    0.7795
   F1 Score:  0.7814

 Confusion Matrix:
   [[807 309   2   0]
[ 71 701  95   0]
[  0  70 467   0]
[  0   0  12   1]]

 ── Eval 10: Assessment period 202106 ──
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Training Logistic Regression...

 Results  (n_test=2,536):
   Accuracy:  0.7169
   Precision: 0.7418
   Recall:    0.7169
   F1 Score:  0.7060

 Confusion Matrix:
   [[805  59  16   0]
[470 549  59   0]
[  4  97 463   0]
[  0   0  13   1]]

 ── Eval 11: Assessment period 202102 ──
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Training Logistic Regression...

 Results  (n_test=2,489):
   Accuracy:  0.7766
   Precision: 0.7839
   Recall:    0.7766
```

```
  F1 Score:  0.7744

 Confusion Matrix:
   [[1007  225    0    0]
[ 123  506   75    0]
[   8   74  415    3]
[   2    0   46    5]]

 —— Eval 12: Assessment period 202010 ——
 Training samples:   12,858
 Test samples:       2,491  (admin2 areas in 202010)
 Training Logistic Regression...

 Results  (n_test=2,491):
   Accuracy:  0.7354
   Precision: 0.7458
   Recall:    0.7354
   F1 Score:  0.7400

 Confusion Matrix:
   [[936 189  10   1]
[192 503 117   2]
[  2 106 389  39]
[  0   0   1   4]]

 —— Eval 13: Assessment period 202006 ——
 Training samples:   11,506
 Test samples:       2,570  (admin2 areas in 202006)
 Training Logistic Regression...

 Results  (n_test=2,570):
   Accuracy:  0.7051
   Precision: 0.7271
   Recall:    0.7051
   F1 Score:  0.6861

 Confusion Matrix:
   [[1077   48    3    0]
[ 482  314   10    0]
[  86   99  370    5]
[   5    0   20   51]]

 —— Eval 14: Assessment period 202002 ——
 Training samples:   12,388
 Test samples:       1,588  (admin2 areas in 202002)
 Training Logistic Regression...

 Results  (n_test=1,588):
   Accuracy:  0.7739
   Precision: 0.7891
   Recall:    0.7739
   F1 Score:  0.7727

 Confusion Matrix:
   [[569  78  21   0]
[ 29 259 164   9]
```

```
[ 13  26 361   2]
[  0   1  16  40]]

 ── Eval 15: Assessment period 201910 ──
 Training samples:   12,533
 Test samples:       2,400  (admin2 areas in 201910)
 Training Logistic Regression...

 Results  (n_test=2,400):
   Accuracy:  0.8025
   Precision: 0.8036
   Recall:    0.8025
   F1 Score:  0.8028

 Confusion Matrix:
   [[1086  108    1    0]
[  86  340  132    1]
[   8   98  453   27]
[   0    0   13   47]]

 ── Eval 16: Assessment period 201906 ──
 Training samples:   12,605
 Test samples:       2,400  (admin2 areas in 201906)
 Training Logistic Regression...

 Results  (n_test=2,400):
   Accuracy:  0.7863
   Precision: 0.7764
   Recall:    0.7863
   F1 Score:  0.7728

 Confusion Matrix:
   [[1157   19    3    0]
[ 258  248   44    0]
[  15  138  428   13]
[   0    0   23   54]]

 ── Eval 17: Assessment period 201902 ──
 Training samples:   12,785
 Test samples:       2,400  (admin2 areas in 201902)
 Training Logistic Regression...

 Results  (n_test=2,400):
   Accuracy:  0.6763
   Precision: 0.6635
   Recall:    0.6763
   F1 Score:  0.6267

 Confusion Matrix:
   [[1340   88    0    0]
[ 237  164   11    0]
[ 353   21  119    0]
[  40    0   27    0]]

 ── Eval 18: Assessment period 201810 ──
 Training samples:   13,957
```

```
 Test samples:       1,081  (admin2 areas in 201810)
 Training Logistic Regression...

 Results  (n_test=1,081):
   Accuracy:  0.8483
   Precision: 0.8444
   Recall:    0.8483
   F1 Score:  0.8446

 Confusion Matrix:
   [[581  63   0   0]
[ 29 216  38   0]
[  3  13 120   1]
[  2   0  15   0]]

 —— Eval 19: Assessment period 201806 ——
 Training samples:   14,374
 Test samples:       2,440  (admin2 areas in 201806)
 Training Logistic Regression...

 Results  (n_test=2,440):
   Accuracy:  0.8098
   Precision: 0.7892
   Recall:    0.8098
   F1 Score:  0.7916

 Confusion Matrix:
   [[1281   40   10    0]
[ 225  229   69    0]
[  22   69  466    2]
[   0    0   27    0]]

 —— Eval 20: Assessment period 201802 ——
 Training samples:   14,614
 Test samples:       2,441  (admin2 areas in 201802)
 Training Logistic Regression...

 Results  (n_test=2,441):
   Accuracy:  0.8906
   Precision: 0.8850
   Recall:    0.8906
   F1 Score:  0.8877

 Confusion Matrix:
   [[1452   56    3    0]
[  76  240   41    0]
[  12   51  482    8]
[   0    3   17    0]]

 —— Eval 21: Assessment period 201710 ——
 Training samples:   14,902
 Test samples:       2,577  (admin2 areas in 201710)
 Training Logistic Regression...

 Results  (n_test=2,577):
   Accuracy:  0.7792
```

```
  Precision: 0.7987
  Recall:    0.7792
  F1 Score:  0.7825

 Confusion Matrix:
   [[1361  281   15    1]
[ 103  183   82    0]
[  25   16  463    1]
[   0    2   43    1]]

 ── Eval 22: Assessment period 201706 ──
 Training samples:   11,953
 Test samples:       2,573  (admin2 areas in 201706)
 Training Logistic Regression...

 Results  (n_test=2,573):
   Accuracy:  0.6277
   Precision: 0.5995
   Recall:    0.6277
   F1 Score:  0.5920

 Confusion Matrix:
   [[1282  101   83    0]
[ 287  196   17    0]
[ 279  111  132    2]
[  14    0   64    5]]

 ── Eval 23: Assessment period 201702 ──
 Training samples:   9,303
 Test samples:       2,449  (admin2 areas in 201702)
 Training Logistic Regression...

 Results  (n_test=2,449):
   Accuracy:  0.8359
   Precision: 0.8216
   Recall:    0.8359
   F1 Score:  0.8253

 Confusion Matrix:
   [[1663   61    0    0    0]
[ 157  239   27    1    0]
[   7  107  143    3    0]
[   0    9   29    2    0]
[   0    0    1    0    0]]

 ── Eval 24: Assessment period 201610 ──
 Training samples:   6,140
 Test samples:       2,748  (admin2 areas in 201610)
 Training Logistic Regression...

 Results  (n_test=2,748):
   Accuracy:  0.8082
   Precision: 0.8009
   Recall:    0.8082
   F1 Score:  0.7992
```

```
  Confusion Matrix:
    [[1653  139   34    1]
 [ 157  193  115    0]
 [   6   26  370    7]
 [   0    3   39    5]]

  ── Eval 25: Assessment period 201606 ──
  Training samples:   3,099
  Test samples:       2,635  (admin2 areas in 201606)
  Training Logistic Regression...

  Results  (n_test=2,635):
    Accuracy:  0.8433
    Precision: 0.8283
    Recall:    0.8433
    F1 Score:  0.8341

  Confusion Matrix:
    [[1657   35    3    1]
 [ 113  210   41    1]
 [  24  133  354    9]
 [   0    2   51    1]]

======================================================================
MODEL: Logistic Regression  |  FEATURE SET: IPC + GDELT + temporal (no geo)
======================================================================
Using 155 features

  ── Eval 1: Assessment period 202406 ──
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Training Logistic Regression...

  Results  (n_test=1,919):
    Accuracy:  0.6483
    Precision: 0.6765
    Recall:    0.6483
    F1 Score:  0.6449

  Confusion Matrix:
    [[428  52   0   0]
 [357 438  34   1]
 [  2 156 292  18]
 [  0   2  53  86]]

  ── Eval 2: Assessment period 202402 ──
  Training samples:   12,107
  Test samples:       2,376  (admin2 areas in 202402)
  Training Logistic Regression...

  Results  (n_test=2,376):
    Accuracy:  0.6772
    Precision: 0.7110
    Recall:    0.6772
    F1 Score:  0.6484
```

```
 Confusion Matrix:
   [[904  53   0   0]
[417 370  47   0]
[103  43 329   1]
[  0   1 102   6]]

 ── Eval 3: Assessment period 202310 ──
 Training samples:   12,645
 Test samples:       1,919  (admin2 areas in 202310)
 Training Logistic Regression...

 Results  (n_test=1,919):
   Accuracy:  0.6253
   Precision: 0.7512
   Recall:    0.6253
   F1 Score:  0.6307

 Confusion Matrix:
   [[489 473   8   0]
[  6 325 131   0]
[  0  40 383   6]
[  0   0  55   3]]

 ── Eval 4: Assessment period 202306 ──
 Training samples:   12,765
 Test samples:       2,613  (admin2 areas in 202306)
 Training Logistic Regression...

 Results  (n_test=2,613):
   Accuracy:  0.4669
   Precision: 0.6023
   Recall:    0.4669
   F1 Score:  0.4358

 Confusion Matrix:
   [[597  36   3   0]
[486 507  64   0]
[620  76 116   1]
[100   1   6   0]]

 ── Eval 5: Assessment period 202302 ──
 Training samples:   13,772
 Test samples:       1,376  (admin2 areas in 202302)
 Training Logistic Regression...

 Results  (n_test=1,376):
   Accuracy:  0.7362
   Precision: 0.7456
   Recall:    0.7362
   F1 Score:  0.7364

 Confusion Matrix:
   [[386 189   5   0]
[ 93 479  38   0]
[  0  37 148   1]
[  0   0   0   0]]
```

```
 ── Eval 6: Assessment period 202210 ──
 Training samples:   13,866
 Test samples:       2,399  (admin2 areas in 202210)
 Training Logistic Regression...

 Results  (n_test=2,399):
   Accuracy:  0.7320
   Precision: 0.7265
   Recall:    0.7320
   F1 Score:  0.7236

 Confusion Matrix:
   [[633 207   2   0]
[167 566  86   1]
[  4  89 548   8]
[  0   0  79   9]]

 ── Eval 7: Assessment period 202206 ──
 Training samples:   13,795
 Test samples:       2,568  (admin2 areas in 202206)
 Training Logistic Regression...

 Results  (n_test=2,568):
   Accuracy:  0.7371
   Precision: 0.7343
   Recall:    0.7371
   F1 Score:  0.7218

 Confusion Matrix:
   [[684 122   0   0]
[196 656  65   1]
[ 49 125 548   2]
[ 31   3  81   5]]

 ── Eval 8: Assessment period 202202 ──
 Training samples:   13,862
 Test samples:       2,535  (admin2 areas in 202202)
 Training Logistic Regression...

 Results  (n_test=2,535):
   Accuracy:  0.7696
   Precision: 0.7695
   Recall:    0.7696
   F1 Score:  0.7635

 Confusion Matrix:
   [[842  74   1   0]
[268 613  27   0]
[  4 173 495   4]
[  0   5  28   1]]

 ── Eval 9: Assessment period 202110 ──
 Training samples:   12,950
 Test samples:       2,535  (admin2 areas in 202110)
 Training Logistic Regression...
```

```
 Results  (n_test=2,535):
   Accuracy:  0.7787
   Precision: 0.8004
   Recall:    0.7787
   F1 Score:  0.7810

 Confusion Matrix:
   [[807 308   3   0]
[ 71 701  94   1]
[  0  70 465   2]
[  0   0  12   1]]

 ── Eval 10: Assessment period 202106 ──
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Training Logistic Regression...

 Results  (n_test=2,536):
   Accuracy:  0.7153
   Precision: 0.7379
   Recall:    0.7153
   F1 Score:  0.7056

 Confusion Matrix:
   [[804  60  16   0]
[469 550  59   0]
[  4  98 457   5]
[  0   0  11   3]]

 ── Eval 11: Assessment period 202102 ──
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Training Logistic Regression...

 Results  (n_test=2,489):
   Accuracy:  0.7774
   Precision: 0.7864
   Recall:    0.7774
   F1 Score:  0.7751

 Confusion Matrix:
   [[1007  225    0    0]
[ 122  507   75    0]
[   8   74  416    2]
[   2    0   46    5]]

 ── Eval 12: Assessment period 202010 ──
 Training samples:   12,858
 Test samples:       2,491  (admin2 areas in 202010)
 Training Logistic Regression...

 Results  (n_test=2,491):
   Accuracy:  0.7350
   Precision: 0.7455
   Recall:    0.7350
```

```
  F1 Score:  0.7396

 Confusion Matrix:
   [[936 190  10   0]
[192 504 116   2]
[  2 107 387  40]
[  0   0   1   4]]

 —— Eval 13: Assessment period 202006 ——
 Training samples:   11,506
 Test samples:       2,570  (admin2 areas in 202006)
 Training Logistic Regression...

 Results  (n_test=2,570):
   Accuracy:  0.7027
   Precision: 0.7230
   Recall:    0.7027
   F1 Score:  0.6828

 Confusion Matrix:
   [[1077   48    3    0]
[ 482  309   15    0]
[  86   98  371    5]
[   5    0   22   49]]

 —— Eval 14: Assessment period 202002 ——
 Training samples:   12,388
 Test samples:       1,588  (admin2 areas in 202002)
 Training Logistic Regression...

 Results  (n_test=1,588):
   Accuracy:  0.7714
   Precision: 0.7868
   Recall:    0.7714
   F1 Score:  0.7704

 Confusion Matrix:
   [[568  79  21   0]
[ 29 258 166   8]
[ 13  28 359   2]
[  0   1  16  40]]

 —— Eval 15: Assessment period 201910 ——
 Training samples:   12,533
 Test samples:       2,400  (admin2 areas in 201910)
 Training Logistic Regression...

 Results  (n_test=2,400):
   Accuracy:  0.8021
   Precision: 0.8031
   Recall:    0.8021
   F1 Score:  0.8023

 Confusion Matrix:
   [[1086  107    2    0]
[  86  339  132    2]
```

```
[   8   99  453   26]
[   0    0   13   47]]

 ── Eval 16: Assessment period 201906 ──
 Training samples:   12,605
 Test samples:       2,400  (admin2 areas in 201906)
 Training Logistic Regression...

 Results  (n_test=2,400):
   Accuracy:  0.7846
   Precision: 0.7739
   Recall:    0.7846
   F1 Score:  0.7705

 Confusion Matrix:
   [[1158   17    4    0]
[ 258  246   46    0]
[  16  137  430   11]
[   0    0   28   49]]

 ── Eval 17: Assessment period 201902 ──
 Training samples:   12,785
 Test samples:       2,400  (admin2 areas in 201902)
 Training Logistic Regression...

 Results  (n_test=2,400):
   Accuracy:  0.6767
   Precision: 0.6841
   Recall:    0.6767
   F1 Score:  0.6279

 Confusion Matrix:
   [[1341   87    0    0]
[ 239  162   11    0]
[ 353   20  119    1]
[  40    0   25    2]]

 ── Eval 18: Assessment period 201810 ──
 Training samples:   13,957
 Test samples:       1,081  (admin2 areas in 201810)
 Training Logistic Regression...

 Results  (n_test=1,081):
   Accuracy:  0.8437
   Precision: 0.8471
   Recall:    0.8437
   F1 Score:  0.8411

 Confusion Matrix:
   [[585  59   0   0]
[ 37 207  39   0]
[  3  15 118   1]
[  2   0  13   2]]

 ── Eval 19: Assessment period 201806 ──
 Training samples:   14,374
```

```
 Test samples:       2,440  (admin2 areas in 201806)
 Training Logistic Regression...

 Results  (n_test=2,440):
   Accuracy:  0.8098
   Precision: 0.7945
   Recall:    0.8098
   F1 Score:  0.7930

 Confusion Matrix:
   [[1283   38   10    0]
[ 227  228   68    0]
[  22   72  462    3]
[   0    1   23    3]]

 —— Eval 20: Assessment period 201802 ——
 Training samples:   14,614
 Test samples:       2,441  (admin2 areas in 201802)
 Training Logistic Regression...

 Results  (n_test=2,441):
   Accuracy:  0.8312
   Precision: 0.8038
   Recall:    0.8312
   F1 Score:  0.7935

 Confusion Matrix:
   [[1495   13    3    0]
[ 260   55   42    0]
[  44   23  478    8]
[   3    0   16    1]]

 —— Eval 21: Assessment period 201710 ——
 Training samples:   14,902
 Test samples:       2,577  (admin2 areas in 201710)
 Training Logistic Regression...

 Results  (n_test=2,577):
   Accuracy:  0.8126
   Precision: 0.7676
   Recall:    0.8126
   F1 Score:  0.7759

 Confusion Matrix:
   [[1586   56   15    1]
[ 237   51   80    0]
[  34   13  452    6]
[   0    3   38    5]]

 —— Eval 22: Assessment period 201706 ——
 Training samples:   11,953
 Test samples:       2,573  (admin2 areas in 201706)
 Training Logistic Regression...

 Results  (n_test=2,573):
   Accuracy:  0.6292
```

```
  Precision: 0.6014
  Recall:    0.6292
  F1 Score:  0.5927

 Confusion Matrix:
   [[1296   87   82    1    0]
[ 301  183   16    0    0]
[ 283  109  129    2    1]
[  14    2   56   11    0]
[   0    0    0    0    0]]

 —— Eval 23: Assessment period 201702 ——
 Training samples:   9,303
 Test samples:       2,449  (admin2 areas in 201702)
 Training Logistic Regression...

 Results  (n_test=2,449):
   Accuracy:  0.8318
   Precision: 0.8211
   Recall:    0.8318
   F1 Score:  0.8229

 Confusion Matrix:
   [[1660   63    1    0    0]
[ 157  234   33    0    0]
[   7  111  136    6    0]
[   0   12   21    7    0]
[   0    0    1    0    0]]

 —— Eval 24: Assessment period 201610 ——
 Training samples:   6,140
 Test samples:       2,748  (admin2 areas in 201610)
 Training Logistic Regression...

 Results  (n_test=2,748):
   Accuracy:  0.8020
   Precision: 0.7961
   Recall:    0.8020
   F1 Score:  0.7961

 Confusion Matrix:
   [[1646  143   36    2]
[ 159  198  105    3]
[   7   39  352   11]
[   0    5   34    8]]

 —— Eval 25: Assessment period 201606 ——
 Training samples:   3,099
 Test samples:       2,635  (admin2 areas in 201606)
 Training Logistic Regression...

 Results  (n_test=2,635):
   Accuracy:  0.8273
   Precision: 0.8195
   Recall:    0.8273
   F1 Score:  0.8211
```

```
  Confusion Matrix:
    [[1641   48    7    0]
 [ 117  199   46    3]
 [  23  147  335   15]
 [   2    4   43    5]]

======================================================================
MODEL: Logistic Regression  |  FEATURE SET: GDELT only (no geo)
======================================================================
Using 154 features

  ── Eval 1: Assessment period 202406 ──
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Training Logistic Regression...

  Results  (n_test=1,919):
    Accuracy:  0.3147
    Precision: 0.4449
    Recall:    0.3147
    F1 Score:  0.2724

  Confusion Matrix:
    [[374  90  16   0]
 [634 169  26   1]
 [334  76  58   0]
 [103  10  25   3]]

  ── Eval 2: Assessment period 202402 ──
  Training samples:   12,107
  Test samples:       2,376  (admin2 areas in 202402)
  Training Logistic Regression...

  Results  (n_test=2,376):
    Accuracy:  0.4226
    Precision: 0.4105
    Recall:    0.4226
    F1 Score:  0.3513

  Confusion Matrix:
    [[802 118  36   1]
 [656 136  42   0]
 [346  63  65   2]
 [ 82   8  18   1]]

  ── Eval 3: Assessment period 202310 ──
  Training samples:   12,645
  Test samples:       1,919  (admin2 areas in 202310)
  Training Logistic Regression...

  Results  (n_test=1,919):
    Accuracy:  0.5055
    Precision: 0.4620
    Recall:    0.5055
    F1 Score:  0.4431
```

```
 Confusion Matrix:
   [[812 126  32   0]
[336  97  29   0]
[315  52  61   1]
[ 44   9   5   0]]

 ── Eval 4: Assessment period 202306 ──
 Training samples:   12,765
 Test samples:       2,613  (admin2 areas in 202306)
 Training Logistic Regression...

 Results  (n_test=2,613):
   Accuracy:  0.4118
   Precision: 0.4100
   Recall:    0.4118
   F1 Score:  0.3068

 Confusion Matrix:
   [[ 33 578  24   1]
[ 41 952  64   0]
[ 16 704  88   5]
[  3  87  14   3]]

 ── Eval 5: Assessment period 202302 ──
 Training samples:   13,772
 Test samples:       1,376  (admin2 areas in 202302)
 Training Logistic Regression...

 Results  (n_test=1,376):
   Accuracy:  0.4462
   Precision: 0.4545
   Recall:    0.4462
   F1 Score:  0.3467

 Confusion Matrix:
   [[537  29  14]
[511  58  41]
[142  25  19]]

 ── Eval 6: Assessment period 202210 ──
 Training samples:   13,866
 Test samples:       2,399  (admin2 areas in 202210)
 Training Logistic Regression...

 Results  (n_test=2,399):
   Accuracy:  0.3718
   Precision: 0.3860
   Recall:    0.3718
   F1 Score:  0.2803

 Confusion Matrix:
   [[743  76  23   0]
[680 103  36   1]
[516  90  42   1]
[ 66  12   6   4]]
```

```
 ── Eval 7: Assessment period 202206 ──
 Training samples:   13,795
 Test samples:       2,568  (admin2 areas in 202206)
 Training Logistic Regression...

 Results  (n_test=2,568):
   Accuracy:  0.3271
   Precision: 0.3689
   Recall:    0.3271
   F1 Score:  0.2252

 Confusion Matrix:
   [[722  71  13   0]
[812  77  29   0]
[608  77  38   1]
[ 89  18  10   3]]

 ── Eval 8: Assessment period 202202 ──
 Training samples:   13,862
 Test samples:       2,535  (admin2 areas in 202202)
 Training Logistic Regression...

 Results  (n_test=2,535):
   Accuracy:  0.3744
   Precision: 0.3709
   Recall:    0.3744
   F1 Score:  0.2616

 Confusion Matrix:
   [[844  62  10   1]
[802  81  25   0]
[579  71  23   3]
[ 24   6   3   1]]

 ── Eval 9: Assessment period 202110 ──
 Training samples:   12,950
 Test samples:       2,535  (admin2 areas in 202110)
 Training Logistic Regression...

 Results  (n_test=2,535):
   Accuracy:  0.4647
   Precision: 0.4687
   Recall:    0.4647
   F1 Score:  0.3558

 Confusion Matrix:
   [[1055   51   12    0]
[ 753   88   24    2]
[ 459   44   34    0]
[   7    4    1    1]]

 ── Eval 10: Assessment period 202106 ──
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Training Logistic Regression...
```

```
 Results  (n_test=2,536):
   Accuracy:  0.3636
   Precision: 0.3901
   Recall:    0.3636
   F1 Score:  0.2514

 Confusion Matrix:
   [[816  47  17   0]
[969  80  29   0]
[487  51  25   1]
[  7   2   4   1]]

 ── Eval 11: Assessment period 202102 ──
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Training Logistic Regression...

 Results  (n_test=2,489):
   Accuracy:  0.4958
   Precision: 0.4326
   Recall:    0.4958
   F1 Score:  0.3713

 Confusion Matrix:
   [[1171   30   31    0]
[ 643   32   29    0]
[ 450   17   30    3]
[  48    3    1    1]]

 ── Eval 12: Assessment period 202010 ──
 Training samples:   12,858
 Test samples:       2,491  (admin2 areas in 202010)
 Training Logistic Regression...

 Results  (n_test=2,491):
   Accuracy:  0.4633
   Precision: 0.4413
   Recall:    0.4633
   F1 Score:  0.3344

 Confusion Matrix:
   [[1086   27   23    0]
[ 753   33   28    0]
[ 484   16   35    1]
[   4    0    1    0]]

 ── Eval 13: Assessment period 202006 ──
 Training samples:   11,506
 Test samples:       2,570  (admin2 areas in 202006)
 Training Logistic Regression...

 Results  (n_test=2,570):
   Accuracy:  0.4479
   Precision: 0.4244
   Recall:    0.4479
```

```
  F1 Score:  0.3109

 Confusion Matrix:
   [[1093   20   15    0]
[ 756   29   21    0]
[ 509   22   28    1]
[  67    2    6    1]]

 ── Eval 14: Assessment period 202002 ──
 Training samples:   12,388
 Test samples:       1,588  (admin2 areas in 202002)
 Training Logistic Regression...

 Results  (n_test=1,588):
   Accuracy:  0.4307
   Precision: 0.4095
   Recall:    0.4307
   F1 Score:  0.2944

 Confusion Matrix:
   [[645  12  10   1]
[434  11  16   0]
[367   6  28   1]
[ 51   0   6   0]]

 ── Eval 15: Assessment period 201910 ──
 Training samples:   12,533
 Test samples:       2,400  (admin2 areas in 201910)
 Training Logistic Regression...

 Results  (n_test=2,400):
   Accuracy:  0.5012
   Precision: 0.4453
   Recall:    0.5012
   F1 Score:  0.3640

 Confusion Matrix:
   [[1159    9   27    0]
[ 535   11   13    0]
[ 544    8   33    1]
[  54    0    6    0]]

 ── Eval 16: Assessment period 201906 ──
 Training samples:   12,605
 Test samples:       2,400  (admin2 areas in 201906)
 Training Logistic Regression...

 Results  (n_test=2,400):
   Accuracy:  0.5000
   Precision: 0.4450
   Recall:    0.5000
   F1 Score:  0.3590

 Confusion Matrix:
   [[1156    6   17    0]
[ 532    6   12    0]
```

```
[ 549    6   38    1]
[  67    0   10    0]]

 ── Eval 17: Assessment period 201902 ──
 Training samples:   12,785
 Test samples:       2,400  (admin2 areas in 201902)
 Training Logistic Regression...

 Results  (n_test=2,400):
   Accuracy:  0.6029
   Precision: 0.5331
   Recall:    0.6029
   F1 Score:  0.4773

 Confusion Matrix:
   [[1406    3   18    1]
[ 394    4   13    1]
[ 452    3   37    1]
[  61    0    6    0]]

 ── Eval 18: Assessment period 201810 ──
 Training samples:   13,957
 Test samples:       1,081  (admin2 areas in 201810)
 Training Logistic Regression...

 Results  (n_test=1,081):
   Accuracy:  0.5967
   Precision: 0.5106
   Recall:    0.5967
   F1 Score:  0.4658

 Confusion Matrix:
   [[633   2   9   0]
[273   2   7   1]
[125   1   9   2]
[ 13   0   3   1]]

 ── Eval 19: Assessment period 201806 ──
 Training samples:   14,374
 Test samples:       2,440  (admin2 areas in 201806)
 Training Logistic Regression...

 Results  (n_test=2,440):
   Accuracy:  0.5533
   Precision: 0.5089
   Recall:    0.5533
   F1 Score:  0.4177

 Confusion Matrix:
   [[1310    3   17    1]
[ 499    7   17    0]
[ 519    6   31    3]
[  24    0    1    2]]

 ── Eval 20: Assessment period 201802 ──
 Training samples:   14,614
```

```
 Test samples:       2,441  (admin2 areas in 201802)
 Training Logistic Regression...

 Results  (n_test=2,441):
   Accuracy:  0.6309
   Precision: 0.5775
   Recall:    0.6309
   F1 Score:  0.5099

 Confusion Matrix:
   [[1499    5    7    0]
[ 347    3    5    2]
[ 499   14   38    2]
[  17    1    2    0]]

 —— Eval 21: Assessment period 201710 ——
 Training samples:   14,902
 Test samples:       2,577  (admin2 areas in 201710)
 Training Logistic Regression...

 Results  (n_test=2,577):
   Accuracy:  0.6453
   Precision: 0.5466
   Recall:    0.6453
   F1 Score:  0.5314

 Confusion Matrix:
   [[1626   16   14    2]
[ 351    6   11    0]
[ 465    9   31    0]
[  38    3    5    0]]

 —— Eval 22: Assessment period 201706 ——
 Training samples:   11,953
 Test samples:       2,573  (admin2 areas in 201706)
 Training Logistic Regression...

 Results  (n_test=2,573):
   Accuracy:  0.5744
   Precision: 0.4955
   Recall:    0.5744
   F1 Score:  0.4449

 Confusion Matrix:
   [[1438   12   15    1    0]
[ 478   13    7    2    0]
[ 483   14   24    2    1]
[  69    5    6    3    0]
[   0    0    0    0    0]]

 —— Eval 23: Assessment period 201702 ——
 Training samples:   9,303
 Test samples:       2,449  (admin2 areas in 201702)
 Training Logistic Regression...

 Results  (n_test=2,449):
```

```
    Accuracy:  0.7011
    Precision: 0.5899
    Recall:    0.7011
    F1 Score:  0.6077

  Confusion Matrix:
    [[1685   28   11    0    0]
 [ 379   20   23    2    0]
 [ 238   10   10    2    0]
 [  28    7    3    2    0]
 [   0    0    1    0    0]]

  ── Eval 24: Assessment period 201610 ──
  Training samples:   6,140
  Test samples:       2,748  (admin2 areas in 201610)
  Training Logistic Regression...

  Results  (n_test=2,748):
    Accuracy:  0.6659
    Precision: 0.5719
    Recall:    0.6659
    F1 Score:  0.5616

  Confusion Matrix:
    [[1785   21   20    1]
 [ 426   31    7    1]
 [ 375   20   12    2]
 [  41    2    2    2]]

  ── Eval 25: Assessment period 201606 ──
  Training samples:   3,099
  Test samples:       2,635  (admin2 areas in 201606)
  Training Logistic Regression...

  Results  (n_test=2,635):
    Accuracy:  0.6307
    Precision: 0.5190
    Recall:    0.6307
    F1 Score:  0.5263

  Confusion Matrix:
    [[1623   49   21    3]
 [ 332   20   12    1]
 [ 468   33   17    2]
 [  48    2    2    2]]

======================================================================
MODEL: Random Forest  |  FEATURE SET: IPC only
======================================================================
Using 3332 features

  ── Eval 1: Assessment period 202406 ──
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Training Random Forest...
```

```
 Results  (n_test=1,919):
   Accuracy:  0.6879
   Precision: 0.7015
   Recall:    0.6879
   F1 Score:  0.6658

 Confusion Matrix:
   [[455  23   2   0]
[164 604  59   3]
[ 17 165 132 154]
[  2   4   6 129]]

 ── Eval 2: Assessment period 202402 ──
 Training samples:   12,107
 Test samples:       2,376  (admin2 areas in 202402)
 Training Random Forest...

 Results  (n_test=2,376):
   Accuracy:  0.6376
   Precision: 0.6751
   Recall:    0.6376
   F1 Score:  0.6314

 Confusion Matrix:
   [[759 190   0   8]
[253 505  49  27]
[ 37  99 148 192]
[  0   0   6 103]]

 ── Eval 3: Assessment period 202310 ──
 Training samples:   12,645
 Test samples:       1,919  (admin2 areas in 202310)
 Training Random Forest...

 Results  (n_test=1,919):
   Accuracy:  0.5456
   Precision: 0.6624
   Recall:    0.5456
   F1 Score:  0.5757

 Confusion Matrix:
   [[607 332   5  26]
[124 239  57  42]
[ 20  29 149 231]
[  0   0   6  52]]

 ── Eval 4: Assessment period 202306 ──
 Training samples:   12,765
 Test samples:       2,613  (admin2 areas in 202306)
 Training Random Forest...

 Results  (n_test=2,613):
   Accuracy:  0.6915
   Precision: 0.7313
   Recall:    0.6915
   F1 Score:  0.6919
```

```
 Confusion Matrix:
   [[527  98   2   9]
[189 787  63  18]
[ 99 181 416 117]
[  5  24   1  77]]

 ── Eval 5: Assessment period 202302 ──
 Training samples:   13,772
 Test samples:       1,376  (admin2 areas in 202302)
 Training Random Forest...

 Results  (n_test=1,376):
   Accuracy:  0.6642
   Precision: 0.7005
   Recall:    0.6642
   F1 Score:  0.6573

 Confusion Matrix:
   [[314 266   0]
[ 80 514  16]
[ 16  84  86]]

 ── Eval 6: Assessment period 202210 ──
 Training samples:   13,866
 Test samples:       2,399  (admin2 areas in 202210)
 Training Random Forest...

 Results  (n_test=2,399):
   Accuracy:  0.6686
   Precision: 0.7243
   Recall:    0.6686
   F1 Score:  0.6798

 Confusion Matrix:
   [[583 240   0  19]
[178 581  28  33]
[ 15 133 367 134]
[  0   0  15  73]]

 ── Eval 7: Assessment period 202206 ──
 Training samples:   13,795
 Test samples:       2,568  (admin2 areas in 202206)
 Training Random Forest...

 Results  (n_test=2,568):
   Accuracy:  0.7360
   Precision: 0.7745
   Recall:    0.7360
   F1 Score:  0.7390

 Confusion Matrix:
   [[646 138   0  22]
[219 678  14   7]
[ 25 149 446 104]
[  0   0   0 120]]
```

```
 ── Eval 8: Assessment period 202202 ──
 Training samples:   13,862
 Test samples:       2,535  (admin2 areas in 202202)
 Training Random Forest...

 Results  (n_test=2,535):
   Accuracy:  0.7495
   Precision: 0.7909
   Recall:    0.7495
   F1 Score:  0.7528

 Confusion Matrix:
   [[816  98   0   3]
[221 665  18   4]
[ 37 157 387  95]
[  0   1   1  32]]

 ── Eval 9: Assessment period 202110 ──
 Training samples:   12,950
 Test samples:       2,535  (admin2 areas in 202110)
 Training Random Forest...

 Results  (n_test=2,535):
   Accuracy:  0.7704
   Precision: 0.8026
   Recall:    0.7704
   F1 Score:  0.7799

 Confusion Matrix:
   [[1000  116    0    2]
[ 199  593   57   18]
[  36   61  348   92]
[   0    0    1   12]]

 ── Eval 10: Assessment period 202106 ──
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Training Random Forest...

 Results  (n_test=2,536):
   Accuracy:  0.7054
   Precision: 0.7466
   Recall:    0.7054
   F1 Score:  0.7052

 Confusion Matrix:
   [[807  58  15   0]
[415 625  38   0]
[ 29 124 343  68]
[  0   0   0  14]]

 ── Eval 11: Assessment period 202102 ──
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Training Random Forest...
```

```
 Results  (n_test=2,489):
   Accuracy:  0.7887
   Precision: 0.7995
   Recall:    0.7887
   F1 Score:  0.7900

 Confusion Matrix:
   [[1063  169    0    0]
[ 158  500   38    8]
[  28   98  347   27]
[   0    0    0   53]]

 ── Eval 12: Assessment period 202010 ──
 Training samples:   12,858
 Test samples:       2,491  (admin2 areas in 202010)
 Training Random Forest...

 Results  (n_test=2,491):
   Accuracy:  0.6965
   Precision: 0.7108
   Recall:    0.6965
   F1 Score:  0.7024

 Confusion Matrix:
   [[899 226  11   0]
[256 461  93   4]
[  8 104 371  53]
[  0   0   1   4]]

 ── Eval 13: Assessment period 202006 ──
 Training samples:   11,506
 Test samples:       2,570  (admin2 areas in 202006)
 Training Random Forest...

 Results  (n_test=2,570):
   Accuracy:  0.7315
   Precision: 0.7442
   Recall:    0.7315
   F1 Score:  0.7271

 Confusion Matrix:
   [[986 133   9   0]
[293 481  26   6]
[ 65 111 342  42]
[  5   0   0  71]]

 ── Eval 14: Assessment period 202002 ──
 Training samples:   12,388
 Test samples:       1,588  (admin2 areas in 202002)
 Training Random Forest...

 Results  (n_test=1,588):
   Accuracy:  0.7330
   Precision: 0.7662
   Recall:    0.7330
```

```
  F1 Score:  0.7382

 Confusion Matrix:
   [[482 184   2   0]
[ 25 281 142  13]
[  0  15 345  42]
[  0   0   1  56]]

 —— Eval 15: Assessment period 201910 ——
 Training samples:   12,533
 Test samples:       2,400  (admin2 areas in 201910)
 Training Random Forest...

 Results  (n_test=2,400):
   Accuracy:  0.7929
   Precision: 0.8041
   Recall:    0.7929
   F1 Score:  0.7939

 Confusion Matrix:
   [[1101   85    9    0]
[ 123  347   76   13]
[   6  118  396   66]
[   0    1    0   59]]

 —— Eval 16: Assessment period 201906 ——
 Training samples:   12,605
 Test samples:       2,400  (admin2 areas in 201906)
 Training Random Forest...

 Results  (n_test=2,400):
   Accuracy:  0.7525
   Precision: 0.7671
   Recall:    0.7525
   F1 Score:  0.7491

 Confusion Matrix:
   [[1077  101    1    0]
[ 210  301   34    5]
[  31  140  357   66]
[   0    1    5   71]]

 —— Eval 17: Assessment period 201902 ——
 Training samples:   12,785
 Test samples:       2,400  (admin2 areas in 201902)
 Training Random Forest...

 Results  (n_test=2,400):
   Accuracy:  0.7917
   Precision: 0.8369
   Recall:    0.7917
   F1 Score:  0.8001

 Confusion Matrix:
   [[1255  173    0    0]
[ 120  276    2   14]
```

```
[  22  106  302   63]
[   0    0    0   67]]

 ── Eval 18: Assessment period 201810 ──
 Training samples:   13,957
 Test samples:       1,081  (admin2 areas in 201810)
 Training Random Forest...

 Results  (n_test=1,081):
   Accuracy:  0.7493
   Precision: 0.8407
   Recall:    0.7493
   F1 Score:  0.7694

 Confusion Matrix:
   [[570  74   0   0]
[ 59 177   0  47]
[  0  17  47  73]
[  0   1   0  16]]

 ── Eval 19: Assessment period 201806 ──
 Training samples:   14,374
 Test samples:       2,440  (admin2 areas in 201806)
 Training Random Forest...

 Results  (n_test=2,440):
   Accuracy:  0.7291
   Precision: 0.7845
   Recall:    0.7291
   F1 Score:  0.7449

 Confusion Matrix:
   [[1156  173    0    2    0]
[ 182  264   20   57    0]
[  18  119  335   77   10]
[   0    1    0   24    2]
[   0    0    0    0    0]]

 ── Eval 20: Assessment period 201802 ──
 Training samples:   14,614
 Test samples:       2,441  (admin2 areas in 201802)
 Training Random Forest...

 Results  (n_test=2,441):
   Accuracy:  0.6993
   Precision: 0.8381
   Recall:    0.6993
   F1 Score:  0.7433

 Confusion Matrix:
   [[1173  335    0    3    0]
[  89  226   12   23    7]
[  10  116  298   90   39]
[   0    0    0   10   10]
[   0    0    0    0    0]]
```

```
 ── Eval 21: Assessment period 201710 ──
 Training samples:   14,902
 Test samples:       2,577  (admin2 areas in 201710)
 Training Random Forest...

 Results  (n_test=2,577):
   Accuracy:  0.7905
   Precision: 0.8719
   Recall:    0.7905
   F1 Score:  0.8146

 Confusion Matrix:
   [[1406  252    0    0    0]
[  53  272   16   25    2]
[   5   93  319   77   11]
[   0    0    1   40    5]
[   0    0    0    0    0]]

 ── Eval 22: Assessment period 201706 ──
 Training samples:   11,953
 Test samples:       2,573  (admin2 areas in 201706)
 Training Random Forest...

 Results  (n_test=2,573):
   Accuracy:  0.7171
   Precision: 0.7187
   Recall:    0.7171
   F1 Score:  0.7150

 Confusion Matrix:
   [[1255   92  119    0    0]
[ 266  208    9   15    2]
[   1  142  313   61    7]
[   0    7    3   69    4]
[   0    0    0    0    0]]

 ── Eval 23: Assessment period 201702 ──
 Training samples:   9,303
 Test samples:       2,449  (admin2 areas in 201702)
 Training Random Forest...

 Results  (n_test=2,449):
   Accuracy:  0.8216
   Precision: 0.8237
   Recall:    0.8216
   F1 Score:  0.8163

 Confusion Matrix:
   [[1664   60    0    0    0]
[ 156  209   35   21    3]
[   6   94  114   34   12]
[   0   10    2   24    4]
[   0    0    0    0    1]]

 ── Eval 24: Assessment period 201610 ──
 Training samples:   6,140
```

```
  Test samples:       2,748  (admin2 areas in 201610)
  Training Random Forest...

  Results  (n_test=2,748):
    Accuracy:  0.8301
    Precision: 0.8395
    Recall:    0.8301
    F1 Score:  0.8320

  Confusion Matrix:
    [[1672  138   16    1]
 [ 140  238   41   46]
 [   8   41  326   34]
 [   0    0    2   45]]

  —— Eval 25: Assessment period 201606 ——
  Training samples:   3,099
  Test samples:       2,635  (admin2 areas in 201606)
  Training Random Forest...

  Results  (n_test=2,635):
    Accuracy:  0.8835
    Precision: 0.8891
    Recall:    0.8835
    F1 Score:  0.8828

  Confusion Matrix:
    [[1663   23   10    0]
 [  81  227   38   19]
 [  11   83  385   41]
 [   0    1    0   53]]

======================================================================
MODEL: Random Forest  |  FEATURE SET: IPC + GDELT base
======================================================================
Using 3451 features

  —— Eval 1: Assessment period 202406 ——
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Training Random Forest...

  Results  (n_test=1,919):
    Accuracy:  0.6754
    Precision: 0.7061
    Recall:    0.6754
    F1 Score:  0.6577

  Confusion Matrix:
    [[470   6   4   0]
 [229 553  45   3]
 [ 30 140 150 148]
 [  2   0  16 123]]

  —— Eval 2: Assessment period 202402 ——
  Training samples:   12,107
```

```
 Test samples:       2,376  (admin2 areas in 202402)
 Training Random Forest...

 Results  (n_test=2,376):
   Accuracy:  0.6721
   Precision: 0.7133
   Recall:    0.6721
   F1 Score:  0.6580

 Confusion Matrix:
   [[886  64   2   5]
[330 441  49  14]
[ 76  62 170 168]
[  0   0   9 100]]

 ── Eval 3: Assessment period 202310 ──
 Training samples:   12,645
 Test samples:       1,919  (admin2 areas in 202310)
 Training Random Forest...

 Results  (n_test=1,919):
   Accuracy:  0.6180
   Precision: 0.6924
   Recall:    0.6180
   F1 Score:  0.6391

 Confusion Matrix:
   [[716 243   5   6]
[131 249  45  37]
[ 39  29 181 180]
[  0   4  14  40]]

 ── Eval 4: Assessment period 202306 ──
 Training samples:   12,765
 Test samples:       2,613  (admin2 areas in 202306)
 Training Random Forest...

 Results  (n_test=2,613):
   Accuracy:  0.6984
   Precision: 0.7449
   Recall:    0.6984
   F1 Score:  0.7053

 Confusion Matrix:
   [[554  77   0   5]
[271 693  67  26]
[ 99  90 512 112]
[ 11   3  27  66]]

 ── Eval 5: Assessment period 202302 ──
 Training samples:   13,772
 Test samples:       1,376  (admin2 areas in 202302)
 Training Random Forest...

 Results  (n_test=1,376):
   Accuracy:  0.7253
```

```
  Precision: 0.7459
  Recall:    0.7253
  F1 Score:  0.7222

 Confusion Matrix:
   [[412 166   0   2]
[ 94 504  11   1]
[ 30  68  82   6]
[  0   0   0   0]]

 ── Eval 6: Assessment period 202210 ──
 Training samples:   13,866
 Test samples:       2,399  (admin2 areas in 202210)
 Training Random Forest...

 Results  (n_test=2,399):
   Accuracy:  0.6882
   Precision: 0.7285
   Recall:    0.6882
   F1 Score:  0.6973

 Confusion Matrix:
   [[635 202   0   5]
[183 564  45  28]
[ 31  96 383 139]
[  0   0  19  69]]

 ── Eval 7: Assessment period 202206 ──
 Training samples:   13,795
 Test samples:       2,568  (admin2 areas in 202206)
 Training Random Forest...

 Results  (n_test=2,568):
   Accuracy:  0.7177
   Precision: 0.7468
   Recall:    0.7177
   F1 Score:  0.7135

 Confusion Matrix:
   [[756  43   0   7]
[331 518  62   7]
[ 70 113 486  55]
[ 29   1   7  83]]

 ── Eval 8: Assessment period 202202 ──
 Training samples:   13,862
 Test samples:       2,535  (admin2 areas in 202202)
 Training Random Forest...

 Results  (n_test=2,535):
   Accuracy:  0.7613
   Precision: 0.7848
   Recall:    0.7613
   F1 Score:  0.7566

 Confusion Matrix:
```

```
   [[892  22   0   3]
[321 557  25   5]
[ 48 145 458  25]
[  0   5   6  23]]

 ── Eval 9: Assessment period 202110 ──
 Training samples:   12,950
 Test samples:       2,535  (admin2 areas in 202110)
 Training Random Forest...

 Results  (n_test=2,535):
   Accuracy:  0.7491
   Precision: 0.7778
   Recall:    0.7491
   F1 Score:  0.7445

 Confusion Matrix:
   [[1069   47    2    0]
[ 332  449   84    2]
[  35   55  373   74]
[   0    0    5    8]]

 ── Eval 10: Assessment period 202106 ──
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Training Random Forest...

 Results  (n_test=2,536):
   Accuracy:  0.6841
   Precision: 0.7307
   Recall:    0.6841
   F1 Score:  0.6809

 Confusion Matrix:
   [[810  54  16   0]
[469 554  55   0]
[ 38 103 358  65]
[  0   0   1  13]]

 ── Eval 11: Assessment period 202102 ──
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Training Random Forest...

 Results  (n_test=2,489):
   Accuracy:  0.7830
   Precision: 0.7858
   Recall:    0.7830
   F1 Score:  0.7738

 Confusion Matrix:
   [[1172   56    1    3]
[ 261  390   48    5]
[  48   84  338   30]
[   2    0    2   49]]
```

```
 —— Eval 12: Assessment period 202010 ——
 Training samples:   12,858
 Test samples:       2,491  (admin2 areas in 202010)
 Training Random Forest...

 Results  (n_test=2,491):
   Accuracy:  0.6905
   Precision: 0.6981
   Recall:    0.6905
   F1 Score:  0.6861

 Confusion Matrix:
   [[992 134  10   0]
[366 375  71   2]
[ 10 126 349  51]
[  0   0   1   4]]

 —— Eval 13: Assessment period 202006 ——
 Training samples:   11,506
 Test samples:       2,570  (admin2 areas in 202006)
 Training Random Forest...

 Results  (n_test=2,570):
   Accuracy:  0.7019
   Precision: 0.7234
   Recall:    0.7019
   F1 Score:  0.6871

 Confusion Matrix:
   [[1055   67    4    2]
[ 414  365   20    7]
[  77  114  318   51]
[   5    0    5   66]]

 —— Eval 14: Assessment period 202002 ——
 Training samples:   12,388
 Test samples:       1,588  (admin2 areas in 202002)
 Training Random Forest...

 Results  (n_test=1,588):
   Accuracy:  0.7210
   Precision: 0.7388
   Recall:    0.7210
   F1 Score:  0.7216

 Confusion Matrix:
   [[552 111   3   2]
[ 82 224 120  35]
[ 11  17 314  60]
[  0   0   2  55]]

 —— Eval 15: Assessment period 201910 ——
 Training samples:   12,533
 Test samples:       2,400  (admin2 areas in 201910)
 Training Random Forest...
```

```
 Results  (n_test=2,400):
   Accuracy:  0.7471
   Precision: 0.7581
   Recall:    0.7471
   F1 Score:  0.7395

 Confusion Matrix:
   [[1117   76    2    0]
[ 199  242   76   42]
[  66   68  376   76]
[   0    1    1   58]]

 ── Eval 16: Assessment period 201906 ──
 Training samples:   12,605
 Test samples:       2,400  (admin2 areas in 201906)
 Training Random Forest...

 Results  (n_test=2,400):
   Accuracy:  0.7450
   Precision: 0.7524
   Recall:    0.7450
   F1 Score:  0.7363

 Confusion Matrix:
   [[1095   80    4    0]
[ 253  257   35    5]
[  42  124  363   65]
[   0    0    4   73]]

 ── Eval 17: Assessment period 201902 ──
 Training samples:   12,785
 Test samples:       2,400  (admin2 areas in 201902)
 Training Random Forest...

 Results  (n_test=2,400):
   Accuracy:  0.8058
   Precision: 0.8166
   Recall:    0.8058
   F1 Score:  0.8018

 Confusion Matrix:
   [[1350   78    0    0]
[ 179  215    5   13]
[  56   58  317   62]
[   0    0   15   52]]

 ── Eval 18: Assessment period 201810 ──
 Training samples:   13,957
 Test samples:       1,081  (admin2 areas in 201810)
 Training Random Forest...

 Results  (n_test=1,081):
   Accuracy:  0.7650
   Precision: 0.8367
   Recall:    0.7650
   F1 Score:  0.7799
```

```
 Confusion Matrix:
   [[596  46   0   2]
[ 72 164   5  42]
[  3  14  52  68]
[  1   1   0  15]]

 —— Eval 19: Assessment period 201806 ——
 Training samples:   14,374
 Test samples:       2,440  (admin2 areas in 201806)
 Training Random Forest...

 Results  (n_test=2,440):
   Accuracy:  0.7430
   Precision: 0.7715
   Recall:    0.7430
   F1 Score:  0.7486

 Confusion Matrix:
   [[1181  131   14    5]
[ 214  223   31   55]
[  47   48  384   80]
[   0    0    2   25]]

 —— Eval 20: Assessment period 201802 ——
 Training samples:   14,614
 Test samples:       2,441  (admin2 areas in 201802)
 Training Random Forest...

 Results  (n_test=2,441):
   Accuracy:  0.8329
   Precision: 0.8764
   Recall:    0.8329
   F1 Score:  0.8482

 Confusion Matrix:
   [[1416   83    4    8    0]
[ 101  216   22   18    0]
[  10   52  384  106    1]
[   0    3    0   17    0]
[   0    0    0    0    0]]

 —— Eval 21: Assessment period 201710 ——
 Training samples:   14,902
 Test samples:       2,577  (admin2 areas in 201710)
 Training Random Forest...

 Results  (n_test=2,577):
   Accuracy:  0.8110
   Precision: 0.8378
   Recall:    0.8110
   F1 Score:  0.8212

 Confusion Matrix:
   [[1504  116   22   16    0]
[  99  209   41   19    0]
```

```
[  18   61  348   78    0]
[   3    1   12   29    1]
[   0    0    0    0    0]]

 ── Eval 22: Assessment period 201706 ──
 Training samples:   11,953
 Test samples:       2,573  (admin2 areas in 201706)
 Training Random Forest...

 Results  (n_test=2,573):
   Accuracy:  0.6992
   Precision: 0.6895
   Recall:    0.6992
   F1 Score:  0.6881

 Confusion Matrix:
   [[1255   72  125   14    0]
[ 268  164   45   23    0]
[  66   73  332   52    1]
[  14    4   17   48    0]
[   0    0    0    0    0]]

 ── Eval 23: Assessment period 201702 ──
 Training samples:   9,303
 Test samples:       2,449  (admin2 areas in 201702)
 Training Random Forest...

 Results  (n_test=2,449):
   Accuracy:  0.8122
   Precision: 0.8101
   Recall:    0.8122
   F1 Score:  0.8083

 Confusion Matrix:
   [[1626   85    2   11    0]
[ 157  197   47   23    0]
[  12   79  143   26    0]
[   2   10    5   23    0]
[   1    0    0    0    0]]

 ── Eval 24: Assessment period 201610 ──
 Training samples:   6,140
 Test samples:       2,748  (admin2 areas in 201610)
 Training Random Forest...

 Results  (n_test=2,748):
   Accuracy:  0.8246
   Precision: 0.8212
   Recall:    0.8246
   F1 Score:  0.8191

 Confusion Matrix:
   [[1690   88   36   13]
[ 167  203   78   17]
[   8   42  336   23]
[   1    1    8   37]]
```

```
  ── Eval 25: Assessment period 201606 ──
  Training samples:   3,099
  Test samples:       2,635  (admin2 areas in 201606)
  Training Random Forest...

  Results  (n_test=2,635):
    Accuracy:  0.8425
    Precision: 0.8317
    Recall:    0.8425
    F1 Score:  0.8293

  Confusion Matrix:
    [[1651   22   15    8]
 [ 162  127   59   17]
 [  43   54  398   25]
 [   0    1    9   44]]

======================================================================
MODEL: Random Forest  |  FEATURE SET: IPC + GDELT + temporal
======================================================================
Using 3486 features

  ── Eval 1: Assessment period 202406 ──
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Training Random Forest...

  Results  (n_test=1,919):
    Accuracy:  0.6686
    Precision: 0.6973
    Recall:    0.6686
    F1 Score:  0.6486

  Confusion Matrix:
    [[463   3  12   2]
 [232 557  37   4]
 [ 37 152 140 139]
 [  0   6  12 123]]

  ── Eval 2: Assessment period 202402 ──
  Training samples:   12,107
  Test samples:       2,376  (admin2 areas in 202402)
  Training Random Forest...

  Results  (n_test=2,376):
    Accuracy:  0.6477
    Precision: 0.6707
    Recall:    0.6477
    F1 Score:  0.6369

  Confusion Matrix:
    [[815 133   6   3]
 [313 450  59  12]
 [ 66  81 176 153]
 [  0   5   6  98]]
```

```
 ── Eval 3: Assessment period 202310 ──
 Training samples:   12,645
 Test samples:       1,919  (admin2 areas in 202310)
 Training Random Forest...

 Results  (n_test=1,919):
   Accuracy:  0.5727
   Precision: 0.6511
   Recall:    0.5727
   F1 Score:  0.5993

 Confusion Matrix:
   [[620 338   9   3]
[123 215  92  32]
[ 26   7 223 173]
[  0   0  17  41]]

 ── Eval 4: Assessment period 202306 ──
 Training samples:   12,765
 Test samples:       2,613  (admin2 areas in 202306)
 Training Random Forest...

 Results  (n_test=2,613):
   Accuracy:  0.6326
   Precision: 0.7220
   Recall:    0.6326
   F1 Score:  0.6389

 Confusion Matrix:
   [[574  54   4   4]
[375 608  63  11]
[245  79 418  71]
[ 40   7   7  53]]

 ── Eval 5: Assessment period 202302 ──
 Training samples:   13,772
 Test samples:       1,376  (admin2 areas in 202302)
 Training Random Forest...

 Results  (n_test=1,376):
   Accuracy:  0.7812
   Precision: 0.7850
   Recall:    0.7812
   F1 Score:  0.7830

 Confusion Matrix:
   [[452 123   5   0]
[102 480  26   2]
[ 25  14 143   4]
[  0   0   0   0]]

 ── Eval 6: Assessment period 202210 ──
 Training samples:   13,866
 Test samples:       2,399  (admin2 areas in 202210)
 Training Random Forest...
```

```
 Results  (n_test=2,399):
   Accuracy:  0.6799
   Precision: 0.7205
   Recall:    0.6799
   F1 Score:  0.6909

 Confusion Matrix:
   [[609 222   2   9]
[196 545  47  32]
[ 17  86 408 138]
[  0   0  19  69]]

 ── Eval 7: Assessment period 202206 ──
 Training samples:   13,795
 Test samples:       2,568  (admin2 areas in 202206)
 Training Random Forest...

 Results  (n_test=2,568):
   Accuracy:  0.7434
   Precision: 0.7594
   Recall:    0.7434
   F1 Score:  0.7422

 Confusion Matrix:
   [[722  79   0   5]
[255 592  64   7]
[ 65 101 515  43]
[ 27   5   8  80]]

 ── Eval 8: Assessment period 202202 ──
 Training samples:   13,862
 Test samples:       2,535  (admin2 areas in 202202)
 Training Random Forest...

 Results  (n_test=2,535):
   Accuracy:  0.7349
   Precision: 0.7650
   Recall:    0.7349
   F1 Score:  0.7308

 Confusion Matrix:
   [[885  30   1   1]
[352 531  20   5]
[ 50 146 433  47]
[  0   6  14  14]]

 ── Eval 9: Assessment period 202110 ──
 Training samples:   12,950
 Test samples:       2,535  (admin2 areas in 202110)
 Training Random Forest...

 Results  (n_test=2,535):
   Accuracy:  0.7625
   Precision: 0.7850
   Recall:    0.7625
```

```
  F1 Score:  0.7628

 Confusion Matrix:
   [[1042   73    2    1]
[ 276  502   84    5]
[  33   53  381   70]
[   0    0    5    8]]

 ── Eval 10: Assessment period 202106 ──
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Training Random Forest...

 Results  (n_test=2,536):
   Accuracy:  0.6684
   Precision: 0.7288
   Recall:    0.6684
   F1 Score:  0.6646

 Confusion Matrix:
   [[816  48  16   0]
[512 510  47   9]
[ 43  93 356  72]
[  0   0   1  13]]

 ── Eval 11: Assessment period 202102 ──
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Training Random Forest...

 Results  (n_test=2,489):
   Accuracy:  0.7638
   Precision: 0.7666
   Recall:    0.7638
   F1 Score:  0.7509

 Confusion Matrix:
   [[1157   69    1    5]
[ 289  331   75    9]
[  47   51  363   39]
[   1    0    2   50]]

 ── Eval 12: Assessment period 202010 ──
 Training samples:   12,858
 Test samples:       2,491  (admin2 areas in 202010)
 Training Random Forest...

 Results  (n_test=2,491):
   Accuracy:  0.7134
   Precision: 0.7368
   Recall:    0.7134
   F1 Score:  0.7167

 Confusion Matrix:
   [[1021   96   12    7]
[ 297  405   79   33]
```

```
[  11  111  347   67]
[   0    0    1    4]]

 ── Eval 13: Assessment period 202006 ──
 Training samples:   11,506
 Test samples:       2,570  (admin2 areas in 202006)
 Training Random Forest...

 Results  (n_test=2,570):
   Accuracy:  0.7074
   Precision: 0.7274
   Recall:    0.7074
   F1 Score:  0.6931

 Confusion Matrix:
   [[1053   64    6    5]
[ 411  363   24    8]
[  67  105  334   54]
[   3    0    5   68]]

 ── Eval 14: Assessment period 202002 ──
 Training samples:   12,388
 Test samples:       1,588  (admin2 areas in 202002)
 Training Random Forest...

 Results  (n_test=1,588):
   Accuracy:  0.7267
   Precision: 0.7500
   Recall:    0.7267
   F1 Score:  0.7283

 Confusion Matrix:
   [[566  92   4   6]
[ 70 225 135  31]
[  7  21 307  67]
[  0   0   1  56]]

 ── Eval 15: Assessment period 201910 ──
 Training samples:   12,533
 Test samples:       2,400  (admin2 areas in 201910)
 Training Random Forest...

 Results  (n_test=2,400):
   Accuracy:  0.7221
   Precision: 0.7329
   Recall:    0.7221
   F1 Score:  0.7149

 Confusion Matrix:
   [[1081   91   12   11]
[ 227  219   84   29]
[  59   57  378   92]
[   0    3    2   55]]

 ── Eval 16: Assessment period 201906 ──
 Training samples:   12,605
```

```
 Test samples:       2,400  (admin2 areas in 201906)
 Training Random Forest...

 Results  (n_test=2,400):
   Accuracy:  0.7396
   Precision: 0.7466
   Recall:    0.7396
   F1 Score:  0.7280

 Confusion Matrix:
   [[1109   60    5    5]
[ 262  237   38   13]
[  52  118  357   67]
[   0    0    5   72]]

 ── Eval 17: Assessment period 201902 ──
 Training samples:   12,785
 Test samples:       2,400  (admin2 areas in 201902)
 Training Random Forest...

 Results  (n_test=2,400):
   Accuracy:  0.7696
   Precision: 0.7910
   Recall:    0.7696
   F1 Score:  0.7756

 Confusion Matrix:
   [[1248  175    0    5]
[ 158  241    7    6]
[  34   78  325   56]
[   0    0   34   33]]

 ── Eval 18: Assessment period 201810 ──
 Training samples:   13,957
 Test samples:       1,081  (admin2 areas in 201810)
 Training Random Forest...

 Results  (n_test=1,081):
   Accuracy:  0.7493
   Precision: 0.8107
   Recall:    0.7493
   F1 Score:  0.7638

 Confusion Matrix:
   [[585  52   1   6]
[ 82 149   9  43]
[  3  14  60  60]
[  0   0   1  16]]

 ── Eval 19: Assessment period 201806 ──
 Training samples:   14,374
 Test samples:       2,440  (admin2 areas in 201806)
 Training Random Forest...

 Results  (n_test=2,440):
   Accuracy:  0.7496
```

```
  Precision: 0.7749
  Recall:    0.7496
  F1 Score:  0.7496

 Confusion Matrix:
   [[1232   80   10    9]
[ 234  200   39   50]
[  47   50  377   85]
[   0    0    7   20]]

 ── Eval 20: Assessment period 201802 ──
 Training samples:   14,614
 Test samples:       2,441  (admin2 areas in 201802)
 Training Random Forest...

 Results  (n_test=2,441):
   Accuracy:  0.8144
   Precision: 0.8576
   Recall:    0.8144
   F1 Score:  0.8297

 Confusion Matrix:
   [[1388  103    3   17    0]
[  96  212   30   19    0]
[  18   82  373   79    1]
[   0    3    2   15    0]
[   0    0    0    0    0]]

 ── Eval 21: Assessment period 201710 ──
 Training samples:   14,902
 Test samples:       2,577  (admin2 areas in 201710)
 Training Random Forest...

 Results  (n_test=2,577):
   Accuracy:  0.7994
   Precision: 0.8309
   Recall:    0.7994
   F1 Score:  0.8101

 Confusion Matrix:
   [[1503  113   10   32    0]
[ 125  188   29   26    0]
[  26   69  338   72    0]
[   1    3   10   31    1]
[   0    0    0    0    0]]

 ── Eval 22: Assessment period 201706 ──
 Training samples:   11,953
 Test samples:       2,573  (admin2 areas in 201706)
 Training Random Forest...

 Results  (n_test=2,573):
   Accuracy:  0.6844
   Precision: 0.6846
   Recall:    0.6844
   F1 Score:  0.6802
```

```
 Confusion Matrix:
   [[1230  114   87   35    0]
[ 282  173   21   24    0]
[  50  108  306   59    1]
[   3    7   21   52    0]
[   0    0    0    0    0]]

 ── Eval 23: Assessment period 201702 ──
 Training samples:   9,303
 Test samples:       2,449  (admin2 areas in 201702)
 Training Random Forest...

 Results  (n_test=2,449):
   Accuracy:  0.8138
   Precision: 0.8214
   Recall:    0.8138
   F1 Score:  0.8123

 Confusion Matrix:
   [[1611   98    3   12    0]
[ 149  228   28   19    0]
[  12   93  125   30    0]
[   1    9    2   28    0]
[   0    0    0    0    1]]

 ── Eval 24: Assessment period 201610 ──
 Training samples:   6,140
 Test samples:       2,748  (admin2 areas in 201610)
 Training Random Forest...

 Results  (n_test=2,748):
   Accuracy:  0.8064
   Precision: 0.8017
   Recall:    0.8064
   F1 Score:  0.8021

 Confusion Matrix:
   [[1667  120   35    5]
[ 164  192   95   14]
[   9   57  321   22]
[   1    3    7   36]]

 ── Eval 25: Assessment period 201606 ──
 Training samples:   3,099
 Test samples:       2,635  (admin2 areas in 201606)
 Training Random Forest...

 Results  (n_test=2,635):
   Accuracy:  0.8452
   Precision: 0.8365
   Recall:    0.8452
   F1 Score:  0.8338

 Confusion Matrix:
   [[1644   22   19   11]
```

```
 [ 161  134   54   16]
 [  35   51  407   27]
 [   0    5    7   42]]

======================================================================
MODEL: Random Forest  |  FEATURE SET: GDELT only
======================================================================
Using 3485 features

  ── Eval 1: Assessment period 202406 ──
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Training Random Forest...

  Results  (n_test=1,919):
    Accuracy:  0.6415
    Precision: 0.6538
    Recall:    0.6415
    F1 Score:  0.6095

  Confusion Matrix:
    [[455   1  19   5]
 [217 564  41   8]
 [ 81 158  88 141]
 [  1   5  11 124]]

  ── Eval 2: Assessment period 202402 ──
  Training samples:   12,107
  Test samples:       2,376  (admin2 areas in 202402)
  Training Random Forest...

  Results  (n_test=2,376):
    Accuracy:  0.5993
    Precision: 0.6215
    Recall:    0.5993
    F1 Score:  0.5906

  Confusion Matrix:
    [[740 176  17  24]
 [281 443  85  25]
 [110  55 141 170]
 [  1   5   3 100]]

  ── Eval 3: Assessment period 202310 ──
  Training samples:   12,645
  Test samples:       1,919  (admin2 areas in 202310)
  Training Random Forest...

  Results  (n_test=1,919):
    Accuracy:  0.5237
    Precision: 0.6009
    Recall:    0.5237
    F1 Score:  0.5403

  Confusion Matrix:
    [[585 346  24  15]
```

```
[152 245  32  33]
[ 91  39 138 161]
[  0   4  17  37]]

 ── Eval 4: Assessment period 202306 ──
 Training samples:   12,765
 Test samples:       2,613  (admin2 areas in 202306)
 Training Random Forest...

 Results  (n_test=2,613):
   Accuracy:  0.6667
   Precision: 0.7241
   Recall:    0.6667
   F1 Score:  0.6744

 Confusion Matrix:
   [[531  78  11  16]
[292 687  51  27]
[132 106 451 124]
[  1   4  29  73]]

 ── Eval 5: Assessment period 202302 ──
 Training samples:   13,772
 Test samples:       1,376  (admin2 areas in 202302)
 Training Random Forest...

 Results  (n_test=1,376):
   Accuracy:  0.5763
   Precision: 0.5927
   Recall:    0.5763
   F1 Score:  0.5461

 Confusion Matrix:
   [[283 292   0   5]
[ 94 495  15   6]
[ 43 123  15   5]
[  0   0   0   0]]

 ── Eval 6: Assessment period 202210 ──
 Training samples:   13,866
 Test samples:       2,399  (admin2 areas in 202210)
 Training Random Forest...

 Results  (n_test=2,399):
   Accuracy:  0.6319
   Precision: 0.6939
   Recall:    0.6319
   F1 Score:  0.6376

 Confusion Matrix:
   [[602 209   4  27]
[220 550  14  36]
[ 84 142 295 128]
[  4   0  15  69]]

 ── Eval 7: Assessment period 202206 ──
```

```
 Training samples:   13,795
 Test samples:       2,568  (admin2 areas in 202206)
 Training Random Forest...

 Results  (n_test=2,568):
   Accuracy:  0.6729
   Precision: 0.7195
   Recall:    0.6729
   F1 Score:  0.6663

 Confusion Matrix:
   [[682 108   6  10]
[250 638  11  19]
[142 207 326  49]
[ 30   1   7  82]]

 —— Eval 8: Assessment period 202202 ——
 Training samples:   13,862
 Test samples:       2,535  (admin2 areas in 202202)
 Training Random Forest...

 Results  (n_test=2,535):
   Accuracy:  0.6682
   Precision: 0.7046
   Recall:    0.6682
   F1 Score:  0.6653

 Confusion Matrix:
   [[804  88   8  17]
[366 498  26  18]
[ 87 188 368  33]
[  2   1   7  24]]

 —— Eval 9: Assessment period 202110 ——
 Training samples:   12,950
 Test samples:       2,535  (admin2 areas in 202110)
 Training Random Forest...

 Results  (n_test=2,535):
   Accuracy:  0.7235
   Precision: 0.7470
   Recall:    0.7235
   F1 Score:  0.7209

 Confusion Matrix:
   [[1022   78   11    7]
[ 322  479   53   13]
[  77   87  328   45]
[   6    1    1    5]]

 —— Eval 10: Assessment period 202106 ——
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Training Random Forest...

 Results  (n_test=2,536):
```

```
  Accuracy:  0.6325
  Precision: 0.6920
  Recall:    0.6325
  F1 Score:  0.6342

 Confusion Matrix:
   [[772  56  32  20]
[452 544  54  28]
[ 81 129 275  79]
[  0   1   0  13]]

 —— Eval 11: Assessment period 202102 ——
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Training Random Forest...

 Results  (n_test=2,489):
  Accuracy:  0.7111
  Precision: 0.7256
  Recall:    0.7111
  F1 Score:  0.7025

 Confusion Matrix:
   [[1108   81    7   36]
[ 296  334   48   26]
[  85   98  280   37]
[   3    0    2   48]]

 —— Eval 12: Assessment period 202010 ——
 Training samples:   12,858
 Test samples:       2,491  (admin2 areas in 202010)
 Training Random Forest...

 Results  (n_test=2,491):
  Accuracy:  0.6210
  Precision: 0.6573
  Recall:    0.6210
  F1 Score:  0.6143

 Confusion Matrix:
   [[994 100  14  28]
[459 254  45  56]
[ 71 104 297  64]
[  2   0   1   2]]

 —— Eval 13: Assessment period 202006 ——
 Training samples:   11,506
 Test samples:       2,570  (admin2 areas in 202006)
 Training Random Forest...

 Results  (n_test=2,570):
  Accuracy:  0.6782
  Precision: 0.7003
  Recall:    0.6782
  F1 Score:  0.6726
```

```
 Confusion Matrix:
   [[988 105  11  24]
[295 411  57  43]
[ 64 137 275  84]
[  2   0   5  69]]

 ── Eval 14: Assessment period 202002 ──
 Training samples:   12,388
 Test samples:       1,588  (admin2 areas in 202002)
 Training Random Forest...

 Results  (n_test=1,588):
   Accuracy:  0.6738
   Precision: 0.7095
   Recall:    0.6738
   F1 Score:  0.6840

 Confusion Matrix:
   [[490 147  12  19]
[ 87 262  85  27]
[ 14  51 263  74]
[  0   1   1  55]]

 ── Eval 15: Assessment period 201910 ──
 Training samples:   12,533
 Test samples:       2,400  (admin2 areas in 201910)
 Training Random Forest...

 Results  (n_test=2,400):
   Accuracy:  0.6846
   Precision: 0.7107
   Recall:    0.6846
   F1 Score:  0.6814

 Confusion Matrix:
   [[1063   77   18   37]
[ 222  235   55   47]
[  93  113  296   84]
[   4    5    2   49]]

 ── Eval 16: Assessment period 201906 ──
 Training samples:   12,605
 Test samples:       2,400  (admin2 areas in 201906)
 Training Random Forest...

 Results  (n_test=2,400):
   Accuracy:  0.6854
   Precision: 0.7044
   Recall:    0.6854
   F1 Score:  0.6778

 Confusion Matrix:
   [[1046  109   12   12]
[ 272  225   29   24]
[  68  153  308   65]
[   3    6    2   66]]
```

```
 ── Eval 17: Assessment period 201902 ──
 Training samples:   12,785
 Test samples:       2,400  (admin2 areas in 201902)
 Training Random Forest...

 Results  (n_test=2,400):
   Accuracy:  0.7542
   Precision: 0.7803
   Recall:    0.7542
   F1 Score:  0.7611

 Confusion Matrix:
   [[1268  136    2   22]
[ 158  219   13   22]
[  26  114  295   58]
[   6    0   33   28]]

 ── Eval 18: Assessment period 201810 ──
 Training samples:   13,957
 Test samples:       1,081  (admin2 areas in 201810)
 Training Random Forest...

 Results  (n_test=1,081):
   Accuracy:  0.6651
   Precision: 0.7415
   Recall:    0.6651
   F1 Score:  0.6734

 Confusion Matrix:
   [[567  54   3  20]
[103 105   4  71]
[ 13  33  33  58]
[  3   0   0  14]]

 ── Eval 19: Assessment period 201806 ──
 Training samples:   14,374
 Test samples:       2,440  (admin2 areas in 201806)
 Training Random Forest...

 Results  (n_test=2,440):
   Accuracy:  0.6697
   Precision: 0.7140
   Recall:    0.6697
   F1 Score:  0.6796

 Confusion Matrix:
   [[1134  161   14   22]
[ 235  190   37   61]
[  62  115  292   90]
[   1    8    0   18]]

 ── Eval 20: Assessment period 201802 ──
 Training samples:   14,614
 Test samples:       2,441  (admin2 areas in 201802)
 Training Random Forest...
```

```
 Results  (n_test=2,441):
   Accuracy:  0.7464
   Precision: 0.8057
   Recall:    0.7464
   F1 Score:  0.7636

 Confusion Matrix:
   [[1313  158   12   28    0]
[ 141  180   14   22    0]
[  67   86  313   86    1]
[   1    3    0   16    0]
[   0    0    0    0    0]]

 ── Eval 21: Assessment period 201710 ──
 Training samples:   14,902
 Test samples:       2,577  (admin2 areas in 201710)
 Training Random Forest...

 Results  (n_test=2,577):
   Accuracy:  0.7652
   Precision: 0.8002
   Recall:    0.7652
   F1 Score:  0.7746

 Confusion Matrix:
   [[1475  123   12   48]
[ 162  161   22   23]
[  60   62  304   79]
[   2    6    6   32]]

 ── Eval 22: Assessment period 201706 ──
 Training samples:   11,953
 Test samples:       2,573  (admin2 areas in 201706)
 Training Random Forest...

 Results  (n_test=2,573):
   Accuracy:  0.6452
   Precision: 0.6376
   Recall:    0.6452
   F1 Score:  0.6259

 Confusion Matrix:
   [[1252  109   46   59]
[ 361   84   15   40]
[ 121   58  276   69]
[   7    9   19   48]]

 ── Eval 23: Assessment period 201702 ──
 Training samples:   9,303
 Test samples:       2,449  (admin2 areas in 201702)
 Training Random Forest...

 Results  (n_test=2,449):
   Accuracy:  0.7264
   Precision: 0.7179
```

```
    Recall:    0.7264
    F1 Score:  0.7065

  Confusion Matrix:
    [[1548  134   12   30    0]
 [ 237  154   15   18    0]
 [ 112   75   51   22    0]
 [   1   12    1   26    0]
 [   1    0    0    0    0]]

  ── Eval 24: Assessment period 201610 ──
  Training samples:   6,140
  Test samples:       2,748  (admin2 areas in 201610)
  Training Random Forest...

  Results  (n_test=2,748):
    Accuracy:  0.7828
    Precision: 0.7757
    Recall:    0.7828
    F1 Score:  0.7767

  Confusion Matrix:
    [[1645  137   26   19]
 [ 229  166   48   22]
 [  29   57  304   19]
 [   2    3    6   36]]

  ── Eval 25: Assessment period 201606 ──
  Training samples:   3,099
  Test samples:       2,635  (admin2 areas in 201606)
  Training Random Forest...

  Results  (n_test=2,635):
    Accuracy:  0.7860
    Precision: 0.7845
    Recall:    0.7860
    F1 Score:  0.7749

  Confusion Matrix:
    [[1586   35   41   34]
 [ 159  139   46   21]
 [ 145   46  303   26]
 [   7    2    2   43]]

======================================================================
MODEL: Random Forest  |  FEATURE SET: IPC only (no geo)
======================================================================
Using 1 features

  ── Eval 1: Assessment period 202406 ──
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Training Random Forest...

  Results  (n_test=1,919):
    Accuracy:  0.6514
```

```
  Precision: 0.6803
  Recall:    0.6514
  F1 Score:  0.6483

 Confusion Matrix:
   [[429  51   0   0]
[359 439  32   0]
[  0 157 292  19]
[  0   2  49  90]]

 ── Eval 2: Assessment period 202402 ──
 Training samples:   12,107
 Test samples:       2,376  (admin2 areas in 202402)
 Training Random Forest...

 Results  (n_test=2,376):
   Accuracy:  0.6940
   Precision: 0.7269
   Recall:    0.6940
   F1 Score:  0.6769

 Confusion Matrix:
   [[906  51   0   0]
[418 370  46   0]
[103  40 324   9]
[  0   1  59  49]]

 ── Eval 3: Assessment period 202310 ──
 Training samples:   12,645
 Test samples:       1,919  (admin2 areas in 202310)
 Training Random Forest...

 Results  (n_test=1,919):
   Accuracy:  0.6165
   Precision: 0.7578
   Recall:    0.6165
   F1 Score:  0.6348

 Confusion Matrix:
   [[493 471   6   0]
[  4 326 132   0]
[  0  40 323  66]
[  0   0  17  41]]

 ── Eval 4: Assessment period 202306 ──
 Training samples:   12,765
 Test samples:       2,613  (admin2 areas in 202306)
 Training Random Forest...

 Results  (n_test=2,613):
   Accuracy:  0.4646
   Precision: 0.6057
   Recall:    0.4646
   F1 Score:  0.4336

 Confusion Matrix:
```

```
   [[598  36   2   0]
[494 500  63   0]
[623  74 116   0]
[102   0   5   0]]

 ── Eval 5: Assessment period 202302 ──
 Training samples:   13,772
 Test samples:       1,376  (admin2 areas in 202302)
 Training Random Forest...

 Results  (n_test=1,376):
   Accuracy:  0.7369
   Precision: 0.7456
   Recall:    0.7369
   F1 Score:  0.7368

 Confusion Matrix:
   [[386 189   5]
[ 94 479  37]
[  0  37 149]]

 ── Eval 6: Assessment period 202210 ──
 Training samples:   13,866
 Test samples:       2,399  (admin2 areas in 202210)
 Training Random Forest...

 Results  (n_test=2,399):
   Accuracy:  0.7253
   Precision: 0.7332
   Recall:    0.7253
   F1 Score:  0.7284

 Confusion Matrix:
   [[636 205   1   0]
[166 570  78   6]
[  4  87 489  69]
[  0   0  43  45]]

 ── Eval 7: Assessment period 202206 ──
 Training samples:   13,795
 Test samples:       2,568  (admin2 areas in 202206)
 Training Random Forest...

 Results  (n_test=2,568):
   Accuracy:  0.7492
   Precision: 0.7603
   Recall:    0.7492
   F1 Score:  0.7431

 Confusion Matrix:
   [[687 119   0   0]
[197 656  65   0]
[ 48 126 549   1]
[ 31   2  55  32]]

 ── Eval 8: Assessment period 202202 ──
```

```
 Training samples:   13,862
 Test samples:       2,535  (admin2 areas in 202202)
 Training Random Forest...

 Results  (n_test=2,535):
   Accuracy:  0.7787
   Precision: 0.7887
   Recall:    0.7787
   F1 Score:  0.7760

 Confusion Matrix:
   [[846  71   0   0]
[268 618  22   0]
[  4 173 498   1]
[  0   5  17  12]]

 —— Eval 9: Assessment period 202110 ——
 Training samples:   12,950
 Test samples:       2,535  (admin2 areas in 202110)
 Training Random Forest...

 Results  (n_test=2,535):
   Accuracy:  0.7815
   Precision: 0.8050
   Recall:    0.7815
   F1 Score:  0.7850

 Confusion Matrix:
   [[809 307   2   0]
[ 70 703  94   0]
[  0  68 462   7]
[  0   0   6   7]]

 —— Eval 10: Assessment period 202106 ——
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Training Random Forest...

 Results  (n_test=2,536):
   Accuracy:  0.7015
   Precision: 0.7365
   Recall:    0.7015
   F1 Score:  0.6966

 Confusion Matrix:
   [[805  59  16   0]
[470 549  59   0]
[  4  96 418  46]
[  0   0   7   7]]

 —— Eval 11: Assessment period 202102 ——
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Training Random Forest...

 Results  (n_test=2,489):
```

```
  Accuracy:  0.7790
  Precision: 0.7901
  Recall:    0.7790
  F1 Score:  0.7763

 Confusion Matrix:
   [[1007  225    0    0]
[ 120  511   73    0]
[   7   75  417    1]
[   2    1   46    4]]

 ── Eval 12: Assessment period 202010 ──
 Training samples:   12,858
 Test samples:       2,491  (admin2 areas in 202010)
 Training Random Forest...

 Results  (n_test=2,491):
  Accuracy:  0.7354
  Precision: 0.7456
  Recall:    0.7354
  F1 Score:  0.7399

 Confusion Matrix:
   [[936 189  11   0]
[190 505 119   0]
[  2 106 387  41]
[  0   0   1   4]]

 ── Eval 13: Assessment period 202006 ──
 Training samples:   11,506
 Test samples:       2,570  (admin2 areas in 202006)
 Training Random Forest...

 Results  (n_test=2,570):
  Accuracy:  0.7054
  Precision: 0.7276
  Recall:    0.7054
  F1 Score:  0.6865

 Confusion Matrix:
   [[1077   48    3    0]
[ 482  314   10    0]
[  86   99  370    5]
[   5    0   19   52]]

 ── Eval 14: Assessment period 202002 ──
 Training samples:   12,388
 Test samples:       1,588  (admin2 areas in 202002)
 Training Random Forest...

 Results  (n_test=1,588):
  Accuracy:  0.7752
  Precision: 0.7906
  Recall:    0.7752
  F1 Score:  0.7737
```

```
 Confusion Matrix:
   [[569  78  21   0]
[ 29 257 167   8]
[ 13  24 363   2]
[  0   1  14  42]]

 ── Eval 15: Assessment period 201910 ──
 Training samples:   12,533
 Test samples:       2,400  (admin2 areas in 201910)
 Training Random Forest...

 Results  (n_test=2,400):
   Accuracy:  0.8017
   Precision: 0.8032
   Recall:    0.8017
   F1 Score:  0.8021

 Confusion Matrix:
   [[1086  108    1    0]
[  85  341  132    1]
[   8  101  449   28]
[   0    0   12   48]]

 ── Eval 16: Assessment period 201906 ──
 Training samples:   12,605
 Test samples:       2,400  (admin2 areas in 201906)
 Training Random Forest...

 Results  (n_test=2,400):
   Accuracy:  0.7892
   Precision: 0.7802
   Recall:    0.7892
   F1 Score:  0.7763

 Confusion Matrix:
   [[1158   19    2    0]
[ 256  252   42    0]
[  14  141  428   11]
[   0    0   21   56]]

 ── Eval 17: Assessment period 201902 ──
 Training samples:   12,785
 Test samples:       2,400  (admin2 areas in 201902)
 Training Random Forest...

 Results  (n_test=2,400):
   Accuracy:  0.6758
   Precision: 0.6861
   Recall:    0.6758
   F1 Score:  0.6291

 Confusion Matrix:
   [[1336   92    0    0]
[ 234  167   11    0]
[ 353   24  109    7]
[  40    0   17   10]]
```

```
 ── Eval 18: Assessment period 201810 ──
 Training samples:   13,957
 Test samples:       1,081  (admin2 areas in 201810)
 Training Random Forest...

 Results  (n_test=1,081):
   Accuracy:  0.8483
   Precision: 0.8560
   Recall:    0.8483
   F1 Score:  0.8513

 Confusion Matrix:
   [[580  64   0   0]
[ 25 220  38   0]
[  3  14 109  11]
[  2   0   7   8]]

 ── Eval 19: Assessment period 201806 ──
 Training samples:   14,374
 Test samples:       2,440  (admin2 areas in 201806)
 Training Random Forest...

 Results  (n_test=2,440):
   Accuracy:  0.8119
   Precision: 0.7984
   Recall:    0.8119
   F1 Score:  0.7987

 Confusion Matrix:
   [[1279   42   10    0    0]
[ 211  244   68    0    0]
[  22   71  458    0    8]
[   0    0   15    0   12]
[   0    0    0    0    0]]

 ── Eval 20: Assessment period 201802 ──
 Training samples:   14,614
 Test samples:       2,441  (admin2 areas in 201802)
 Training Random Forest...

 Results  (n_test=2,441):
   Accuracy:  0.8845
   Precision: 0.8940
   Recall:    0.8845
   F1 Score:  0.8888

 Confusion Matrix:
   [[1449   59    3    0    0]
[  64  255   38    0    0]
[  11   51  455    0   36]
[   0    3    9    0    8]
[   0    0    0    0    0]]

 ── Eval 21: Assessment period 201710 ──
 Training samples:   14,902
```

```
 Test samples:       2,577  (admin2 areas in 201710)
 Training Random Forest...

 Results  (n_test=2,577):
   Accuracy:  0.7575
   Precision: 0.7991
   Recall:    0.7575
   F1 Score:  0.7747

 Confusion Matrix:
   [[1352  291   15    0    0]
[  93  193   81    0    1]
[  25   15  407    0   58]
[   0    1   21    0   24]
[   0    0    0    0    0]]

 —— Eval 22: Assessment period 201706 ——
 Training samples:   11,953
 Test samples:       2,573  (admin2 areas in 201706)
 Training Random Forest...

 Results  (n_test=2,573):
   Accuracy:  0.6211
   Precision: 0.5811
   Recall:    0.6211
   F1 Score:  0.5864

 Confusion Matrix:
   [[1275  108   83    0    0]
[ 283  200   17    0    0]
[ 276  115  123    0   10]
[  14    1   37    0   31]
[   0    0    0    0    0]]

 —— Eval 23: Assessment period 201702 ——
 Training samples:   9,303
 Test samples:       2,449  (admin2 areas in 201702)
 Training Random Forest...

 Results  (n_test=2,449):
   Accuracy:  0.8383
   Precision: 0.8301
   Recall:    0.8383
   F1 Score:  0.8301

 Confusion Matrix:
   [[1664   60    0    0    0]
[ 156  249   17    0    2]
[   6  110  139    0    5]
[   0    9   12    0   19]
[   0    0    0    0    1]]

 —— Eval 24: Assessment period 201610 ——
 Training samples:   6,140
 Test samples:       2,748  (admin2 areas in 201610)
 Training Random Forest...
```

```
  Results  (n_test=2,748):
    Accuracy:  0.8173
    Precision: 0.8136
    Recall:    0.8173
    F1 Score:  0.8124

  Confusion Matrix:
    [[1653  139   35    0]
 [ 152  200  113    0]
 [   6   22  360   21]
 [   0    2   12   33]]

  —— Eval 25: Assessment period 201606 ——
  Training samples:   3,099
  Test samples:       2,635  (admin2 areas in 201606)
  Training Random Forest...

  Results  (n_test=2,635):
    Accuracy:  0.7347
    Precision: 0.6744
    Recall:    0.7347
    F1 Score:  0.6973

  Confusion Matrix:
    [[1659   37    0    0]
 [ 113  227    0   25]
 [  24  138    0  358]
 [   0    4    0   50]]

======================================================================
MODEL: Random Forest  |  FEATURE SET: IPC + GDELT base (no geo)
======================================================================
Using 120 features

  —— Eval 1: Assessment period 202406 ——
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Training Random Forest...

  Results  (n_test=1,919):
    Accuracy:  0.6451
    Precision: 0.6723
    Recall:    0.6451
    F1 Score:  0.6422

  Confusion Matrix:
    [[422  58   0   0]
 [357 440  33   0]
 [  0 156 291  21]
 [  0   2  54  85]]

  —— Eval 2: Assessment period 202402 ——
  Training samples:   12,107
  Test samples:       2,376  (admin2 areas in 202402)
  Training Random Forest...
```

```
 Results  (n_test=2,376):
   Accuracy:  0.6881
   Precision: 0.7124
   Recall:    0.6881
   F1 Score:  0.6725

 Confusion Matrix:
   [[888  68   0   1]
[407 379  48   0]
[100  43 320  13]
[  0   2  59  48]]

 ── Eval 3: Assessment period 202310 ──
 Training samples:   12,645
 Test samples:       1,919  (admin2 areas in 202310)
 Training Random Forest...

 Results  (n_test=1,919):
   Accuracy:  0.6045
   Precision: 0.7488
   Recall:    0.6045
   F1 Score:  0.6217

 Confusion Matrix:
   [[472 492   6   0]
[  9 320 131   2]
[  1  39 329  60]
[  0   0  19  39]]

 ── Eval 4: Assessment period 202306 ──
 Training samples:   12,765
 Test samples:       2,613  (admin2 areas in 202306)
 Training Random Forest...

 Results  (n_test=2,613):
   Accuracy:  0.4684
   Precision: 0.6073
   Recall:    0.4684
   F1 Score:  0.4408

 Confusion Matrix:
   [[581  52   2   1]
[474 514  69   0]
[595  89 125   4]
[ 94   2   7   4]]

 ── Eval 5: Assessment period 202302 ──
 Training samples:   13,772
 Test samples:       1,376  (admin2 areas in 202302)
 Training Random Forest...

 Results  (n_test=1,376):
   Accuracy:  0.7333
   Precision: 0.7442
   Recall:    0.7333
```

```
  F1 Score:  0.7337

 Confusion Matrix:
   [[383 192   5   0]
[ 90 481  39   0]
[  1  38 145   2]
[  0   0   0   0]]

 —— Eval 6: Assessment period 202210 ——
 Training samples:   13,866
 Test samples:       2,399  (admin2 areas in 202210)
 Training Random Forest...

 Results  (n_test=2,399):
   Accuracy:  0.7211
   Precision: 0.7305
   Recall:    0.7211
   F1 Score:  0.7247

 Confusion Matrix:
   [[628 213   1   0]
[165 570  78   7]
[  4  88 486  71]
[  0   0  42  46]]

 —— Eval 7: Assessment period 202206 ——
 Training samples:   13,795
 Test samples:       2,568  (admin2 areas in 202206)
 Training Random Forest...

 Results  (n_test=2,568):
   Accuracy:  0.7445
   Precision: 0.7501
   Recall:    0.7445
   F1 Score:  0.7385

 Confusion Matrix:
   [[678 126   0   2]
[194 657  67   0]
[ 48 125 546   5]
[ 31   2  56  31]]

 —— Eval 8: Assessment period 202202 ——
 Training samples:   13,862
 Test samples:       2,535  (admin2 areas in 202202)
 Training Random Forest...

 Results  (n_test=2,535):
   Accuracy:  0.7748
   Precision: 0.7833
   Recall:    0.7748
   F1 Score:  0.7725

 Confusion Matrix:
   [[838  78   1   0]
[266 618  23   1]
```

```
[  4 173 495   4]
[  0   5  16  13]]

 ── Eval 9: Assessment period 202110 ──
 Training samples:   12,950
 Test samples:       2,535  (admin2 areas in 202110)
 Training Random Forest...

 Results  (n_test=2,535):
   Accuracy:  0.7795
   Precision: 0.8034
   Recall:    0.7795
   F1 Score:  0.7832

 Confusion Matrix:
   [[806 310   2   0]
[ 72 700  94   1]
[  0  67 462   8]
[  0   0   5   8]]

 ── Eval 10: Assessment period 202106 ──
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Training Random Forest...

 Results  (n_test=2,536):
   Accuracy:  0.6976
   Precision: 0.7303
   Recall:    0.6976
   F1 Score:  0.6936

 Confusion Matrix:
   [[793  71  16   0]
[465 553  60   0]
[  4  96 417  47]
[  0   0   8   6]]

 ── Eval 11: Assessment period 202102 ──
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Training Random Forest...

 Results  (n_test=2,489):
   Accuracy:  0.7730
   Precision: 0.7844
   Recall:    0.7730
   F1 Score:  0.7707

 Confusion Matrix:
   [[1004  228    0    0]
[ 122  498   84    0]
[   8   74  417    1]
[   2    0   46    5]]

 ── Eval 12: Assessment period 202010 ──
 Training samples:   12,858
```

```
 Test samples:       2,491  (admin2 areas in 202010)
 Training Random Forest...

 Results  (n_test=2,491):
   Accuracy:  0.7302
   Precision: 0.7412
   Recall:    0.7302
   F1 Score:  0.7351

 Confusion Matrix:
   [[924 198  14   0]
[190 501 121   2]
[  2 104 390  40]
[  0   0   1   4]]

 —— Eval 13: Assessment period 202006 ——
 Training samples:   11,506
 Test samples:       2,570  (admin2 areas in 202006)
 Training Random Forest...

 Results  (n_test=2,570):
   Accuracy:  0.7019
   Precision: 0.7187
   Recall:    0.7019
   F1 Score:  0.6834

 Confusion Matrix:
   [[1069   54    5    0]
[ 472  319   15    0]
[  80  104  368    8]
[   5    0   23   48]]

 —— Eval 14: Assessment period 202002 ——
 Training samples:   12,388
 Test samples:       1,588  (admin2 areas in 202002)
 Training Random Forest...

 Results  (n_test=1,588):
   Accuracy:  0.7720
   Precision: 0.7875
   Recall:    0.7720
   F1 Score:  0.7709

 Confusion Matrix:
   [[566  78  22   2]
[ 28 258 163  12]
[ 13  24 360   5]
[  0   1  14  42]]

 —— Eval 15: Assessment period 201910 ——
 Training samples:   12,533
 Test samples:       2,400  (admin2 areas in 201910)
 Training Random Forest...

 Results  (n_test=2,400):
   Accuracy:  0.7971
```

```
  Precision: 0.7986
  Recall:    0.7971
  F1 Score:  0.7975

 Confusion Matrix:
   [[1081  113    1    0]
[  87  334  136    2]
[   9   97  451   29]
[   0    0   13   47]]

 ── Eval 16: Assessment period 201906 ──
 Training samples:   12,605
 Test samples:       2,400  (admin2 areas in 201906)
 Training Random Forest...

 Results  (n_test=2,400):
   Accuracy:  0.7821
   Precision: 0.7721
   Recall:    0.7821
   F1 Score:  0.7698

 Confusion Matrix:
   [[1146   29    4    0]
[ 252  252   46    0]
[  15  139  427   13]
[   0    0   25   52]]

 ── Eval 17: Assessment period 201902 ──
 Training samples:   12,785
 Test samples:       2,400  (admin2 areas in 201902)
 Training Random Forest...

 Results  (n_test=2,400):
   Accuracy:  0.6746
   Precision: 0.6807
   Recall:    0.6746
   F1 Score:  0.6280

 Confusion Matrix:
   [[1331   96    1    0]
[ 229  172   11    0]
[ 347   32  108    6]
[  40    0   19    8]]

 ── Eval 18: Assessment period 201810 ──
 Training samples:   13,957
 Test samples:       1,081  (admin2 areas in 201810)
 Training Random Forest...

 Results  (n_test=1,081):
   Accuracy:  0.8409
   Precision: 0.8486
   Recall:    0.8409
   F1 Score:  0.8439

 Confusion Matrix:
```

```
   [[578  66   0   0]
[ 28 217  38   0]
[  3  15 107  12]
[  2   0   8   7]]

 ── Eval 19: Assessment period 201806 ──
 Training samples:   14,374
 Test samples:       2,440  (admin2 areas in 201806)
 Training Random Forest...

 Results  (n_test=2,440):
   Accuracy:  0.8119
   Precision: 0.8000
   Recall:    0.8119
   F1 Score:  0.7997

 Confusion Matrix:
   [[1275   46   10    0]
[ 214  241   68    0]
[  22   71  453   13]
[   0    0   15   12]]

 ── Eval 20: Assessment period 201802 ──
 Training samples:   14,614
 Test samples:       2,441  (admin2 areas in 201802)
 Training Random Forest...

 Results  (n_test=2,441):
   Accuracy:  0.8878
   Precision: 0.8945
   Recall:    0.8878
   F1 Score:  0.8905

 Confusion Matrix:
   [[1443   64    4    0]
[  63  257   37    0]
[  11   53  459   30]
[   0    3    9    8]]

 ── Eval 21: Assessment period 201710 ──
 Training samples:   14,902
 Test samples:       2,577  (admin2 areas in 201710)
 Training Random Forest...

 Results  (n_test=2,577):
   Accuracy:  0.7598
   Precision: 0.7993
   Recall:    0.7598
   F1 Score:  0.7752

 Confusion Matrix:
   [[1341  300   16    1]
[  95  188   82    3]
[  25   21  408   51]
[   0    3   22   21]]
```

```
 ── Eval 22: Assessment period 201706 ──
 Training samples:   11,953
 Test samples:       2,573  (admin2 areas in 201706)
 Training Random Forest...

 Results  (n_test=2,573):
   Accuracy:  0.6296
   Precision: 0.6027
   Recall:    0.6296
   F1 Score:  0.6000

 Confusion Matrix:
   [[1268  116   81    1]
[ 285  194   21    0]
[ 274  112  128   10]
[  14    3   36   30]]

 ── Eval 23: Assessment period 201702 ──
 Training samples:   9,303
 Test samples:       2,449  (admin2 areas in 201702)
 Training Random Forest...

 Results  (n_test=2,449):
   Accuracy:  0.8391
   Precision: 0.8370
   Recall:    0.8391
   F1 Score:  0.8335

 Confusion Matrix:
   [[1657   65    1    1    0]
[ 153  245   14   12    0]
[   6  109  134   11    0]
[   0   10   11   19    0]
[   1    0    0    0    0]]

 ── Eval 24: Assessment period 201610 ──
 Training samples:   6,140
 Test samples:       2,748  (admin2 areas in 201610)
 Training Random Forest...

 Results  (n_test=2,748):
   Accuracy:  0.8133
   Precision: 0.8105
   Recall:    0.8133
   F1 Score:  0.8097

 Confusion Matrix:
   [[1653  140   34    0]
[ 152  205  106    2]
[   7   29  347   26]
[   0    3   14   30]]

 ── Eval 25: Assessment period 201606 ──
 Training samples:   3,099
 Test samples:       2,635  (admin2 areas in 201606)
 Training Random Forest...
```

```
  Results  (n_test=2,635):
    Accuracy:  0.8159
    Precision: 0.8307
    Recall:    0.8159
    F1 Score:  0.8156

  Confusion Matrix:
    [[1647   43    6    0]
 [ 113  217   31    4]
 [  25  134  265   96]
 [   0    3   30   21]]

======================================================================
MODEL: Random Forest  |  FEATURE SET: IPC + GDELT + temporal (no geo)
======================================================================
Using 155 features

  ── Eval 1: Assessment period 202406 ──
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Training Random Forest...

  Results  (n_test=1,919):
    Accuracy:  0.6472
    Precision: 0.6731
    Recall:    0.6472
    F1 Score:  0.6447

  Confusion Matrix:
    [[418  61   1   0]
 [349 448  33   0]
 [  1 154 289  24]
 [  0   2  52  87]]

  ── Eval 2: Assessment period 202402 ──
  Training samples:   12,107
  Test samples:       2,376  (admin2 areas in 202402)
  Training Random Forest...

  Results  (n_test=2,376):
    Accuracy:  0.6759
    Precision: 0.6918
    Recall:    0.6759
    F1 Score:  0.6616

  Confusion Matrix:
    [[868  87   1   1]
 [400 383  50   1]
 [ 96  50 306  24]
 [  0   0  60  49]]

  ── Eval 3: Assessment period 202310 ──
  Training samples:   12,645
  Test samples:       1,919  (admin2 areas in 202310)
  Training Random Forest...
```

```
 Results  (n_test=1,919):
   Accuracy:  0.6050
   Precision: 0.7487
   Recall:    0.6050
   F1 Score:  0.6233

 Confusion Matrix:
   [[479 483   8   0]
[  9 324 126   3]
[  1  39 320  69]
[  0   0  20  38]]

 ── Eval 4: Assessment period 202306 ──
 Training samples:   12,765
 Test samples:       2,613  (admin2 areas in 202306)
 Training Random Forest...

 Results  (n_test=2,613):
   Accuracy:  0.4784
   Precision: 0.6113
   Recall:    0.4784
   F1 Score:  0.4508

 Confusion Matrix:
   [[581  53   2   0]
[461 534  61   1]
[577  98 132   6]
[ 92   5   7   3]]

 ── Eval 5: Assessment period 202302 ──
 Training samples:   13,772
 Test samples:       1,376  (admin2 areas in 202302)
 Training Random Forest...

 Results  (n_test=1,376):
   Accuracy:  0.7289
   Precision: 0.7437
   Recall:    0.7289
   F1 Score:  0.7306

 Confusion Matrix:
   [[378 197   5   0]
[ 87 482  39   2]
[  1  37 143   5]
[  0   0   0   0]]

 ── Eval 6: Assessment period 202210 ──
 Training samples:   13,866
 Test samples:       2,399  (admin2 areas in 202210)
 Training Random Forest...

 Results  (n_test=2,399):
   Accuracy:  0.7178
   Precision: 0.7255
   Recall:    0.7178
```

```
  F1 Score:  0.7209

 Confusion Matrix:
   [[624 215   3   0]
[167 566  80   7]
[  4  88 490  67]
[  0   0  46  42]]

 —— Eval 7: Assessment period 202206 ——
 Training samples:   13,795
 Test samples:       2,568  (admin2 areas in 202206)
 Training Random Forest...

 Results  (n_test=2,568):
   Accuracy:  0.7348
   Precision: 0.7341
   Recall:    0.7348
   F1 Score:  0.7285

 Confusion Matrix:
   [[682 124   0   0]
[201 643  73   1]
[ 48 125 533  18]
[ 30   3  58  29]]

 —— Eval 8: Assessment period 202202 ——
 Training samples:   13,862
 Test samples:       2,535  (admin2 areas in 202202)
 Training Random Forest...

 Results  (n_test=2,535):
   Accuracy:  0.7665
   Precision: 0.7740
   Recall:    0.7665
   F1 Score:  0.7637

 Confusion Matrix:
   [[834  82   1   0]
[275 608  25   0]
[ 13 163 491   9]
[  0   5  19  10]]

 —— Eval 9: Assessment period 202110 ——
 Training samples:   12,950
 Test samples:       2,535  (admin2 areas in 202110)
 Training Random Forest...

 Results  (n_test=2,535):
   Accuracy:  0.7716
   Precision: 0.7936
   Recall:    0.7716
   F1 Score:  0.7754

 Confusion Matrix:
   [[803 311   4   0]
[ 86 687  90   4]
```

```
[  4  63 459  11]
[  0   0   6   7]]

 ── Eval 10: Assessment period 202106 ──
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Training Random Forest...

 Results  (n_test=2,536):
   Accuracy:  0.6987
   Precision: 0.7351
   Recall:    0.6987
   F1 Score:  0.6969

 Confusion Matrix:
   [[792  71  16   1]
[452 563  59   4]
[  4  95 408  57]
[  0   0   5   9]]

 ── Eval 11: Assessment period 202102 ──
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Training Random Forest...

 Results  (n_test=2,489):
   Accuracy:  0.7662
   Precision: 0.7737
   Recall:    0.7662
   F1 Score:  0.7669

 Confusion Matrix:
   [[988 241   2   1]
[119 498  84   3]
[  6  71 412  11]
[  2   0  42   9]]

 ── Eval 12: Assessment period 202010 ──
 Training samples:   12,858
 Test samples:       2,491  (admin2 areas in 202010)
 Training Random Forest...

 Results  (n_test=2,491):
   Accuracy:  0.7214
   Precision: 0.7375
   Recall:    0.7214
   F1 Score:  0.7286

 Confusion Matrix:
   [[923 202  11   0]
[194 493 116  11]
[  3 107 377  49]
[  0   0   1   4]]

 ── Eval 13: Assessment period 202006 ──
 Training samples:   11,506
```

```
 Test samples:       2,570  (admin2 areas in 202006)
 Training Random Forest...

 Results  (n_test=2,570):
   Accuracy:  0.6938
   Precision: 0.7049
   Recall:    0.6938
   F1 Score:  0.6770

 Confusion Matrix:
   [[1047   73    7    1]
[ 460  327   17    2]
[  74  108  362   16]
[   4    1   24   47]]

 —— Eval 14: Assessment period 202002 ——
 Training samples:   12,388
 Test samples:       1,588  (admin2 areas in 202002)
 Training Random Forest...

 Results  (n_test=1,588):
   Accuracy:  0.7538
   Precision: 0.7725
   Recall:    0.7538
   F1 Score:  0.7559

 Confusion Matrix:
   [[564  82  21   1]
[ 31 260 153  17]
[ 14  25 331  32]
[  0   1  14  42]]

 —— Eval 15: Assessment period 201910 ——
 Training samples:   12,533
 Test samples:       2,400  (admin2 areas in 201910)
 Training Random Forest...

 Results  (n_test=2,400):
   Accuracy:  0.7817
   Precision: 0.7908
   Recall:    0.7817
   F1 Score:  0.7845

 Confusion Matrix:
   [[1070  121    1    3]
[  88  335  128    8]
[  13   93  422   58]
[   0    0   11   49]]

 —— Eval 16: Assessment period 201906 ——
 Training samples:   12,605
 Test samples:       2,400  (admin2 areas in 201906)
 Training Random Forest...

 Results  (n_test=2,400):
   Accuracy:  0.7762
```

```
  Precision: 0.7664
  Recall:    0.7762
  F1 Score:  0.7641

 Confusion Matrix:
   [[1145   26    7    1]
[ 247  256   47    0]
[  15  140  411   28]
[   0    0   26   51]]

 ── Eval 17: Assessment period 201902 ──
 Training samples:   12,785
 Test samples:       2,400  (admin2 areas in 201902)
 Training Random Forest...

 Results  (n_test=2,400):
   Accuracy:  0.6758
   Precision: 0.6793
   Recall:    0.6758
   F1 Score:  0.6315

 Confusion Matrix:
   [[1327   98    2    1]
[ 226  174   12    0]
[ 336   37  112    8]
[  36    4   18    9]]

 ── Eval 18: Assessment period 201810 ──
 Training samples:   13,957
 Test samples:       1,081  (admin2 areas in 201810)
 Training Random Forest...

 Results  (n_test=1,081):
   Accuracy:  0.8363
   Precision: 0.8468
   Recall:    0.8363
   F1 Score:  0.8407

 Confusion Matrix:
   [[579  65   0   0]
[ 28 217  37   1]
[  3  15 101  18]
[  2   0   8   7]]

 ── Eval 19: Assessment period 201806 ──
 Training samples:   14,374
 Test samples:       2,440  (admin2 areas in 201806)
 Training Random Forest...

 Results  (n_test=2,440):
   Accuracy:  0.8025
   Precision: 0.7936
   Recall:    0.8025
   F1 Score:  0.7927

 Confusion Matrix:
```

```
   [[1262   58   11    0]
 [ 211  243   66    3]
 [  23   72  439   25]
 [   0    0   13   14]]

 ── Eval 20: Assessment period 201802 ──
 Training samples:   14,614
 Test samples:       2,441  (admin2 areas in 201802)
 Training Random Forest...

 Results  (n_test=2,441):
   Accuracy:  0.8755
   Precision: 0.8897
   Recall:    0.8755
   F1 Score:  0.8810

 Confusion Matrix:
   [[1430   77    3    1]
 [  67  253   34    3]
 [  11   56  444   42]
 [   0    3    7   10]]

 ── Eval 21: Assessment period 201710 ──
 Training samples:   14,902
 Test samples:       2,577  (admin2 areas in 201710)
 Training Random Forest...

 Results  (n_test=2,577):
   Accuracy:  0.7167
   Precision: 0.7438
   Recall:    0.7167
   F1 Score:  0.7154

 Confusion Matrix:
   [[1340   94  224    0]
 [  94   74  192    8]
 [  26   26  418   35]
 [   0    3   28   15]]

 ── Eval 22: Assessment period 201706 ──
 Training samples:   11,953
 Test samples:       2,573  (admin2 areas in 201706)
 Training Random Forest...

 Results  (n_test=2,573):
   Accuracy:  0.6113
   Precision: 0.5719
   Recall:    0.6113
   F1 Score:  0.5770

 Confusion Matrix:
   [[1264   77  117    8]
 [ 277   85  135    3]
 [ 263   50  198   13]
 [  13    4   40   26]]
```

```
  ── Eval 23: Assessment period 201702 ──
  Training samples:   9,303
  Test samples:       2,449  (admin2 areas in 201702)
  Training Random Forest...

  Results  (n_test=2,449):
    Accuracy:  0.8261
    Precision: 0.8269
    Recall:    0.8261
    F1 Score:  0.8206

  Confusion Matrix:
    [[1636   85    1    2    0]
 [ 152  257   10    5    0]
 [   6  125  118   11    0]
 [   0   11   18   11    0]
 [   0    0    0    0    1]]

  ── Eval 24: Assessment period 201610 ──
  Training samples:   6,140
  Test samples:       2,748  (admin2 areas in 201610)
  Training Random Forest...

  Results  (n_test=2,748):
    Accuracy:  0.7507
    Precision: 0.7580
    Recall:    0.7507
    F1 Score:  0.7381

  Confusion Matrix:
    [[1611   72  142    2]
 [ 149   94  215    7]
 [   7   42  346   14]
 [   0    1   34   12]]

  ── Eval 25: Assessment period 201606 ──
  Training samples:   3,099
  Test samples:       2,635  (admin2 areas in 201606)
  Training Random Forest...

  Results  (n_test=2,635):
    Accuracy:  0.8110
    Precision: 0.8322
    Recall:    0.8110
    F1 Score:  0.8110

  Confusion Matrix:
    [[1649   44    1    2]
 [ 117  217   27    4]
 [  27  132  248  113]
 [   0    3   28   23]]

======================================================================
MODEL: Random Forest  |  FEATURE SET: GDELT only (no geo)
======================================================================
Using 154 features
```

```
 ── Eval 1: Assessment period 202406 ──
 Training samples:   11,437
 Test samples:       1,919  (admin2 areas in 202406)
 Training Random Forest...

 Results  (n_test=1,919):
   Accuracy:  0.3007
   Precision: 0.4233
   Recall:    0.3007
   F1 Score:  0.2528

 Confusion Matrix:
   [[381  49  47   3]
[658  99  64   9]
[341  25  81  21]
[103   5  17  16]]

 ── Eval 2: Assessment period 202402 ──
 Training samples:   12,107
 Test samples:       2,376  (admin2 areas in 202402)
 Training Random Forest...

 Results  (n_test=2,376):
   Accuracy:  0.4146
   Precision: 0.4049
   Recall:    0.4146
   F1 Score:  0.3479

 Confusion Matrix:
   [[783  97  70   7]
[630 100  92  12]
[334  28  91  23]
[ 82   2  14  11]]

 ── Eval 3: Assessment period 202310 ──
 Training samples:   12,645
 Test samples:       1,919  (admin2 areas in 202310)
 Training Random Forest...

 Results  (n_test=1,919):
   Accuracy:  0.4971
   Precision: 0.4454
   Recall:    0.4971
   F1 Score:  0.4350

 Confusion Matrix:
   [[812  90  57  11]
[330  55  67  10]
[313  14  81  21]
[ 43   2   7   6]]

 ── Eval 4: Assessment period 202306 ──
 Training samples:   12,765
 Test samples:       2,613  (admin2 areas in 202306)
 Training Random Forest...
```

```
 Results  (n_test=2,613):
   Accuracy:  0.3039
   Precision: 0.4467
   Recall:    0.3039
   F1 Score:  0.2504

 Confusion Matrix:
   [[540  50  38   8]
[828 130  86  13]
[632  41 113  27]
[ 78   0  18  11]]

 ── Eval 5: Assessment period 202302 ──
 Training samples:   13,772
 Test samples:       1,376  (admin2 areas in 202302)
 Training Random Forest...

 Results  (n_test=1,376):
   Accuracy:  0.4390
   Precision: 0.4547
   Recall:    0.4390
   F1 Score:  0.3475

 Confusion Matrix:
   [[523  34  19   4]
[492  53  50  15]
[134  18  28   6]
[  0   0   0   0]]

 ── Eval 6: Assessment period 202210 ──
 Training samples:   13,866
 Test samples:       2,399  (admin2 areas in 202210)
 Training Random Forest...

 Results  (n_test=2,399):
   Accuracy:  0.3747
   Precision: 0.3973
   Recall:    0.3747
   F1 Score:  0.2789

 Confusion Matrix:
   [[759  35  37  11]
[686  57  68   9]
[524  32  73  20]
[ 66   3   9  10]]

 ── Eval 7: Assessment period 202206 ──
 Training samples:   13,795
 Test samples:       2,568  (admin2 areas in 202206)
 Training Random Forest...

 Results  (n_test=2,568):
   Accuracy:  0.3318
   Precision: 0.3814
   Recall:    0.3318
```

```
  F1 Score:  0.2373

 Confusion Matrix:
   [[714  28  48  16]
[785  54  67  12]
[603  22  77  22]
[ 86  10  17   7]]

 —— Eval 8: Assessment period 202202 ——
 Training samples:   13,862
 Test samples:       2,535  (admin2 areas in 202202)
 Training Random Forest...

 Results  (n_test=2,535):
   Accuracy:  0.3842
   Precision: 0.4104
   Recall:    0.3842
   F1 Score:  0.2914

 Confusion Matrix:
   [[818  40  46  13]
[767  62  73   6]
[537  35  92  12]
[ 23   3   6   2]]

 —— Eval 9: Assessment period 202110 ——
 Training samples:   12,950
 Test samples:       2,535  (admin2 areas in 202110)
 Training Random Forest...

 Results  (n_test=2,535):
   Accuracy:  0.4481
   Precision: 0.4585
   Recall:    0.4481
   F1 Score:  0.3549

 Confusion Matrix:
   [[997  45  57  19]
[713  69  65  20]
[444  16  67  10]
[  8   1   1   3]]

 —— Eval 10: Assessment period 202106 ——
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Training Random Forest...

 Results  (n_test=2,536):
   Accuracy:  0.3478
   Precision: 0.4015
   Recall:    0.3478
   F1 Score:  0.2527

 Confusion Matrix:
   [[759  41  49  31]
[914  63  65  36]
```

```
[457  25  55  27]
[  7   1   1   5]]

 ── Eval 11: Assessment period 202102 ──
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Training Random Forest...

 Results  (n_test=2,489):
   Accuracy:  0.4624
   Precision: 0.4120
   Recall:    0.4624
   F1 Score:  0.3832

 Confusion Matrix:
   [[1039   64   72   57]
[ 562   58   53   31]
[ 383   37   47   33]
[  43    0    3    7]]

 ── Eval 12: Assessment period 202010 ──
 Training samples:   12,858
 Test samples:       2,491  (admin2 areas in 202010)
 Training Random Forest...

 Results  (n_test=2,491):
   Accuracy:  0.4400
   Precision: 0.4367
   Recall:    0.4400
   F1 Score:  0.3417

 Confusion Matrix:
   [[1006   36   47   47]
[ 670   41   45   58]
[ 439   13   47   37]
[   2    1    0    2]]

 ── Eval 13: Assessment period 202006 ──
 Training samples:   11,506
 Test samples:       2,570  (admin2 areas in 202006)
 Training Random Forest...

 Results  (n_test=2,570):
   Accuracy:  0.4218
   Precision: 0.4006
   Recall:    0.4218
   F1 Score:  0.3255

 Confusion Matrix:
   [[983  44  42  59]
[676  47  30  53]
[429  33  44  54]
[ 61   0   5  10]]

 ── Eval 14: Assessment period 202002 ──
 Training samples:   12,388
```

```
 Test samples:       1,588  (admin2 areas in 202002)
 Training Random Forest...

 Results  (n_test=1,588):
   Accuracy:  0.4232
   Precision: 0.4275
   Recall:    0.4232
   F1 Score:  0.3339

 Confusion Matrix:
   [[582  26  32  28]
[378  36  23  24]
[311  17  42  32]
[ 41   2   2  12]]

 —— Eval 15: Assessment period 201910 ——
 Training samples:   12,533
 Test samples:       2,400  (admin2 areas in 201910)
 Training Random Forest...

 Results  (n_test=2,400):
   Accuracy:  0.4854
   Precision: 0.4525
   Recall:    0.4854
   F1 Score:  0.3904

 Confusion Matrix:
   [[1064   37   37   57]
[ 471   32   16   40]
[ 462   30   50   44]
[  36    2    3   19]]

 —— Eval 16: Assessment period 201906 ——
 Training samples:   12,605
 Test samples:       2,400  (admin2 areas in 201906)
 Training Random Forest...

 Results  (n_test=2,400):
   Accuracy:  0.4783
   Precision: 0.4145
   Recall:    0.4783
   F1 Score:  0.3833

 Confusion Matrix:
   [[1050   58   37   34]
[ 469   35   32   14]
[ 474   39   53   28]
[  54    3   10   10]]

 —— Eval 17: Assessment period 201902 ——
 Training samples:   12,785
 Test samples:       2,400  (admin2 areas in 201902)
 Training Random Forest...

 Results  (n_test=2,400):
   Accuracy:  0.5775
```

```
  Precision: 0.5059
  Recall:    0.5775
  F1 Score:  0.5053

 Confusion Matrix:
   [[1275   85   49   19]
[ 332   42   29    9]
[ 391   24   62   16]
[  46    2   12    7]]

 ── Eval 18: Assessment period 201810 ──
 Training samples:   13,957
 Test samples:       1,081  (admin2 areas in 201810)
 Training Random Forest...

 Results  (n_test=1,081):
  Accuracy:  0.5763
  Precision: 0.5020
  Recall:    0.5763
  F1 Score:  0.4975

 Confusion Matrix:
   [[584  30  22   8]
[217  27  30   9]
[112   7   7  11]
[  7   2   3   5]]

 ── Eval 19: Assessment period 201806 ──
 Training samples:   14,374
 Test samples:       2,440  (admin2 areas in 201806)
 Training Random Forest...

 Results  (n_test=2,440):
  Accuracy:  0.5225
  Precision: 0.4641
  Recall:    0.5225
  F1 Score:  0.4487

 Confusion Matrix:
   [[1153  113   50   15]
[ 420   53   36   14]
[ 433   41   65   20]
[  21    2    0    4]]

 ── Eval 20: Assessment period 201802 ──
 Training samples:   14,614
 Test samples:       2,441  (admin2 areas in 201802)
 Training Random Forest...

 Results  (n_test=2,441):
  Accuracy:  0.5760
  Precision: 0.5353
  Recall:    0.5760
  F1 Score:  0.5219

 Confusion Matrix:
```

```
   [[1290  134   59   28]
[ 289   40   17   11]
[ 409   51   72   21]
[  15    1    0    4]]

 ── Eval 21: Assessment period 201710 ──
 Training samples:   14,902
 Test samples:       2,577  (admin2 areas in 201710)
 Training Random Forest...

 Results  (n_test=2,577):
   Accuracy:  0.5906
   Precision: 0.5406
   Recall:    0.5906
   F1 Score:  0.5386

 Confusion Matrix:
   [[1407  172   44   35]
[ 274   69   12   13]
[ 384   54   36   31]
[  25    6    5   10]]

 ── Eval 22: Assessment period 201706 ──
 Training samples:   11,953
 Test samples:       2,573  (admin2 areas in 201706)
 Training Random Forest...

 Results  (n_test=2,573):
   Accuracy:  0.5309
   Precision: 0.5094
   Recall:    0.5309
   F1 Score:  0.4661

 Confusion Matrix:
   [[1226  191   11   38]
[ 386   90    9   15]
[ 400   65   33   26]
[  49   12    5   17]]

 ── Eval 23: Assessment period 201702 ──
 Training samples:   9,303
 Test samples:       2,449  (admin2 areas in 201702)
 Training Random Forest...

 Results  (n_test=2,449):
   Accuracy:  0.6529
   Precision: 0.6092
   Recall:    0.6529
   F1 Score:  0.6202

 Confusion Matrix:
   [[1461  221   27   15    0]
[ 291  115    8   10    0]
[ 189   48   15    8    0]
[  19   12    1    8    0]
[   1    0    0    0    0]]
```

```
  ── Eval 24: Assessment period 201610 ──
  Training samples:   6,140
  Test samples:       2,748  (admin2 areas in 201610)
  Training Random Forest...

  Results  (n_test=2,748):
    Accuracy:  0.6325
    Precision: 0.5648
    Recall:    0.6325
    F1 Score:  0.5749

  Confusion Matrix:
    [[1616  150   28   33]
 [ 352   90   15    8]
 [ 324   50   22   13]
 [  29    4    4   10]]

  ── Eval 25: Assessment period 201606 ──
  Training samples:   3,099
  Test samples:       2,635  (admin2 areas in 201606)
  Training Random Forest...

  Results  (n_test=2,635):
    Accuracy:  0.6209
    Precision: 0.5485
    Recall:    0.6209
    F1 Score:  0.5413

  Confusion Matrix:
    [[1563   35   53   45]
 [ 303   29   13   20]
 [ 439   17   39   25]
 [  43    1    5    5]]

======================================================================
MODEL: CatBoost  |  FEATURE SET: IPC only
======================================================================
Using 3332 features

  ── Eval 1: Assessment period 202406 ──
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Training CatBoost...

  Results  (n_test=1,919):
    Accuracy:  0.6368
    Precision: 0.6795
    Recall:    0.6368
    F1 Score:  0.6283

  Confusion Matrix:
    [[429  51   0   0]
 [359 439  32   0]
 [  0 157 223  88]
 [  0   2   8 131]]
```

```
 ── Eval 2: Assessment period 202402 ──
 Training samples:   12,107
 Test samples:       2,376  (admin2 areas in 202402)
 Training CatBoost...

 Results  (n_test=2,376):
   Accuracy:  0.6860
   Precision: 0.7242
   Recall:    0.6860
   F1 Score:  0.6691

 Confusion Matrix:
   [[906  51   0   0]
[386 402  46   0]
[ 62  68 216 130]
[  0   0   3 106]]

 ── Eval 3: Assessment period 202310 ──
 Training samples:   12,645
 Test samples:       1,919  (admin2 areas in 202310)
 Training CatBoost...

 Results  (n_test=1,919):
   Accuracy:  0.5878
   Precision: 0.7533
   Recall:    0.5878
   F1 Score:  0.6197

 Confusion Matrix:
   [[522 442   5   1]
[  4 318 112  28]
[  0  22 234 173]
[  0   0   4  54]]

 ── Eval 4: Assessment period 202306 ──
 Training samples:   12,765
 Test samples:       2,613  (admin2 areas in 202306)
 Training CatBoost...

 Results  (n_test=2,613):
   Accuracy:  0.4646
   Precision: 0.6057
   Recall:    0.4646
   F1 Score:  0.4336

 Confusion Matrix:
   [[598  36   2   0]
[494 500  63   0]
[623  74 116   0]
[102   0   5   0]]

 ── Eval 5: Assessment period 202302 ──
 Training samples:   13,772
 Test samples:       1,376  (admin2 areas in 202302)
 Training CatBoost...
```

```
 Results  (n_test=1,376):
   Accuracy:  0.7369
   Precision: 0.7456
   Recall:    0.7369
   F1 Score:  0.7368

 Confusion Matrix:
   [[386 189   5]
[ 94 479  37]
[  0  37 149]]

 ── Eval 6: Assessment period 202210 ──
 Training samples:   13,866
 Test samples:       2,399  (admin2 areas in 202210)
 Training CatBoost...

 Results  (n_test=2,399):
   Accuracy:  0.7165
   Precision: 0.7465
   Recall:    0.7165
   F1 Score:  0.7232

 Confusion Matrix:
   [[636 205   0   1]
[166 564  74  16]
[  4  87 432 126]
[  0   0   1  87]]

 ── Eval 7: Assessment period 202206 ──
 Training samples:   13,795
 Test samples:       2,568  (admin2 areas in 202206)
 Training CatBoost...

 Results  (n_test=2,568):
   Accuracy:  0.7422
   Precision: 0.7591
   Recall:    0.7422
   F1 Score:  0.7430

 Confusion Matrix:
   [[687 119   0   0]
[197 655  66   0]
[ 48 112 475  89]
[ 31   0   0  89]]

 ── Eval 8: Assessment period 202202 ──
 Training samples:   13,862
 Test samples:       2,535  (admin2 areas in 202202)
 Training CatBoost...

 Results  (n_test=2,535):
   Accuracy:  0.7590
   Precision: 0.7884
   Recall:    0.7590
   F1 Score:  0.7595
```

```
 Confusion Matrix:
   [[846  71   0   0]
[268 618  22   0]
[  4 173 426  73]
[  0   0   0  34]]

 ── Eval 9: Assessment period 202110 ──
 Training samples:   12,950
 Test samples:       2,535  (admin2 areas in 202110)
 Training CatBoost...

 Results  (n_test=2,535):
   Accuracy:  0.7661
   Precision: 0.8036
   Recall:    0.7661
   F1 Score:  0.7749

 Confusion Matrix:
   [[809 307   2   0]
[ 67 706  94   0]
[  0  68 414  55]
[  0   0   0  13]]

 ── Eval 10: Assessment period 202106 ──
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Training CatBoost...

 Results  (n_test=2,536):
   Accuracy:  0.7374
   Precision: 0.7614
   Recall:    0.7374
   F1 Score:  0.7328

 Confusion Matrix:
   [[805  59  16   0]
[406 613  59   0]
[  4  95 438  27]
[  0   0   0  14]]

 ── Eval 11: Assessment period 202102 ──
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Training CatBoost...

 Results  (n_test=2,489):
   Accuracy:  0.7987
   Precision: 0.8099
   Recall:    0.7987
   F1 Score:  0.8024

 Confusion Matrix:
   [[1007  225    0    0]
[ 119  511   66    8]
[   6   75  417    2]
```

```
[   0    0    0   53]]

 ── Eval 12: Assessment period 202010 ──
 Training samples:   12,858
 Test samples:       2,491  (admin2 areas in 202010)
 Training CatBoost...

 Results  (n_test=2,491):
   Accuracy:  0.7371
   Precision: 0.7471
   Recall:    0.7371
   F1 Score:  0.7414

 Confusion Matrix:
   [[936 189  11   0]
[190 505 119   0]
[  2 102 391  41]
[  0   0   1   4]]

 ── Eval 13: Assessment period 202006 ──
 Training samples:   11,506
 Test samples:       2,570  (admin2 areas in 202006)
 Training CatBoost...

 Results  (n_test=2,570):
   Accuracy:  0.7019
   Precision: 0.7280
   Recall:    0.7019
   F1 Score:  0.6815

 Confusion Matrix:
   [[1077   48    3    0]
[ 482  310   14    0]
[  86   97  347   30]
[   5    0    1   70]]

 ── Eval 14: Assessment period 202002 ──
 Training samples:   12,388
 Test samples:       1,588  (admin2 areas in 202002)
 Training CatBoost...

 Results  (n_test=1,588):
   Accuracy:  0.7550
   Precision: 0.7780
   Recall:    0.7550
   F1 Score:  0.7537

 Confusion Matrix:
   [[569  78  21   0]
[ 29 235 183  14]
[ 13  12 339  38]
[  0   0   1  56]]

 ── Eval 15: Assessment period 201910 ──
 Training samples:   12,533
 Test samples:       2,400  (admin2 areas in 201910)
```

```
 Training CatBoost...

 Results  (n_test=2,400):
   Accuracy:  0.7875
   Precision: 0.8002
   Recall:    0.7875
   F1 Score:  0.7904

 Confusion Matrix:
   [[1086  108    1    0]
[  85  341  126    7]
[   3  105  403   75]
[   0    0    0   60]]

 ── Eval 16: Assessment period 201906 ──
 Training samples:   12,605
 Test samples:       2,400  (admin2 areas in 201906)
 Training CatBoost...

 Results  (n_test=2,400):
   Accuracy:  0.7771
   Precision: 0.7784
   Recall:    0.7771
   F1 Score:  0.7639

 Confusion Matrix:
   [[1158   19    2    0]
[ 254  252   41    3]
[  14  139  380   61]
[   0    0    2   75]]

 ── Eval 17: Assessment period 201902 ──
 Training samples:   12,785
 Test samples:       2,400  (admin2 areas in 201902)
 Training CatBoost...

 Results  (n_test=2,400):
   Accuracy:  0.6704
   Precision: 0.7167
   Recall:    0.6704
   F1 Score:  0.6317

 Confusion Matrix:
   [[1336   92    0    0]
[ 224  177    8    3]
[ 102  273   69   49]
[  40    0    0   27]]

 ── Eval 18: Assessment period 201810 ──
 Training samples:   13,957
 Test samples:       1,081  (admin2 areas in 201810)
 Training CatBoost...

 Results  (n_test=1,081):
   Accuracy:  0.8039
   Precision: 0.8860
```

```
  Recall:    0.8039
  F1 Score:  0.8244

 Confusion Matrix:
   [[578  66   0   0]
[ 24 221   1  37]
[  3  14  55  65]
[  2   0   0  15]]

 ── Eval 19: Assessment period 201806 ──
 Training samples:   14,374
 Test samples:       2,440  (admin2 areas in 201806)
 Training CatBoost...

 Results  (n_test=2,440):
   Accuracy:  0.8004
   Precision: 0.8339
   Recall:    0.8004
   F1 Score:  0.8055

 Confusion Matrix:
   [[1279   42   10    0]
[ 177  278   27   41]
[  21   72  370   96]
[   0    0    1   26]]

 ── Eval 20: Assessment period 201802 ──
 Training samples:   14,614
 Test samples:       2,441  (admin2 areas in 201802)
 Training CatBoost...

 Results  (n_test=2,441):
   Accuracy:  0.8525
   Precision: 0.9059
   Recall:    0.8525
   F1 Score:  0.8704

 Confusion Matrix:
   [[1449   59    1    2    0]
[  64  255   18   20    0]
[  11   51  360  129    2]
[   0    3    0   17    0]
[   0    0    0    0    0]]

 ── Eval 21: Assessment period 201710 ──
 Training samples:   14,902
 Test samples:       2,577  (admin2 areas in 201710)
 Training CatBoost...

 Results  (n_test=2,577):
   Accuracy:  0.7602
   Precision: 0.8143
   Recall:    0.7602
   F1 Score:  0.7798

 Confusion Matrix:
```

```
   [[1352  291   15    0]
[  93  193   60   22]
[  25   15  370   95]
[   0    1    1   44]]

 ── Eval 22: Assessment period 201706 ──
 Training samples:   11,953
 Test samples:       2,573  (admin2 areas in 201706)
 Training CatBoost...

 Results  (n_test=2,573):
   Accuracy:  0.6506
   Precision: 0.6572
   Recall:    0.6506
   F1 Score:  0.6373

 Confusion Matrix:
   [[1275  108   83    0    0]
[ 248  235   14    3    0]
[   1  390  102   29    2]
[  14    1    6   62    0]
[   0    0    0    0    0]]

 ── Eval 23: Assessment period 201702 ──
 Training samples:   9,303
 Test samples:       2,449  (admin2 areas in 201702)
 Training CatBoost...

 Results  (n_test=2,449):
   Accuracy:  0.8469
   Precision: 0.8518
   Recall:    0.8469
   F1 Score:  0.8402

 Confusion Matrix:
   [[1664   60    0    0    0]
[ 156  258    6    4    0]
[   6  109  120   25    0]
[   0    4    4   31    1]
[   0    0    0    0    1]]

 ── Eval 24: Assessment period 201610 ──
 Training samples:   6,140
 Test samples:       2,748  (admin2 areas in 201610)
 Training CatBoost...

 Results  (n_test=2,748):
   Accuracy:  0.8184
   Precision: 0.8184
   Recall:    0.8184
   F1 Score:  0.8150

 Confusion Matrix:
   [[1652  140   34    1]
[ 149  202   92   22]
[   5   23  351   30]
```

```
 [   0    1    2   44]]

  ── Eval 25: Assessment period 201606 ──
  Training samples:   3,099
  Test samples:       2,635  (admin2 areas in 201606)
  Training CatBoost...

  Results  (n_test=2,635):
    Accuracy:  0.9021
    Precision: 0.9118
    Recall:    0.9021
    F1 Score:  0.9030

  Confusion Matrix:
    [[1652   43    1    0]
 [  53  288   22    2]
 [  11  101  383   25]
 [   0    0    0   54]]

======================================================================
MODEL: CatBoost  |  FEATURE SET: IPC + GDELT base
======================================================================
Using 3451 features

  ── Eval 1: Assessment period 202406 ──
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Training CatBoost...

  Results  (n_test=1,919):
    Accuracy:  0.6378
    Precision: 0.6798
    Recall:    0.6378
    F1 Score:  0.6294

  Confusion Matrix:
    [[429  51   0   0]
 [355 443  32   0]
 [  0 157 222  89]
 [  0   2   9 130]]

  ── Eval 2: Assessment period 202402 ──
  Training samples:   12,107
  Test samples:       2,376  (admin2 areas in 202402)
  Training CatBoost...

  Results  (n_test=2,376):
    Accuracy:  0.6721
    Precision: 0.7283
    Recall:    0.6721
    F1 Score:  0.6535

  Confusion Matrix:
    [[904  53   0   0]
 [417 371  46   0]
 [103  27 216 130]
```

```
[  0   0   3 106]]

 ── Eval 3: Assessment period 202310 ──
 Training samples:   12,645
 Test samples:       1,919  (admin2 areas in 202310)
 Training CatBoost...

 Results  (n_test=1,919):
   Accuracy:  0.5758
   Precision: 0.7441
   Recall:    0.5758
   F1 Score:  0.6042

 Confusion Matrix:
   [[493 471   6   0]
[  4 319 128  11]
[  0  24 241 164]
[  0   0   6  52]]

 ── Eval 4: Assessment period 202306 ──
 Training samples:   12,765
 Test samples:       2,613  (admin2 areas in 202306)
 Training CatBoost...

 Results  (n_test=2,613):
   Accuracy:  0.4703
   Precision: 0.6072
   Recall:    0.4703
   F1 Score:  0.4392

 Confusion Matrix:
   [[598  36   2   0]
[479 515  63   0]
[621  76 116   0]
[102   0   5   0]]

 ── Eval 5: Assessment period 202302 ──
 Training samples:   13,772
 Test samples:       1,376  (admin2 areas in 202302)
 Training CatBoost...

 Results  (n_test=1,376):
   Accuracy:  0.7384
   Precision: 0.7474
   Recall:    0.7384
   F1 Score:  0.7382

 Confusion Matrix:
   [[386 189   5]
[ 92 481  37]
[  0  37 149]]

 ── Eval 6: Assessment period 202210 ──
 Training samples:   13,866
 Test samples:       2,399  (admin2 areas in 202210)
 Training CatBoost...
```

```
 Results  (n_test=2,399):
   Accuracy:  0.7186
   Precision: 0.7495
   Recall:    0.7186
   F1 Score:  0.7254

 Confusion Matrix:
   [[636 205   0   1]
[166 569  69  16]
[  4  87 432 126]
[  0   0   1  87]]

 ── Eval 7: Assessment period 202206 ──
 Training samples:   13,795
 Test samples:       2,568  (admin2 areas in 202206)
 Training CatBoost...

 Results  (n_test=2,568):
   Accuracy:  0.7418
   Precision: 0.7585
   Recall:    0.7418
   F1 Score:  0.7426

 Confusion Matrix:
   [[687 119   0   0]
[197 655  66   0]
[ 48 112 475  89]
[ 31   0   1  88]]

 ── Eval 8: Assessment period 202202 ──
 Training samples:   13,862
 Test samples:       2,535  (admin2 areas in 202202)
 Training CatBoost...

 Results  (n_test=2,535):
   Accuracy:  0.7602
   Precision: 0.7882
   Recall:    0.7602
   F1 Score:  0.7607

 Confusion Matrix:
   [[846  71   0   0]
[268 618  22   0]
[  4 172 431  69]
[  0   2   0  32]]

 ── Eval 9: Assessment period 202110 ──
 Training samples:   12,950
 Test samples:       2,535  (admin2 areas in 202110)
 Training CatBoost...

 Results  (n_test=2,535):
   Accuracy:  0.7649
   Precision: 0.8021
   Recall:    0.7649
```

```
  F1 Score:  0.7739

 Confusion Matrix:
   [[809 307   2   0]
[ 69 704  94   0]
[  0  68 414  55]
[  0   0   1  12]]

 ── Eval 10: Assessment period 202106 ──
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Training CatBoost...

 Results  (n_test=2,536):
   Accuracy:  0.7370
   Precision: 0.7609
   Recall:    0.7370
   F1 Score:  0.7325

 Confusion Matrix:
   [[804  60  16   0]
[406 613  59   0]
[  4  95 438  27]
[  0   0   0  14]]

 ── Eval 11: Assessment period 202102 ──
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Training CatBoost...

 Results  (n_test=2,489):
   Accuracy:  0.7987
   Precision: 0.8097
   Recall:    0.7987
   F1 Score:  0.8024

 Confusion Matrix:
   [[1007  225    0    0]
[ 120  511   66    7]
[   7   75  417    1]
[   0    0    0   53]]

 ── Eval 12: Assessment period 202010 ──
 Training samples:   12,858
 Test samples:       2,491  (admin2 areas in 202010)
 Training CatBoost...

 Results  (n_test=2,491):
   Accuracy:  0.7363
   Precision: 0.7464
   Recall:    0.7363
   F1 Score:  0.7407

 Confusion Matrix:
   [[936 189  11   0]
[190 505 119   0]
```

```
[  2 104 389  41]
[  0   0   1   4]]

 ── Eval 13: Assessment period 202006 ──
 Training samples:   11,506
 Test samples:       2,570  (admin2 areas in 202006)
 Training CatBoost...

 Results  (n_test=2,570):
   Accuracy:  0.7008
   Precision: 0.7272
   Recall:    0.7008
   F1 Score:  0.6803

 Confusion Matrix:
   [[1077   48    3    0]
[ 481  311   14    0]
[  85   98  343   34]
[   5    0    1   70]]

 ── Eval 14: Assessment period 202002 ──
 Training samples:   12,388
 Test samples:       1,588  (admin2 areas in 202002)
 Training CatBoost...

 Results  (n_test=1,588):
   Accuracy:  0.7550
   Precision: 0.7762
   Recall:    0.7550
   F1 Score:  0.7541

 Confusion Matrix:
   [[569  78  21   0]
[ 29 239 180  13]
[ 13  17 337  35]
[  0   0   3  54]]

 ── Eval 15: Assessment period 201910 ──
 Training samples:   12,533
 Test samples:       2,400  (admin2 areas in 201910)
 Training CatBoost...

 Results  (n_test=2,400):
   Accuracy:  0.7875
   Precision: 0.7990
   Recall:    0.7875
   F1 Score:  0.7901

 Confusion Matrix:
   [[1084  110    1    0]
[  85  341  126    7]
[   8  100  406   72]
[   0    0    1   59]]

 ── Eval 16: Assessment period 201906 ──
 Training samples:   12,605
```

```
 Test samples:       2,400  (admin2 areas in 201906)
 Training CatBoost...

 Results  (n_test=2,400):
   Accuracy:  0.7783
   Precision: 0.7795
   Recall:    0.7783
   F1 Score:  0.7654

 Confusion Matrix:
   [[1158   19    2    0]
[ 251  255   42    2]
[  14  139  380   61]
[   0    0    2   75]]

 —— Eval 17: Assessment period 201902 ——
 Training samples:   12,785
 Test samples:       2,400  (admin2 areas in 201902)
 Training CatBoost...

 Results  (n_test=2,400):
   Accuracy:  0.6742
   Precision: 0.7068
   Recall:    0.6742
   F1 Score:  0.6238

 Confusion Matrix:
   [[1336   92    0    0]
[ 225  176    9    2]
[ 342   24   79   48]
[  40    0    0   27]]

 —— Eval 18: Assessment period 201810 ——
 Training samples:   13,957
 Test samples:       1,081  (admin2 areas in 201810)
 Training CatBoost...

 Results  (n_test=1,081):
   Accuracy:  0.8048
   Precision: 0.8809
   Recall:    0.8048
   F1 Score:  0.8241

 Confusion Matrix:
   [[580  64   0   0]
[ 24 221   4  34]
[  3  14  54  66]
[  2   0   0  15]]

 —— Eval 19: Assessment period 201806 ——
 Training samples:   14,374
 Test samples:       2,440  (admin2 areas in 201806)
 Training CatBoost...

 Results  (n_test=2,440):
   Accuracy:  0.7980
```

```
  Precision: 0.8332
  Recall:    0.7980
  F1 Score:  0.8034

 Confusion Matrix:
   [[1279   42   10    0]
[ 178  277   27   41]
[  22   71  365  101]
[   0    0    1   26]]

 ── Eval 20: Assessment period 201802 ──
 Training samples:   14,614
 Test samples:       2,441  (admin2 areas in 201802)
 Training CatBoost...

 Results  (n_test=2,441):
   Accuracy:  0.8611
   Precision: 0.9055
   Recall:    0.8611
   F1 Score:  0.8766

 Confusion Matrix:
   [[1449   59    1    2    0]
[  64  255   19   19    0]
[  11   51  382  108    1]
[   0    3    1   16    0]
[   0    0    0    0    0]]

 ── Eval 21: Assessment period 201710 ──
 Training samples:   14,902
 Test samples:       2,577  (admin2 areas in 201710)
 Training CatBoost...

 Results  (n_test=2,577):
   Accuracy:  0.7656
   Precision: 0.8161
   Recall:    0.7656
   F1 Score:  0.7840

 Confusion Matrix:
   [[1352  291   15    0]
[  93  193   58   24]
[  25   15  385   80]
[   0    1    2   43]]

 ── Eval 22: Assessment period 201706 ──
 Training samples:   11,953
 Test samples:       2,573  (admin2 areas in 201706)
 Training CatBoost...

 Results  (n_test=2,573):
   Accuracy:  0.6483
   Precision: 0.6487
   Recall:    0.6483
   F1 Score:  0.6328
```

```
 Confusion Matrix:
   [[1275  108   83    0    0]
[ 248  235   15    2    0]
[  20  371  100   32    1]
[  14    1   10   58    0]
[   0    0    0    0    0]]

 —— Eval 23: Assessment period 201702 ——
 Training samples:   9,303
 Test samples:       2,449  (admin2 areas in 201702)
 Training CatBoost...

 Results  (n_test=2,449):
   Accuracy:  0.8448
   Precision: 0.8455
   Recall:    0.8448
   F1 Score:  0.8384

 Confusion Matrix:
   [[1663   61    0    0    0]
[ 155  252   13    4    0]
[   6  109  125   20    0]
[   0    7    5   28    0]
[   0    0    0    0    1]]

 —— Eval 24: Assessment period 201610 ——
 Training samples:   6,140
 Test samples:       2,748  (admin2 areas in 201610)
 Training CatBoost...

 Results  (n_test=2,748):
   Accuracy:  0.8191
   Precision: 0.8180
   Recall:    0.8191
   F1 Score:  0.8152

 Confusion Matrix:
   [[1653  140   34    0]
[ 152  201   91   21]
[   6   22  352   29]
[   0    0    2   45]]

 —— Eval 25: Assessment period 201606 ——
 Training samples:   3,099
 Test samples:       2,635  (admin2 areas in 201606)
 Training CatBoost...

 Results  (n_test=2,635):
   Accuracy:  0.9051
   Precision: 0.9112
   Recall:    0.9051
   F1 Score:  0.9057

 Confusion Matrix:
   [[1652   43    1    0]
[  61  279   24    1]
```

```
 [  11   96  400   13]
 [   0    0    0   54]]

======================================================================
MODEL: CatBoost  |  FEATURE SET: IPC + GDELT + temporal
======================================================================
Using 3486 features

  ── Eval 1: Assessment period 202406 ──
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Training CatBoost...

  Results  (n_test=1,919):
    Accuracy:  0.6415
    Precision: 0.6832
    Recall:    0.6415
    F1 Score:  0.6333

  Confusion Matrix:
    [[429  51   0   0]
 [351 447  32   0]
 [  0 156 224  88]
 [  0   2   8 131]]

  ── Eval 2: Assessment period 202402 ──
  Training samples:   12,107
  Test samples:       2,376  (admin2 areas in 202402)
  Training CatBoost...

  Results  (n_test=2,376):
    Accuracy:  0.6734
    Precision: 0.7294
    Recall:    0.6734
    F1 Score:  0.6555

  Confusion Matrix:
    [[904  53   0   0]
 [407 381  46   0]
 [103  27 210 136]
 [  0   0   4 105]]

  ── Eval 3: Assessment period 202310 ──
  Training samples:   12,645
  Test samples:       1,919  (admin2 areas in 202310)
  Training CatBoost...

  Results  (n_test=1,919):
    Accuracy:  0.5690
    Precision: 0.7424
    Recall:    0.5690
    F1 Score:  0.5985

  Confusion Matrix:
    [[494 470   6   0]
 [  4 324 120  14]
```

```
[  0  34 222 173]
[  0   0   6  52]]

 ── Eval 4: Assessment period 202306 ──
 Training samples:   12,765
 Test samples:       2,613  (admin2 areas in 202306)
 Training CatBoost...

 Results  (n_test=2,613):
   Accuracy:  0.4692
   Precision: 0.6072
   Recall:    0.4692
   F1 Score:  0.4381

 Confusion Matrix:
   [[598  36   2   0]
[482 512  63   0]
[622  75 116   0]
[102   0   5   0]]

 ── Eval 5: Assessment period 202302 ──
 Training samples:   13,772
 Test samples:       1,376  (admin2 areas in 202302)
 Training CatBoost...

 Results  (n_test=1,376):
   Accuracy:  0.7369
   Precision: 0.7456
   Recall:    0.7369
   F1 Score:  0.7368

 Confusion Matrix:
   [[386 189   5]
[ 94 479  37]
[  0  37 149]]

 ── Eval 6: Assessment period 202210 ──
 Training samples:   13,866
 Test samples:       2,399  (admin2 areas in 202210)
 Training CatBoost...

 Results  (n_test=2,399):
   Accuracy:  0.7216
   Precision: 0.7501
   Recall:    0.7216
   F1 Score:  0.7278

 Confusion Matrix:
   [[636 205   0   1]
[166 569  70  15]
[  4  87 439 119]
[  0   0   1  87]]

 ── Eval 7: Assessment period 202206 ──
 Training samples:   13,795
 Test samples:       2,568  (admin2 areas in 202206)
```

```
 Training CatBoost...

 Results  (n_test=2,568):
   Accuracy:  0.7477
   Precision: 0.7614
   Recall:    0.7477
   F1 Score:  0.7482

 Confusion Matrix:
   [[687 119   0   0]
[197 656  65   0]
[ 48 113 490  73]
[ 31   0   2  87]]

 ── Eval 8: Assessment period 202202 ──
 Training samples:   13,862
 Test samples:       2,535  (admin2 areas in 202202)
 Training CatBoost...

 Results  (n_test=2,535):
   Accuracy:  0.7598
   Precision: 0.7873
   Recall:    0.7598
   F1 Score:  0.7602

 Confusion Matrix:
   [[846  71   0   0]
[268 617  23   0]
[  4 172 431  69]
[  0   1   1  32]]

 ── Eval 9: Assessment period 202110 ──
 Training samples:   12,950
 Test samples:       2,535  (admin2 areas in 202110)
 Training CatBoost...

 Results  (n_test=2,535):
   Accuracy:  0.7653
   Precision: 0.8030
   Recall:    0.7653
   F1 Score:  0.7744

 Confusion Matrix:
   [[809 307   2   0]
[ 68 705  93   1]
[  0  68 414  55]
[  0   0   1  12]]

 ── Eval 10: Assessment period 202106 ──
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Training CatBoost...

 Results  (n_test=2,536):
   Accuracy:  0.7350
   Precision: 0.7585
```

```
  Recall:    0.7350
  F1 Score:  0.7308

 Confusion Matrix:
   [[804  60  16   0]
[406 613  59   0]
[  4  95 438  27]
[  0   0   5   9]]

 —— Eval 11: Assessment period 202102 ——
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Training CatBoost...

 Results  (n_test=2,489):
   Accuracy:  0.7991
   Precision: 0.8101
   Recall:    0.7991
   F1 Score:  0.8027

 Confusion Matrix:
   [[1007  225    0    0]
[ 120  511   66    7]
[   6   74  418    2]
[   0    0    0   53]]

 —— Eval 12: Assessment period 202010 ——
 Training samples:   12,858
 Test samples:       2,491  (admin2 areas in 202010)
 Training CatBoost...

 Results  (n_test=2,491):
   Accuracy:  0.7371
   Precision: 0.7471
   Recall:    0.7371
   F1 Score:  0.7414

 Confusion Matrix:
   [[936 189  11   0]
[190 505 119   0]
[  2 102 391  41]
[  0   0   1   4]]

 —— Eval 13: Assessment period 202006 ——
 Training samples:   11,506
 Test samples:       2,570  (admin2 areas in 202006)
 Training CatBoost...

 Results  (n_test=2,570):
   Accuracy:  0.7008
   Precision: 0.7285
   Recall:    0.7008
   F1 Score:  0.6806

 Confusion Matrix:
   [[1077   48    3    0]
```

```
[ 478  314   12    2]
[  85   98  340   37]
[   5    0    1   70]]

 ── Eval 14: Assessment period 202002 ──
 Training samples:   12,388
 Test samples:       1,588  (admin2 areas in 202002)
 Training CatBoost...

 Results  (n_test=1,588):
   Accuracy:  0.7626
   Precision: 0.7808
   Recall:    0.7626
   F1 Score:  0.7602

 Confusion Matrix:
   [[569  78  21   0]
[ 29 239 181  12]
[ 13  17 347  25]
[  0   0   1  56]]

 ── Eval 15: Assessment period 201910 ──
 Training samples:   12,533
 Test samples:       2,400  (admin2 areas in 201910)
 Training CatBoost...

 Results  (n_test=2,400):
   Accuracy:  0.7921
   Precision: 0.8004
   Recall:    0.7921
   F1 Score:  0.7938

 Confusion Matrix:
   [[1086  108    1    0]
[  85  341  129    4]
[   8  101  415   62]
[   0    0    1   59]]

 ── Eval 16: Assessment period 201906 ──
 Training samples:   12,605
 Test samples:       2,400  (admin2 areas in 201906)
 Training CatBoost...

 Results  (n_test=2,400):
   Accuracy:  0.7783
   Precision: 0.7789
   Recall:    0.7783
   F1 Score:  0.7652

 Confusion Matrix:
   [[1158   19    2    0]
[ 253  253   42    2]
[  14  139  382   59]
[   0    0    2   75]]

 ── Eval 17: Assessment period 201902 ──
```

```
 Training samples:   12,785
 Test samples:       2,400  (admin2 areas in 201902)
 Training CatBoost...

 Results  (n_test=2,400):
   Accuracy:  0.6804
   Precision: 0.7131
   Recall:    0.6804
   F1 Score:  0.6347

 Confusion Matrix:
   [[1336   92    0    0]
[ 225  176    9    2]
[ 329   23   94   47]
[  40    0    0   27]]

 ── Eval 18: Assessment period 201810 ──
 Training samples:   13,957
 Test samples:       1,081  (admin2 areas in 201810)
 Training CatBoost...

 Results  (n_test=1,081):
   Accuracy:  0.8039
   Precision: 0.8749
   Recall:    0.8039
   F1 Score:  0.8222

 Confusion Matrix:
   [[580  64   0   0]
[ 24 221   7  31]
[  3  14  53  67]
[  2   0   0  15]]

 ── Eval 19: Assessment period 201806 ──
 Training samples:   14,374
 Test samples:       2,440  (admin2 areas in 201806)
 Training CatBoost...

 Results  (n_test=2,440):
   Accuracy:  0.7885
   Precision: 0.8186
   Recall:    0.7885
   F1 Score:  0.7900

 Confusion Matrix:
   [[1279   42   10    0]
[ 209  246   27   41]
[  22   71  373   93]
[   0    0    1   26]]

 ── Eval 20: Assessment period 201802 ──
 Training samples:   14,614
 Test samples:       2,441  (admin2 areas in 201802)
 Training CatBoost...

 Results  (n_test=2,441):
```

```
  Accuracy:  0.8628
  Precision: 0.9057
  Recall:    0.8628
  F1 Score:  0.8778

 Confusion Matrix:
   [[1449   59    1    2    0]
[  64  255   19   19    0]
[  11   51  386  104    1]
[   0    3    1   16    0]
[   0    0    0    0    0]]

 ── Eval 21: Assessment period 201710 ──
 Training samples:   14,902
 Test samples:       2,577  (admin2 areas in 201710)
 Training CatBoost...

 Results  (n_test=2,577):
   Accuracy:  0.7610
   Precision: 0.8128
   Recall:    0.7610
   F1 Score:  0.7797

 Confusion Matrix:
   [[1351  292   15    0]
[  93  193   65   17]
[  25   15  373   92]
[   0    1    1   44]]

 ── Eval 22: Assessment period 201706 ──
 Training samples:   11,953
 Test samples:       2,573  (admin2 areas in 201706)
 Training CatBoost...

 Results  (n_test=2,573):
   Accuracy:  0.6487
   Precision: 0.6502
   Recall:    0.6487
   F1 Score:  0.6332

 Confusion Matrix:
   [[1274  109   83    0]
[ 248  235   14    3]
[  18  373  100   33]
[  14    1    8   60]]

 ── Eval 23: Assessment period 201702 ──
 Training samples:   9,303
 Test samples:       2,449  (admin2 areas in 201702)
 Training CatBoost...

 Results  (n_test=2,449):
   Accuracy:  0.8448
   Precision: 0.8462
   Recall:    0.8448
   F1 Score:  0.8384
```

```
  Confusion Matrix:
    [[1663   61    0    0    0]
 [ 156  252   12    4    0]
 [   6  110  125   19    0]
 [   0    8    4   28    0]
 [   0    0    0    0    1]]

  ── Eval 24: Assessment period 201610 ──
  Training samples:   6,140
  Test samples:       2,748  (admin2 areas in 201610)
  Training CatBoost...

  Results  (n_test=2,748):
    Accuracy:  0.8202
    Precision: 0.8186
    Recall:    0.8202
    F1 Score:  0.8165

  Confusion Matrix:
    [[1653  142   32    0]
 [ 149  206   99   11]
 [   6   22  351   30]
 [   0    1    2   44]]

  ── Eval 25: Assessment period 201606 ──
  Training samples:   3,099
  Test samples:       2,635  (admin2 areas in 201606)
  Training CatBoost...

  Results  (n_test=2,635):
    Accuracy:  0.9025
    Precision: 0.9116
    Recall:    0.9025
    F1 Score:  0.9035

  Confusion Matrix:
    [[1650   45    1    0]
 [  52  288   24    1]
 [  11  105  386   18]
 [   0    0    0   54]]

======================================================================
MODEL: CatBoost  |  FEATURE SET: GDELT only
======================================================================
Using 3485 features

  ── Eval 1: Assessment period 202406 ──
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Training CatBoost...

  Results  (n_test=1,919):
    Accuracy:  0.6389
    Precision: 0.6792
    Recall:    0.6389
```

```
  F1 Score:  0.6138

 Confusion Matrix:
   [[446   2   8  24]
[241 542  35  12]
[ 57 157 107 147]
[  0   2   8 131]]

 ── Eval 2: Assessment period 202402 ──
 Training samples:   12,107
 Test samples:       2,376  (admin2 areas in 202402)
 Training CatBoost...

 Results  (n_test=2,376):
   Accuracy:  0.6073
   Precision: 0.6471
   Recall:    0.6073
   F1 Score:  0.6006

 Confusion Matrix:
   [[685 245   3  24]
[253 509  45  27]
[ 68 101 142 165]
[  0   1   1 107]]

 ── Eval 3: Assessment period 202310 ──
 Training samples:   12,645
 Test samples:       1,919  (admin2 areas in 202310)
 Training CatBoost...

 Results  (n_test=1,919):
   Accuracy:  0.5430
   Precision: 0.6633
   Recall:    0.5430
   F1 Score:  0.5587

 Confusion Matrix:
   [[597 341   5  27]
[128 280  16  38]
[ 69  55 115 190]
[  0   0   8  50]]

 ── Eval 4: Assessment period 202306 ──
 Training samples:   12,765
 Test samples:       2,613  (admin2 areas in 202306)
 Training CatBoost...

 Results  (n_test=2,613):
   Accuracy:  0.6728
   Precision: 0.7499
   Recall:    0.6728
   F1 Score:  0.6814

 Confusion Matrix:
   [[522  88   4  22]
[290 725  19  23]
```

```
[110 122 419 162]
[  0   2  13  92]]

 ── Eval 5: Assessment period 202302 ──
 Training samples:   13,772
 Test samples:       1,376  (admin2 areas in 202302)
 Training CatBoost...

 Results  (n_test=1,376):
   Accuracy:  0.5974
   Precision: 0.6312
   Recall:    0.5974
   F1 Score:  0.5606

 Confusion Matrix:
   [[299 278   3   0]
[ 97 511   2   0]
[ 40 129  12   5]
[  0   0   0   0]]

 ── Eval 6: Assessment period 202210 ──
 Training samples:   13,866
 Test samples:       2,399  (admin2 areas in 202210)
 Training CatBoost...

 Results  (n_test=2,399):
   Accuracy:  0.6407
   Precision: 0.7194
   Recall:    0.6407
   F1 Score:  0.6463

 Confusion Matrix:
   [[563 256   0  23]
[192 587   4  37]
[ 76 140 300 133]
[  0   0   1  87]]

 ── Eval 7: Assessment period 202206 ──
 Training samples:   13,795
 Test samples:       2,568  (admin2 areas in 202206)
 Training CatBoost...

 Results  (n_test=2,568):
   Accuracy:  0.6558
   Precision: 0.7248
   Recall:    0.6558
   F1 Score:  0.6495

 Confusion Matrix:
   [[632 150   2  22]
[251 642  10  15]
[ 84 204 290 146]
[  0   0   0 120]]

 ── Eval 8: Assessment period 202202 ──
 Training samples:   13,862
```

```
 Test samples:       2,535  (admin2 areas in 202202)
 Training CatBoost...

 Results  (n_test=2,535):
   Accuracy:  0.6655
   Precision: 0.7140
   Recall:    0.6655
   F1 Score:  0.6600

 Confusion Matrix:
   [[831  74   5   7]
[377 490  28  13]
[ 91 176 333  76]
[  0   0   1  33]]

 ── Eval 9: Assessment period 202110 ──
 Training samples:   12,950
 Test samples:       2,535  (admin2 areas in 202110)
 Training CatBoost...

 Results  (n_test=2,535):
   Accuracy:  0.7148
   Precision: 0.7490
   Recall:    0.7148
   F1 Score:  0.7089

 Confusion Matrix:
   [[1053   54    6    5]
[ 376  437   50    4]
[  65   87  310   75]
[   0    1    0   12]]

 ── Eval 10: Assessment period 202106 ──
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Training CatBoost...

 Results  (n_test=2,536):
   Accuracy:  0.6514
   Precision: 0.7108
   Recall:    0.6514
   F1 Score:  0.6484

 Confusion Matrix:
   [[795  82   3   0]
[491 559  21   7]
[ 58 152 284  70]
[  0   0   0  14]]

 ── Eval 11: Assessment period 202102 ──
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Training CatBoost...

 Results  (n_test=2,489):
   Accuracy:  0.7545
```

```
  Precision: 0.7578
  Recall:    0.7545
  F1 Score:  0.7455

 Confusion Matrix:
   [[1135   92    2    3]
[ 257  391   47    9]
[  75   99  299   27]
[   0    0    0   53]]

 ── Eval 12: Assessment period 202010 ──
 Training samples:   12,858
 Test samples:       2,491  (admin2 areas in 202010)
 Training CatBoost...

 Results  (n_test=2,491):
   Accuracy:  0.6688
   Precision: 0.6905
   Recall:    0.6688
   F1 Score:  0.6618

 Confusion Matrix:
   [[1011  115    7    3]
[ 420  341   45    8]
[  51  114  310   61]
[   0    0    1    4]]

 ── Eval 13: Assessment period 202006 ──
 Training samples:   11,506
 Test samples:       2,570  (admin2 areas in 202006)
 Training CatBoost...

 Results  (n_test=2,570):
   Accuracy:  0.7012
   Precision: 0.7082
   Recall:    0.7012
   F1 Score:  0.6953

 Confusion Matrix:
   [[972 146  10   0]
[271 454  63  18]
[ 33 179 300  48]
[  0   0   0  76]]

 ── Eval 14: Assessment period 202002 ──
 Training samples:   12,388
 Test samples:       1,588  (admin2 areas in 202002)
 Training CatBoost...

 Results  (n_test=1,588):
   Accuracy:  0.6417
   Precision: 0.7109
   Recall:    0.6417
   F1 Score:  0.6505

 Confusion Matrix:
```

```
   [[360 306   2   0]
[ 48 316  87  10]
[  5  68 287  42]
[  0   0   1  56]]

 ── Eval 15: Assessment period 201910 ──
 Training samples:   12,533
 Test samples:       2,400  (admin2 areas in 201910)
 Training CatBoost...

 Results  (n_test=2,400):
   Accuracy:  0.7154
   Precision: 0.7320
   Recall:    0.7154
   F1 Score:  0.7095

 Confusion Matrix:
   [[1078  105   12    0]
[ 205  276   49   29]
[  92  112  308   74]
[   0    5    0   55]]

 ── Eval 16: Assessment period 201906 ──
 Training samples:   12,605
 Test samples:       2,400  (admin2 areas in 201906)
 Training CatBoost...

 Results  (n_test=2,400):
   Accuracy:  0.7050
   Precision: 0.7443
   Recall:    0.7050
   F1 Score:  0.7109

 Confusion Matrix:
   [[959 209  11   0]
[190 320  22  18]
[ 38 154 338  64]
[  0   1   1  75]]

 ── Eval 17: Assessment period 201902 ──
 Training samples:   12,785
 Test samples:       2,400  (admin2 areas in 201902)
 Training CatBoost...

 Results  (n_test=2,400):
   Accuracy:  0.7588
   Precision: 0.8212
   Recall:    0.7588
   F1 Score:  0.7733

 Confusion Matrix:
   [[1197  230    0    1]
[ 123  259    3   27]
[  21   98  298   76]
[   0    0    0   67]]
```

```
 ── Eval 18: Assessment period 201810 ──
 Training samples:   13,957
 Test samples:       1,081  (admin2 areas in 201810)
 Training CatBoost...

 Results  (n_test=1,081):
   Accuracy:  0.6707
   Precision: 0.7891
   Recall:    0.6707
   F1 Score:  0.6909

 Confusion Matrix:
   [[560  82   0   2]
[ 70 120   0  93]
[  0  36  29  72]
[  0   1   0  16]]

 ── Eval 19: Assessment period 201806 ──
 Training samples:   14,374
 Test samples:       2,440  (admin2 areas in 201806)
 Training CatBoost...

 Results  (n_test=2,440):
   Accuracy:  0.6832
   Precision: 0.7581
   Recall:    0.6832
   F1 Score:  0.6992

 Confusion Matrix:
   [[1130  199    0    2]
[ 204  239   10   70]
[  53  117  273  116]
[   0    1    1   25]]

 ── Eval 20: Assessment period 201802 ──
 Training samples:   14,614
 Test samples:       2,441  (admin2 areas in 201802)
 Training CatBoost...

 Results  (n_test=2,441):
   Accuracy:  0.7145
   Precision: 0.8429
   Recall:    0.7145
   F1 Score:  0.7495

 Confusion Matrix:
   [[1233  275    0    3    0]
[ 101  221    0   35    0]
[  25  103  270  154    1]
[   0    0    0   20    0]
[   0    0    0    0    0]]

 ── Eval 21: Assessment period 201710 ──
 Training samples:   14,902
 Test samples:       2,577  (admin2 areas in 201710)
 Training CatBoost...
```

```
 Results  (n_test=2,577):
   Accuracy:  0.7497
   Precision: 0.8340
   Recall:    0.7497
   F1 Score:  0.7731

 Confusion Matrix:
   [[1383  267    7    1    0]
[ 104  228   11   25    0]
[  21   81  277  126    0]
[   0    0    1   44    1]
[   0    0    0    0    0]]

 ── Eval 22: Assessment period 201706 ──
 Training samples:   11,953
 Test samples:       2,573  (admin2 areas in 201706)
 Training CatBoost...

 Results  (n_test=2,573):
   Accuracy:  0.6763
   Precision: 0.6632
   Recall:    0.6763
   F1 Score:  0.6605

 Confusion Matrix:
   [[1249  149   66    2]
[ 337  130   16   17]
[ 106   81  287   50]
[   0    5    4   74]]

 ── Eval 23: Assessment period 201702 ──
 Training samples:   9,303
 Test samples:       2,449  (admin2 areas in 201702)
 Training CatBoost...

 Results  (n_test=2,449):
   Accuracy:  0.6905
   Precision: 0.7628
   Recall:    0.6905
   F1 Score:  0.6991

 Confusion Matrix:
   [[1338  375    8    3    0]
[ 126  272   11   15    0]
[  57  133   48   22    0]
[   0    8    0   32    0]
[   0    0    0    0    1]]

 ── Eval 24: Assessment period 201610 ──
 Training samples:   6,140
 Test samples:       2,748  (admin2 areas in 201610)
 Training CatBoost...

 Results  (n_test=2,748):
   Accuracy:  0.7911
```

```
    Precision: 0.8083
    Recall:    0.7911
    F1 Score:  0.7969

  Confusion Matrix:
    [[1577  234   14    2]
 [ 174  252   25   14]
 [  27   52  300   30]
 [   0    0    2   45]]

  —— Eval 25: Assessment period 201606 ——
  Training samples:   3,099
  Test samples:       2,635  (admin2 areas in 201606)
  Training CatBoost...

  Results  (n_test=2,635):
    Accuracy:  0.8414
    Precision: 0.8502
    Recall:    0.8414
    F1 Score:  0.8421

  Confusion Matrix:
    [[1564   88   32   12]
 [  93  240   20   12]
 [  66   78  359   17]
 [   0    0    0   54]]

======================================================================
MODEL: CatBoost  |  FEATURE SET: IPC only (no geo)
======================================================================
Using 1 features

  —— Eval 1: Assessment period 202406 ——
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Training CatBoost...

  Results  (n_test=1,919):
    Accuracy:  0.6514
    Precision: 0.6803
    Recall:    0.6514
    F1 Score:  0.6483

  Confusion Matrix:
    [[429  51   0   0]
 [359 439  32   0]
 [  0 157 292  19]
 [  0   2  49  90]]

  —— Eval 2: Assessment period 202402 ——
  Training samples:   12,107
  Test samples:       2,376  (admin2 areas in 202402)
  Training CatBoost...

  Results  (n_test=2,376):
    Accuracy:  0.6940
```

```
  Precision: 0.7269
  Recall:    0.6940
  F1 Score:  0.6769

 Confusion Matrix:
   [[906  51   0   0]
[418 370  46   0]
[103  40 324   9]
[  0   1  59  49]]

 ── Eval 3: Assessment period 202310 ──
 Training samples:   12,645
 Test samples:       1,919  (admin2 areas in 202310)
 Training CatBoost...

 Results  (n_test=1,919):
  Accuracy:  0.6165
  Precision: 0.7578
  Recall:    0.6165
  F1 Score:  0.6348

 Confusion Matrix:
   [[493 471   6   0]
[  4 326 132   0]
[  0  40 323  66]
[  0   0  17  41]]

 ── Eval 4: Assessment period 202306 ──
 Training samples:   12,765
 Test samples:       2,613  (admin2 areas in 202306)
 Training CatBoost...

 Results  (n_test=2,613):
  Accuracy:  0.4646
  Precision: 0.6057
  Recall:    0.4646
  F1 Score:  0.4336

 Confusion Matrix:
   [[598  36   2   0]
[494 500  63   0]
[623  74 116   0]
[102   0   5   0]]

 ── Eval 5: Assessment period 202302 ──
 Training samples:   13,772
 Test samples:       1,376  (admin2 areas in 202302)
 Training CatBoost...

 Results  (n_test=1,376):
  Accuracy:  0.7369
  Precision: 0.7456
  Recall:    0.7369
  F1 Score:  0.7368

 Confusion Matrix:
```

```
   [[386 189   5]
[ 94 479  37]
[  0  37 149]]

 ── Eval 6: Assessment period 202210 ──
 Training samples:   13,866
 Test samples:       2,399  (admin2 areas in 202210)
 Training CatBoost...

 Results  (n_test=2,399):
   Accuracy:  0.7253
   Precision: 0.7332
   Recall:    0.7253
   F1 Score:  0.7284

 Confusion Matrix:
   [[636 205   1   0]
[166 570  78   6]
[  4  87 489  69]
[  0   0  43  45]]

 ── Eval 7: Assessment period 202206 ──
 Training samples:   13,795
 Test samples:       2,568  (admin2 areas in 202206)
 Training CatBoost...

 Results  (n_test=2,568):
   Accuracy:  0.7492
   Precision: 0.7603
   Recall:    0.7492
   F1 Score:  0.7431

 Confusion Matrix:
   [[687 119   0   0]
[197 656  65   0]
[ 48 126 549   1]
[ 31   2  55  32]]

 ── Eval 8: Assessment period 202202 ──
 Training samples:   13,862
 Test samples:       2,535  (admin2 areas in 202202)
 Training CatBoost...

 Results  (n_test=2,535):
   Accuracy:  0.7787
   Precision: 0.7887
   Recall:    0.7787
   F1 Score:  0.7760

 Confusion Matrix:
   [[846  71   0   0]
[268 618  22   0]
[  4 173 498   1]
[  0   5  17  12]]

 ── Eval 9: Assessment period 202110 ──
```

```
 Training samples:   12,950
 Test samples:       2,535  (admin2 areas in 202110)
 Training CatBoost...

 Results  (n_test=2,535):
   Accuracy:  0.7815
   Precision: 0.8050
   Recall:    0.7815
   F1 Score:  0.7850

 Confusion Matrix:
   [[809 307   2   0]
[ 70 703  94   0]
[  0  68 462   7]
[  0   0   6   7]]

 —— Eval 10: Assessment period 202106 ——
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Training CatBoost...

 Results  (n_test=2,536):
   Accuracy:  0.7015
   Precision: 0.7365
   Recall:    0.7015
   F1 Score:  0.6966

 Confusion Matrix:
   [[805  59  16   0]
[470 549  59   0]
[  4  96 418  46]
[  0   0   7   7]]

 —— Eval 11: Assessment period 202102 ——
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Training CatBoost...

 Results  (n_test=2,489):
   Accuracy:  0.7790
   Precision: 0.7901
   Recall:    0.7790
   F1 Score:  0.7763

 Confusion Matrix:
   [[1007  225    0    0]
[ 120  511   73    0]
[   7   75  417    1]
[   2    1   46    4]]

 —— Eval 12: Assessment period 202010 ——
 Training samples:   12,858
 Test samples:       2,491  (admin2 areas in 202010)
 Training CatBoost...

 Results  (n_test=2,491):
```

```
  Accuracy:  0.7354
  Precision: 0.7456
  Recall:    0.7354
  F1 Score:  0.7399

 Confusion Matrix:
   [[936 189  11   0]
[190 505 119   0]
[  2 106 387  41]
[  0   0   1   4]]

 ── Eval 13: Assessment period 202006 ──
 Training samples:   11,506
 Test samples:       2,570  (admin2 areas in 202006)
 Training CatBoost...

 Results  (n_test=2,570):
  Accuracy:  0.7054
  Precision: 0.7276
  Recall:    0.7054
  F1 Score:  0.6865

 Confusion Matrix:
   [[1077   48    3    0]
[ 482  314   10    0]
[  86   99  370    5]
[   5    0   19   52]]

 ── Eval 14: Assessment period 202002 ──
 Training samples:   12,388
 Test samples:       1,588  (admin2 areas in 202002)
 Training CatBoost...

 Results  (n_test=1,588):
  Accuracy:  0.7752
  Precision: 0.7906
  Recall:    0.7752
  F1 Score:  0.7737

 Confusion Matrix:
   [[569  78  21   0]
[ 29 257 167   8]
[ 13  24 363   2]
[  0   1  14  42]]

 ── Eval 15: Assessment period 201910 ──
 Training samples:   12,533
 Test samples:       2,400  (admin2 areas in 201910)
 Training CatBoost...

 Results  (n_test=2,400):
  Accuracy:  0.8017
  Precision: 0.8032
  Recall:    0.8017
  F1 Score:  0.8021
```

```
 Confusion Matrix:
   [[1086  108    1    0]
[  85  341  132    1]
[   8  101  449   28]
[   0    0   12   48]]

 ── Eval 16: Assessment period 201906 ──
 Training samples:   12,605
 Test samples:       2,400  (admin2 areas in 201906)
 Training CatBoost...

 Results  (n_test=2,400):
   Accuracy:  0.7892
   Precision: 0.7802
   Recall:    0.7892
   F1 Score:  0.7763

 Confusion Matrix:
   [[1158   19    2    0]
[ 256  252   42    0]
[  14  141  428   11]
[   0    0   21   56]]

 ── Eval 17: Assessment period 201902 ──
 Training samples:   12,785
 Test samples:       2,400  (admin2 areas in 201902)
 Training CatBoost...

 Results  (n_test=2,400):
   Accuracy:  0.6758
   Precision: 0.6861
   Recall:    0.6758
   F1 Score:  0.6291

 Confusion Matrix:
   [[1336   92    0    0]
[ 234  167   11    0]
[ 353   24  109    7]
[  40    0   17   10]]

 ── Eval 18: Assessment period 201810 ──
 Training samples:   13,957
 Test samples:       1,081  (admin2 areas in 201810)
 Training CatBoost...

 Results  (n_test=1,081):
   Accuracy:  0.8483
   Precision: 0.8560
   Recall:    0.8483
   F1 Score:  0.8513

 Confusion Matrix:
   [[580  64   0   0]
[ 25 220  38   0]
[  3  14 109  11]
[  2   0   7   8]]
```

```
── Eval 19: Assessment period 201806 ──
Training samples:   14,374
Test samples:       2,440  (admin2 areas in 201806)
Training CatBoost...

Results  (n_test=2,440):
  Accuracy:  0.8119
  Precision: 0.7984
  Recall:    0.8119
  F1 Score:  0.7987

Confusion Matrix:
  [[1279   42   10    0    0]
[ 211  244   68    0    0]
[  22   71  458    0    8]
[   0    0   15    0   12]
[   0    0    0    0    0]]

── Eval 20: Assessment period 201802 ──
Training samples:   14,614
Test samples:       2,441  (admin2 areas in 201802)
Training CatBoost...

Results  (n_test=2,441):
  Accuracy:  0.8845
  Precision: 0.8940
  Recall:    0.8845
  F1 Score:  0.8888

Confusion Matrix:
  [[1449   59    3    0    0]
[  64  255   38    0    0]
[  11   51  455    0   36]
[   0    3    9    0    8]
[   0    0    0    0    0]]

── Eval 21: Assessment period 201710 ──
Training samples:   14,902
Test samples:       2,577  (admin2 areas in 201710)
Training CatBoost...

Results  (n_test=2,577):
  Accuracy:  0.7575
  Precision: 0.7991
  Recall:    0.7575
  F1 Score:  0.7747

Confusion Matrix:
  [[1352  291   15    0    0]
[  93  193   81    0    1]
[  25   15  407    0   58]
[   0    1   21    0   24]
[   0    0    0    0    0]]

── Eval 22: Assessment period 201706 ──
```

```
 Training samples:   11,953
 Test samples:       2,573  (admin2 areas in 201706)
 Training CatBoost...

 Results  (n_test=2,573):
   Accuracy:  0.6215
   Precision: 0.5815
   Recall:    0.6215
   F1 Score:  0.5869

 Confusion Matrix:
   [[1275  108   83    0    0]
[ 283  200   17    0    0]
[ 276  115  124    0    9]
[  14    1   37    0   31]
[   0    0    0    0    0]]

 ── Eval 23: Assessment period 201702 ──
 Training samples:   9,303
 Test samples:       2,449  (admin2 areas in 201702)
 Training CatBoost...

 Results  (n_test=2,449):
   Accuracy:  0.8383
   Precision: 0.8301
   Recall:    0.8383
   F1 Score:  0.8301

 Confusion Matrix:
   [[1664   60    0    0    0]
[ 156  249   17    0    2]
[   6  110  139    0    5]
[   0    9   12    0   19]
[   0    0    0    0    1]]

 ── Eval 24: Assessment period 201610 ──
 Training samples:   6,140
 Test samples:       2,748  (admin2 areas in 201610)
 Training CatBoost...

 Results  (n_test=2,748):
   Accuracy:  0.8173
   Precision: 0.8136
   Recall:    0.8173
   F1 Score:  0.8124

 Confusion Matrix:
   [[1653  139   35    0]
[ 152  200  113    0]
[   6   22  360   21]
[   0    2   12   33]]

 ── Eval 25: Assessment period 201606 ──
 Training samples:   3,099
 Test samples:       2,635  (admin2 areas in 201606)
 Training CatBoost...
```

```
  Results  (n_test=2,635):
    Accuracy:  0.7347
    Precision: 0.6744
    Recall:    0.7347
    F1 Score:  0.6973

  Confusion Matrix:
    [[1659   37    0    0]
 [ 113  227    0   25]
 [  24  138    0  358]
 [   0    4    0   50]]

======================================================================
MODEL: CatBoost  |  FEATURE SET: IPC + GDELT base (no geo)
======================================================================
Using 120 features

  ── Eval 1: Assessment period 202406 ──
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Training CatBoost...

  Results  (n_test=1,919):
    Accuracy:  0.6477
    Precision: 0.6755
    Recall:    0.6477
    F1 Score:  0.6441

  Confusion Matrix:
    [[429  51   0   0]
 [359 438  33   0]
 [  0 155 286  27]
 [  0   2  49  90]]

  ── Eval 2: Assessment period 202402 ──
  Training samples:   12,107
  Test samples:       2,376  (admin2 areas in 202402)
  Training CatBoost...

  Results  (n_test=2,376):
    Accuracy:  0.6919
    Precision: 0.7227
    Recall:    0.6919
    F1 Score:  0.6750

  Confusion Matrix:
    [[906  51   0   0]
 [417 371  46   0]
 [103  40 316  17]
 [  0   1  57  51]]

  ── Eval 3: Assessment period 202310 ──
  Training samples:   12,645
  Test samples:       1,919  (admin2 areas in 202310)
  Training CatBoost...
```

```
 Results  (n_test=1,919):
   Accuracy:  0.6149
   Precision: 0.7577
   Recall:    0.6149
   F1 Score:  0.6339

 Confusion Matrix:
   [[493 471   6   0]
[  4 326 129   3]
[  0  40 320  69]
[  0   0  17  41]]

 ── Eval 4: Assessment period 202306 ──
 Training samples:   12,765
 Test samples:       2,613  (admin2 areas in 202306)
 Training CatBoost...

 Results  (n_test=2,613):
   Accuracy:  0.4642
   Precision: 0.6080
   Recall:    0.4642
   F1 Score:  0.4330

 Confusion Matrix:
   [[597  37   2   0]
[492 502  58   5]
[620  77 114   2]
[102   0   5   0]]

 ── Eval 5: Assessment period 202302 ──
 Training samples:   13,772
 Test samples:       1,376  (admin2 areas in 202302)
 Training CatBoost...

 Results  (n_test=1,376):
   Accuracy:  0.7362
   Precision: 0.7462
   Recall:    0.7362
   F1 Score:  0.7366

 Confusion Matrix:
   [[386 189   5   0]
[ 93 480  37   0]
[  0  37 147   2]
[  0   0   0   0]]

 ── Eval 6: Assessment period 202210 ──
 Training samples:   13,866
 Test samples:       2,399  (admin2 areas in 202210)
 Training CatBoost...

 Results  (n_test=2,399):
   Accuracy:  0.7224
   Precision: 0.7326
   Recall:    0.7224
```

```
  F1 Score:  0.7262

 Confusion Matrix:
   [[636 205   1   0]
[166 568  79   7]
[  4  87 481  77]
[  0   0  40  48]]

 —— Eval 7: Assessment period 202206 ——
 Training samples:   13,795
 Test samples:       2,568  (admin2 areas in 202206)
 Training CatBoost...

 Results  (n_test=2,568):
   Accuracy:  0.7465
   Precision: 0.7500
   Recall:    0.7465
   F1 Score:  0.7409

 Confusion Matrix:
   [[687 119   0   0]
[197 656  62   3]
[ 48 126 541   9]
[ 31   2  54  33]]

 —— Eval 8: Assessment period 202202 ——
 Training samples:   13,862
 Test samples:       2,535  (admin2 areas in 202202)
 Training CatBoost...

 Results  (n_test=2,535):
   Accuracy:  0.7783
   Precision: 0.7872
   Recall:    0.7783
   F1 Score:  0.7760

 Confusion Matrix:
   [[846  71   0   0]
[267 617  22   2]
[  4 173 496   3]
[  0   5  15  14]]

 —— Eval 9: Assessment period 202110 ——
 Training samples:   12,950
 Test samples:       2,535  (admin2 areas in 202110)
 Training CatBoost...

 Results  (n_test=2,535):
   Accuracy:  0.7819
   Precision: 0.8064
   Recall:    0.7819
   F1 Score:  0.7857

 Confusion Matrix:
   [[809 307   2   0]
[ 68 705  93   1]
```

```
[  0  68 460   9]
[  0   0   5   8]]

 ── Eval 10: Assessment period 202106 ──
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Training CatBoost...

 Results  (n_test=2,536):
   Accuracy:  0.7027
   Precision: 0.7376
   Recall:    0.7027
   F1 Score:  0.6978

 Confusion Matrix:
   [[805  59  16   0]
[468 551  59   0]
[  4  96 418  46]
[  0   0   6   8]]

 ── Eval 11: Assessment period 202102 ──
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Training CatBoost...

 Results  (n_test=2,489):
   Accuracy:  0.7786
   Precision: 0.7882
   Recall:    0.7786
   F1 Score:  0.7765

 Confusion Matrix:
   [[1006  226    0    0]
[ 120  511   73    0]
[   7   75  416    2]
[   2    1   45    5]]

 ── Eval 12: Assessment period 202010 ──
 Training samples:   12,858
 Test samples:       2,491  (admin2 areas in 202010)
 Training CatBoost...

 Results  (n_test=2,491):
   Accuracy:  0.7354
   Precision: 0.7465
   Recall:    0.7354
   F1 Score:  0.7403

 Confusion Matrix:
   [[936 188  12   0]
[190 505 116   3]
[  2 106 387  41]
[  0   0   1   4]]

 ── Eval 13: Assessment period 202006 ──
 Training samples:   11,506
```

```
 Test samples:       2,570  (admin2 areas in 202006)
 Training CatBoost...

 Results  (n_test=2,570):
   Accuracy:  0.7054
   Precision: 0.7265
   Recall:    0.7054
   F1 Score:  0.6867

 Confusion Matrix:
   [[1076   49    3    0]
[ 480  316   10    0]
[  83  102  369    6]
[   5    0   19   52]]

 ── Eval 14: Assessment period 202002 ──
 Training samples:   12,388
 Test samples:       1,588  (admin2 areas in 202002)
 Training CatBoost...

 Results  (n_test=1,588):
   Accuracy:  0.7739
   Precision: 0.7888
   Recall:    0.7739
   F1 Score:  0.7725

 Confusion Matrix:
   [[569  78  21   0]
[ 29 257 165  10]
[ 13  24 361   4]
[  0   1  14  42]]

 ── Eval 15: Assessment period 201910 ──
 Training samples:   12,533
 Test samples:       2,400  (admin2 areas in 201910)
 Training CatBoost...

 Results  (n_test=2,400):
   Accuracy:  0.7996
   Precision: 0.8017
   Recall:    0.7996
   F1 Score:  0.8002

 Confusion Matrix:
   [[1086  108    1    0]
[  85  341  132    1]
[   8  101  444   33]
[   0    0   12   48]]

 ── Eval 16: Assessment period 201906 ──
 Training samples:   12,605
 Test samples:       2,400  (admin2 areas in 201906)
 Training CatBoost...

 Results  (n_test=2,400):
   Accuracy:  0.7883
```

```
  Precision: 0.7794
  Recall:    0.7883
  F1 Score:  0.7756

 Confusion Matrix:
   [[1157   20    2    0]
[ 256  252   42    0]
[  14  141  427   12]
[   0    0   21   56]]

 ── Eval 17: Assessment period 201902 ──
 Training samples:   12,785
 Test samples:       2,400  (admin2 areas in 201902)
 Training CatBoost...

 Results  (n_test=2,400):
   Accuracy:  0.6754
   Precision: 0.6849
   Recall:    0.6754
   F1 Score:  0.6284

 Confusion Matrix:
   [[1336   92    0    0]
[ 234  167   11    0]
[ 353   24  108    8]
[  40    0   17   10]]

 ── Eval 18: Assessment period 201810 ──
 Training samples:   13,957
 Test samples:       1,081  (admin2 areas in 201810)
 Training CatBoost...

 Results  (n_test=1,081):
   Accuracy:  0.8474
   Precision: 0.8558
   Recall:    0.8474
   F1 Score:  0.8507

 Confusion Matrix:
   [[580  64   0   0]
[ 25 220  37   1]
[  3  14 108  12]
[  2   0   7   8]]

 ── Eval 19: Assessment period 201806 ──
 Training samples:   14,374
 Test samples:       2,440  (admin2 areas in 201806)
 Training CatBoost...

 Results  (n_test=2,440):
   Accuracy:  0.8127
   Precision: 0.8023
   Recall:    0.8127
   F1 Score:  0.8010

 Confusion Matrix:
```

```
   [[1279   42   10    0]
[ 211  244   67    1]
[  22   71  448   18]
[   0    0   15   12]]

 ── Eval 20: Assessment period 201802 ──
 Training samples:   14,614
 Test samples:       2,441  (admin2 areas in 201802)
 Training CatBoost...

 Results  (n_test=2,441):
   Accuracy:  0.8841
   Precision: 0.8949
   Recall:    0.8841
   F1 Score:  0.8884

 Confusion Matrix:
   [[1449   59    3    0    0]
[  64  255   38    0    0]
[  11   51  446   44    1]
[   0    3    9    8    0]
[   0    0    0    0    0]]

 ── Eval 21: Assessment period 201710 ──
 Training samples:   14,902
 Test samples:       2,577  (admin2 areas in 201710)
 Training CatBoost...

 Results  (n_test=2,577):
   Accuracy:  0.7633
   Precision: 0.8051
   Recall:    0.7633
   F1 Score:  0.7795

 Confusion Matrix:
   [[1350  293   15    0]
[  93  193   77    5]
[  25   15  396   69]
[   0    1   17   28]]

 ── Eval 22: Assessment period 201706 ──
 Training samples:   11,953
 Test samples:       2,573  (admin2 areas in 201706)
 Training CatBoost...

 Results  (n_test=2,573):
   Accuracy:  0.6300
   Precision: 0.5981
   Recall:    0.6300
   F1 Score:  0.5981

 Confusion Matrix:
   [[1274  109   82    1]
[ 283  200   17    0]
[ 275  116  114   19]
[  14    1   35   33]]
```

```
  ── Eval 23: Assessment period 201702 ──
  Training samples:   9,303
  Test samples:       2,449  (admin2 areas in 201702)
  Training CatBoost...

  Results  (n_test=2,449):
    Accuracy:  0.8436
    Precision: 0.8403
    Recall:    0.8436
    F1 Score:  0.8372

  Confusion Matrix:
    [[1663   61    0    0    0]
 [ 156  249   17    2    0]
 [   6  110  131   13    0]
 [   0    8   10   22    0]
 [   0    0    0    0    1]]

  ── Eval 24: Assessment period 201610 ──
  Training samples:   6,140
  Test samples:       2,748  (admin2 areas in 201610)
  Training CatBoost...

  Results  (n_test=2,748):
    Accuracy:  0.8144
    Precision: 0.8117
    Recall:    0.8144
    F1 Score:  0.8104

  Confusion Matrix:
    [[1653  141   33    0]
 [ 152  202  109    2]
 [   6   23  351   29]
 [   0    2   13   32]]

  ── Eval 25: Assessment period 201606 ──
  Training samples:   3,099
  Test samples:       2,635  (admin2 areas in 201606)
  Training CatBoost...

  Results  (n_test=2,635):
    Accuracy:  0.7400
    Precision: 0.8041
    Recall:    0.7400
    F1 Score:  0.7108

  Confusion Matrix:
    [[1657   39    0    0]
 [ 111  227    7   20]
 [  24  137   19  340]
 [   0    4    3   47]]

======================================================================
MODEL: CatBoost  |  FEATURE SET: IPC + GDELT + temporal (no geo)
======================================================================
```

```
Using 155 features

  ── Eval 1: Assessment period 202406 ──
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Training CatBoost...

  Results  (n_test=1,919):
    Accuracy:  0.6477
    Precision: 0.6756
    Recall:    0.6477
    F1 Score:  0.6440

  Confusion Matrix:
    [[429  51   0   0]
 [359 438  33   0]
 [  0 155 284  29]
 [  0   2  47  92]]

  ── Eval 2: Assessment period 202402 ──
  Training samples:   12,107
  Test samples:       2,376  (admin2 areas in 202402)
  Training CatBoost...

  Results  (n_test=2,376):
    Accuracy:  0.6919
    Precision: 0.7226
    Recall:    0.6919
    F1 Score:  0.6751

  Confusion Matrix:
    [[906  51   0   0]
 [417 371  45   1]
 [103  39 315  19]
 [  0   1  56  52]]

  ── Eval 3: Assessment period 202310 ──
  Training samples:   12,645
  Test samples:       1,919  (admin2 areas in 202310)
  Training CatBoost...

  Results  (n_test=1,919):
    Accuracy:  0.6081
    Precision: 0.7553
    Recall:    0.6081
    F1 Score:  0.6289

  Confusion Matrix:
    [[493 471   6   0]
 [  4 326 126   6]
 [  0  40 307  82]
 [  0   0  17  41]]

  ── Eval 4: Assessment period 202306 ──
  Training samples:   12,765
  Test samples:       2,613  (admin2 areas in 202306)
```

```
 Training CatBoost...

 Results  (n_test=2,613):
   Accuracy:  0.4646
   Precision: 0.6126
   Recall:    0.4646
   F1 Score:  0.4335

 Confusion Matrix:
   [[598  36   2   0]
[492 502  59   4]
[619  77 113   4]
[102   0   4   1]]

 ── Eval 5: Assessment period 202302 ──
 Training samples:   13,772
 Test samples:       1,376  (admin2 areas in 202302)
 Training CatBoost...

 Results  (n_test=1,376):
   Accuracy:  0.7340
   Precision: 0.7463
   Recall:    0.7340
   F1 Score:  0.7356

 Confusion Matrix:
   [[386 189   5   0]
[ 93 480  36   1]
[  0  37 144   5]
[  0   0   0   0]]

 ── Eval 6: Assessment period 202210 ──
 Training samples:   13,866
 Test samples:       2,399  (admin2 areas in 202210)
 Training CatBoost...

 Results  (n_test=2,399):
   Accuracy:  0.7220
   Precision: 0.7334
   Recall:    0.7220
   F1 Score:  0.7262

 Confusion Matrix:
   [[636 205   1   0]
[166 569  78   7]
[  4  87 477  81]
[  0   0  38  50]]

 ── Eval 7: Assessment period 202206 ──
 Training samples:   13,795
 Test samples:       2,568  (admin2 areas in 202206)
 Training CatBoost...

 Results  (n_test=2,568):
   Accuracy:  0.7438
   Precision: 0.7452
```

```
  Recall:    0.7438
  F1 Score:  0.7391

 Confusion Matrix:
   [[686 120   0   0]
[197 655  62   4]
[ 48 126 533  17]
[ 31   2  51  36]]

 —— Eval 8: Assessment period 202202 ——
 Training samples:   13,862
 Test samples:       2,535  (admin2 areas in 202202)
 Training CatBoost...

 Results  (n_test=2,535):
   Accuracy:  0.7744
   Precision: 0.7832
   Recall:    0.7744
   F1 Score:  0.7721

 Confusion Matrix:
   [[846  71   0   0]
[268 616  23   1]
[  4 172 487  13]
[  0   5  15  14]]

 —— Eval 9: Assessment period 202110 ——
 Training samples:   12,950
 Test samples:       2,535  (admin2 areas in 202110)
 Training CatBoost...

 Results  (n_test=2,535):
   Accuracy:  0.7807
   Precision: 0.8058
   Recall:    0.7807
   F1 Score:  0.7847

 Confusion Matrix:
   [[808 308   2   0]
[ 68 705  93   1]
[  0  68 458  11]
[  0   0   5   8]]

 —— Eval 10: Assessment period 202106 ——
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Training CatBoost...

 Results  (n_test=2,536):
   Accuracy:  0.7023
   Precision: 0.7373
   Recall:    0.7023
   F1 Score:  0.6973

 Confusion Matrix:
   [[805  59  16   0]
```

```
[467 550  61   0]
[  4  96 417  47]
[  0   0   5   9]]

 ── Eval 11: Assessment period 202102 ──
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Training CatBoost...

 Results  (n_test=2,489):
   Accuracy:  0.7794
   Precision: 0.7913
   Recall:    0.7794
   F1 Score:  0.7772

 Confusion Matrix:
   [[1006  226    0    0]
[ 119  512   73    0]
[   7   75  417    1]
[   2    1   45    5]]

 ── Eval 12: Assessment period 202010 ──
 Training samples:   12,858
 Test samples:       2,491  (admin2 areas in 202010)
 Training CatBoost...

 Results  (n_test=2,491):
   Accuracy:  0.7346
   Precision: 0.7458
   Recall:    0.7346
   F1 Score:  0.7396

 Confusion Matrix:
   [[935 190  11   0]
[190 505 117   2]
[  2 106 386  42]
[  0   0   1   4]]

 ── Eval 13: Assessment period 202006 ──
 Training samples:   11,506
 Test samples:       2,570  (admin2 areas in 202006)
 Training CatBoost...

 Results  (n_test=2,570):
   Accuracy:  0.7058
   Precision: 0.7263
   Recall:    0.7058
   F1 Score:  0.6872

 Confusion Matrix:
   [[1076   49    3    0]
[ 479  317   10    0]
[  82  103  369    6]
[   4    1   19   52]]

 ── Eval 14: Assessment period 202002 ──
```

```
 Training samples:   12,388
 Test samples:       1,588  (admin2 areas in 202002)
 Training CatBoost...

 Results  (n_test=1,588):
   Accuracy:  0.7695
   Precision: 0.7844
   Recall:    0.7695
   F1 Score:  0.7686

 Confusion Matrix:
   [[569  78  21   0]
[ 29 257 161  14]
[ 13  24 354  11]
[  0   1  14  42]]

 —— Eval 15: Assessment period 201910 ——
 Training samples:   12,533
 Test samples:       2,400  (admin2 areas in 201910)
 Training CatBoost...

 Results  (n_test=2,400):
   Accuracy:  0.7971
   Precision: 0.8008
   Recall:    0.7971
   F1 Score:  0.7981

 Confusion Matrix:
   [[1086  108    1    0]
[  85  341  129    4]
[   8  101  437   40]
[   0    0   11   49]]

 —— Eval 16: Assessment period 201906 ——
 Training samples:   12,605
 Test samples:       2,400  (admin2 areas in 201906)
 Training CatBoost...

 Results  (n_test=2,400):
   Accuracy:  0.7879
   Precision: 0.7792
   Recall:    0.7879
   F1 Score:  0.7752

 Confusion Matrix:
   [[1158   19    2    0]
[ 253  255   42    0]
[  14  141  421   18]
[   0    0   20   57]]

 —— Eval 17: Assessment period 201902 ——
 Training samples:   12,785
 Test samples:       2,400  (admin2 areas in 201902)
 Training CatBoost...

 Results  (n_test=2,400):
```

```
  Accuracy:  0.6746
  Precision: 0.6853
  Recall:    0.6746
  F1 Score:  0.6273

 Confusion Matrix:
   [[1336   92    0    0]
[ 234  167   10    1]
[ 353   24  104   12]
[  40    0   15   12]]

 —— Eval 18: Assessment period 201810 ——
 Training samples:   13,957
 Test samples:       1,081  (admin2 areas in 201810)
 Training CatBoost...

 Results  (n_test=1,081):
  Accuracy:  0.8409
  Precision: 0.8531
  Recall:    0.8409
  F1 Score:  0.8459

 Confusion Matrix:
   [[580  64   0   0]
[ 25 220  37   1]
[  3  14 100  20]
[  2   0   6   9]]

 —— Eval 19: Assessment period 201806 ——
 Training samples:   14,374
 Test samples:       2,440  (admin2 areas in 201806)
 Training CatBoost...

 Results  (n_test=2,440):
  Accuracy:  0.8107
  Precision: 0.8038
  Recall:    0.8107
  F1 Score:  0.8004

 Confusion Matrix:
   [[1279   42   10    0]
[ 210  245   62    6]
[  22   71  440   26]
[   0    0   13   14]]

 —— Eval 20: Assessment period 201802 ——
 Training samples:   14,614
 Test samples:       2,441  (admin2 areas in 201802)
 Training CatBoost...

 Results  (n_test=2,441):
  Accuracy:  0.8816
  Precision: 0.8949
  Recall:    0.8816
  F1 Score:  0.8869
```

```
 Confusion Matrix:
   [[1449   59    3    0    0]
[  64  255   38    0    0]
[  11   51  439   50    2]
[   0    3    8    9    0]
[   0    0    0    0    0]]

 ── Eval 21: Assessment period 201710 ──
 Training samples:   14,902
 Test samples:       2,577  (admin2 areas in 201710)
 Training CatBoost...

 Results  (n_test=2,577):
   Accuracy:  0.7579
   Precision: 0.8048
   Recall:    0.7579
   F1 Score:  0.7761

 Confusion Matrix:
   [[1351  292   15    0]
[  93  193   73    9]
[  25   15  380   85]
[   0    1   16   29]]

 ── Eval 22: Assessment period 201706 ──
 Training samples:   11,953
 Test samples:       2,573  (admin2 areas in 201706)
 Training CatBoost...

 Results  (n_test=2,573):
   Accuracy:  0.6316
   Precision: 0.6011
   Recall:    0.6316
   F1 Score:  0.5991

 Confusion Matrix:
   [[1274  109   76    7]
[ 283  200   14    3]
[ 275  115  112   22]
[  14    1   29   39]]

 ── Eval 23: Assessment period 201702 ──
 Training samples:   9,303
 Test samples:       2,449  (admin2 areas in 201702)
 Training CatBoost...

 Results  (n_test=2,449):
   Accuracy:  0.8424
   Precision: 0.8413
   Recall:    0.8424
   F1 Score:  0.8358

 Confusion Matrix:
   [[1664   60    0    0    0]
[ 156  249   14    5    0]
[   6  110  125   19    0]
```

```
 [   0    8    8   24    0]
 [   0    0    0    0    1]]

  ── Eval 24: Assessment period 201610 ──
  Training samples:   6,140
  Test samples:       2,748  (admin2 areas in 201610)
  Training CatBoost...

  Results  (n_test=2,748):
    Accuracy:  0.8144
    Precision: 0.8135
    Recall:    0.8144
    F1 Score:  0.8117

  Confusion Matrix:
    [[1648  145   33    1]
 [ 151  210  101    3]
 [   6   26  343   34]
 [   0    2    8   37]]

  ── Eval 25: Assessment period 201606 ──
  Training samples:   3,099
  Test samples:       2,635  (admin2 areas in 201606)
  Training CatBoost...

  Results  (n_test=2,635):
    Accuracy:  0.8148
    Precision: 0.8404
    Recall:    0.8148
    F1 Score:  0.8147

  Confusion Matrix:
    [[1657   39    0    0]
 [ 111  229   22    3]
 [  23  137  237  123]
 [   0    4   26   24]]

======================================================================
MODEL: CatBoost  |  FEATURE SET: GDELT only (no geo)
======================================================================
Using 154 features

  ── Eval 1: Assessment period 202406 ──
  Training samples:   11,437
  Test samples:       1,919  (admin2 areas in 202406)
  Training CatBoost...

  Results  (n_test=1,919):
    Accuracy:  0.3048
    Precision: 0.4438
    Recall:    0.3048
    F1 Score:  0.2563

  Confusion Matrix:
    [[389  39  46   6]
 [669 103  42  16]
```

```
[342  30  70  26]
[105   4   9  23]]

 ── Eval 2: Assessment period 202402 ──
 Training samples:   12,107
 Test samples:       2,376  (admin2 areas in 202402)
 Training CatBoost...

 Results  (n_test=2,376):
   Accuracy:  0.4112
   Precision: 0.3952
   Recall:    0.4112
   F1 Score:  0.3351

 Confusion Matrix:
   [[807  80  59  11]
[655  88  69  22]
[334  37  66  39]
[ 81   3   9  16]]

 ── Eval 3: Assessment period 202310 ──
 Training samples:   12,645
 Test samples:       1,919  (admin2 areas in 202310)
 Training CatBoost...

 Results  (n_test=1,919):
   Accuracy:  0.4940
   Precision: 0.4576
   Recall:    0.4940
   F1 Score:  0.4261

 Confusion Matrix:
   [[829  66  48  27]
[335  54  46  27]
[318  10  57  44]
[ 44   2   4   8]]

 ── Eval 4: Assessment period 202306 ──
 Training samples:   12,765
 Test samples:       2,613  (admin2 areas in 202306)
 Training CatBoost...

 Results  (n_test=2,613):
   Accuracy:  0.2813
   Precision: 0.4389
   Recall:    0.2813
   F1 Score:  0.2131

 Confusion Matrix:
   [[540  40  29  27]
[856  92  63  46]
[643  28  77  65]
[ 76   0   5  26]]

 ── Eval 5: Assessment period 202302 ──
 Training samples:   13,772
```

```
 Test samples:       1,376  (admin2 areas in 202302)
 Training CatBoost...

 Results  (n_test=1,376):
   Accuracy:  0.4375
   Precision: 0.4788
   Recall:    0.4375
   F1 Score:  0.3414

 Confusion Matrix:
   [[533  25  13   9]
[499  50  33  28]
[132  15  19  20]
[  0   0   0   0]]

 ── Eval 6: Assessment period 202210 ──
 Training samples:   13,866
 Test samples:       2,399  (admin2 areas in 202210)
 Training CatBoost...

 Results  (n_test=2,399):
   Accuracy:  0.3681
   Precision: 0.4170
   Recall:    0.3681
   F1 Score:  0.2664

 Confusion Matrix:
   [[757  27  29  29]
[686  49  49  36]
[521  22  54  52]
[ 63   0   2  23]]

 ── Eval 7: Assessment period 202206 ──
 Training samples:   13,795
 Test samples:       2,568  (admin2 areas in 202206)
 Training CatBoost...

 Results  (n_test=2,568):
   Accuracy:  0.3283
   Precision: 0.3943
   Recall:    0.3283
   F1 Score:  0.2334

 Confusion Matrix:
   [[707  27  45  27]
[787  47  56  28]
[596  15  71  42]
[ 85   7  10  18]]

 ── Eval 8: Assessment period 202202 ──
 Training samples:   13,862
 Test samples:       2,535  (admin2 areas in 202202)
 Training CatBoost...

 Results  (n_test=2,535):
   Accuracy:  0.3755
```

```
  Precision: 0.4079
  Recall:    0.3755
  F1 Score:  0.2815

 Confusion Matrix:
   [[809  38  48  22]
[769  57  61  21]
[540  30  80  26]
[ 21   3   4   6]]

 ── Eval 9: Assessment period 202110 ──
 Training samples:   12,950
 Test samples:       2,535  (admin2 areas in 202110)
 Training CatBoost...

 Results  (n_test=2,535):
  Accuracy:  0.4422
  Precision: 0.4487
  Recall:    0.4422
  F1 Score:  0.3493

 Confusion Matrix:
   [[990  51  46  31]
[717  64  55  31]
[437  17  63  20]
[  7   1   1   4]]

 ── Eval 10: Assessment period 202106 ──
 Training samples:   12,859
 Test samples:       2,536  (admin2 areas in 202106)
 Training CatBoost...

 Results  (n_test=2,536):
  Accuracy:  0.3482
  Precision: 0.4281
  Recall:    0.3482
  F1 Score:  0.2518

 Confusion Matrix:
   [[764  28  46  42]
[907  58  66  47]
[449  21  55  39]
[  7   0   1   6]]

 ── Eval 11: Assessment period 202102 ──
 Training samples:   12,839
 Test samples:       2,489  (admin2 areas in 202102)
 Training CatBoost...

 Results  (n_test=2,489):
  Accuracy:  0.4600
  Precision: 0.4345
  Recall:    0.4600
  F1 Score:  0.3775

 Confusion Matrix:
```

```
   [[1044   43   61   84]
[ 569   40   48   47]
[ 385   15   53   47]
[  43    0    2    8]]

 ── Eval 12: Assessment period 202010 ──
 Training samples:   12,858
 Test samples:       2,491  (admin2 areas in 202010)
 Training CatBoost...

 Results  (n_test=2,491):
   Accuracy:  0.4376
   Precision: 0.4474
   Recall:    0.4376
   F1 Score:  0.3448

 Confusion Matrix:
   [[994  38  46  58]
[661  42  40  71]
[431  11  52  42]
[  2   1   0   2]]

 ── Eval 13: Assessment period 202006 ──
 Training samples:   11,506
 Test samples:       2,570  (admin2 areas in 202006)
 Training CatBoost...

 Results  (n_test=2,570):
   Accuracy:  0.4284
   Precision: 0.4403
   Recall:    0.4284
   F1 Score:  0.3323

 Confusion Matrix:
   [[995  33  32  68]
[675  46  28  57]
[424  23  50  63]
[ 61   0   5  10]]

 ── Eval 14: Assessment period 202002 ──
 Training samples:   12,388
 Test samples:       1,588  (admin2 areas in 202002)
 Training CatBoost...

 Results  (n_test=1,588):
   Accuracy:  0.4081
   Precision: 0.4023
   Recall:    0.4081
   F1 Score:  0.3135

 Confusion Matrix:
   [[576  22  32  38]
[374  35  20  32]
[317  17  24  44]
[ 41   1   2  13]]
```

```
 ── Eval 15: Assessment period 201910 ──
 Training samples:   12,533
 Test samples:       2,400  (admin2 areas in 201910)
 Training CatBoost...

 Results  (n_test=2,400):
   Accuracy:  0.4671
   Precision: 0.4126
   Recall:    0.4671
   F1 Score:  0.3686

 Confusion Matrix:
   [[1043   51   26   75]
[ 466   34   16   43]
[ 465   32   23   66]
[  36    2    1   21]]

 ── Eval 16: Assessment period 201906 ──
 Training samples:   12,605
 Test samples:       2,400  (admin2 areas in 201906)
 Training CatBoost...

 Results  (n_test=2,400):
   Accuracy:  0.4637
   Precision: 0.4130
   Recall:    0.4637
   F1 Score:  0.3751

 Confusion Matrix:
   [[1014   78   28   59]
[ 449   53   21   27]
[ 464   41   30   59]
[  52    5    4   16]]

 ── Eval 17: Assessment period 201902 ──
 Training samples:   12,785
 Test samples:       2,400  (admin2 areas in 201902)
 Training CatBoost...

 Results  (n_test=2,400):
   Accuracy:  0.5471
   Precision: 0.4843
   Recall:    0.5471
   F1 Score:  0.4867

 Confusion Matrix:
   [[1208  102   66   52]
[ 314   49   19   30]
[ 378   31   45   39]
[  45    3    8   11]]

 ── Eval 18: Assessment period 201810 ──
 Training samples:   13,957
 Test samples:       1,081  (admin2 areas in 201810)
 Training CatBoost...
```

```
 Results  (n_test=1,081):
   Accuracy:  0.5476
   Precision: 0.4752
   Recall:    0.5476
   F1 Score:  0.4751

 Confusion Matrix:
   [[561  30  27  26]
[209  19  26  29]
[109   8   5  15]
[  6   1   3   7]]

 ── Eval 19: Assessment period 201806 ──
 Training samples:   14,374
 Test samples:       2,440  (admin2 areas in 201806)
 Training CatBoost...

 Results  (n_test=2,440):
   Accuracy:  0.5057
   Precision: 0.4512
   Recall:    0.5057
   F1 Score:  0.4393

 Confusion Matrix:
   [[1119  113   65   34]
[ 406   47   34   36]
[ 417   38   63   41]
[  21    1    0    5]]

 ── Eval 20: Assessment period 201802 ──
 Training samples:   14,614
 Test samples:       2,441  (admin2 areas in 201802)
 Training CatBoost...

 Results  (n_test=2,441):
   Accuracy:  0.5473
   Precision: 0.5121
   Recall:    0.5473
   F1 Score:  0.5002

 Confusion Matrix:
   [[1240  153   60   58]
[ 282   38   19   18]
[ 398   48   54   53]
[  15    1    0    4]]

 ── Eval 21: Assessment period 201710 ──
 Training samples:   14,902
 Test samples:       2,577  (admin2 areas in 201710)
 Training CatBoost...

 Results  (n_test=2,577):
   Accuracy:  0.5716
   Precision: 0.5402
   Recall:    0.5716
   F1 Score:  0.5239
```

```
 Confusion Matrix:
   [[1369  182   31   76]
[ 271   63    6   28]
[ 376   44   26   59]
[  23    4    4   15]]

 —— Eval 22: Assessment period 201706 ——
 Training samples:   11,953
 Test samples:       2,573  (admin2 areas in 201706)
 Training CatBoost...

 Results  (n_test=2,573):
   Accuracy:  0.5084
   Precision: 0.4771
   Recall:    0.5084
   F1 Score:  0.4427

 Confusion Matrix:
   [[1187  191   12   75    1]
[ 387   74    7   32    0]
[ 397   43   17   67    0]
[  48    3    2   30    0]
[   0    0    0    0    0]]

 —— Eval 23: Assessment period 201702 ——
 Training samples:   9,303
 Test samples:       2,449  (admin2 areas in 201702)
 Training CatBoost...

 Results  (n_test=2,449):
   Accuracy:  0.6513
   Precision: 0.6189
   Recall:    0.6513
   F1 Score:  0.6181

 Confusion Matrix:
   [[1464  187    9   64    0]
[ 285  104    9   26    0]
[ 187   40    9   24    0]
[  18    5    0   17    0]
[   0    0    0    0    1]]

 —— Eval 24: Assessment period 201610 ——
 Training samples:   6,140
 Test samples:       2,748  (admin2 areas in 201610)
 Training CatBoost...

 Results  (n_test=2,748):
   Accuracy:  0.6183
   Precision: 0.5625
   Recall:    0.6183
   F1 Score:  0.5653

 Confusion Matrix:
   [[1575  188   18   46]
```

```
 [ 339   96   10   20]
 [ 320   52   15   22]
 [  29    3    2   13]]

  ── Eval 25: Assessment period 201606 ──
  Training samples:   3,099
  Test samples:       2,635  (admin2 areas in 201606)
  Training CatBoost...

  Results  (n_test=2,635):
    Accuracy:  0.6118
    Precision: 0.5376
    Recall:    0.6118
    F1 Score:  0.5360

  Confusion Matrix:
    [[1538   69   40   49]
 [ 296   34   15   20]
 [ 432   23   31   34]
 [  41    3    1    9]]

======================================================================
SUMMARY: AVERAGE RESULTS BY MODEL & FEATURE SET
======================================================================

Logistic Regression — IPC only:
----------------------------------------------------------------------
  Average Metrics (across 25 assessments, 58,058 total test samples):
    Accuracy:  0.8376
    Precision: 0.8435
    Recall:    0.8376
    F1 Score:  0.8358

  Per assessment:
    202406:  n=1,919  acc=0.709  f1=0.709
    202402:  n=2,376  acc=0.791  f1=0.792
    202310:  n=1,919  acc=0.760  f1=0.768
    202306:  n=2,613  acc=0.778  f1=0.780
    202302:  n=1,376  acc=0.818  f1=0.820
    202210:  n=2,399  acc=0.782  f1=0.780
    202206:  n=2,568  acc=0.832  f1=0.824
    202202:  n=2,535  acc=0.821  f1=0.818
    202110:  n=2,535  acc=0.866  f1=0.863
    202106:  n=2,536  acc=0.797  f1=0.797
    202102:  n=2,489  acc=0.863  f1=0.861
    202010:  n=2,491  acc=0.823  f1=0.826
    202006:  n=2,570  acc=0.805  f1=0.802
    202002:  n=1,588  acc=0.849  f1=0.846
    201910:  n=2,400  acc=0.845  f1=0.837
    201906:  n=2,400  acc=0.840  f1=0.835
    201902:  n=2,400  acc=0.863  f1=0.861
    201810:  n=1,081  acc=0.887  f1=0.885
    201806:  n=2,440  acc=0.816  f1=0.804
    201802:  n=2,441  acc=0.906  f1=0.907
    201710:  n=2,577  acc=0.884  f1=0.883
    201706:  n=2,573  acc=0.777  f1=0.767
```

```
    201702:  n=2,449  acc=0.900  f1=0.894
    201610:  n=2,748  acc=0.927  f1=0.935
    201606:  n=2,635  acc=1.000  f1=1.000

Logistic Regression — IPC + GDELT base:
------------------------------------------------------------------------
  Average Metrics (across 25 assessments, 58,058 total test samples):
    Accuracy:  0.8350
    Precision: 0.8408
    Recall:    0.8350
    F1 Score:  0.8333

  Per assessment:
    202406:  n=1,919  acc=0.711  f1=0.711
    202402:  n=2,376  acc=0.791  f1=0.792
    202310:  n=1,919  acc=0.760  f1=0.769
    202306:  n=2,613  acc=0.782  f1=0.784
    202302:  n=1,376  acc=0.814  f1=0.815
    202210:  n=2,399  acc=0.779  f1=0.777
    202206:  n=2,568  acc=0.831  f1=0.824
    202202:  n=2,535  acc=0.804  f1=0.801
    202110:  n=2,535  acc=0.864  f1=0.861
    202106:  n=2,536  acc=0.795  f1=0.795
    202102:  n=2,489  acc=0.856  f1=0.855
    202010:  n=2,491  acc=0.820  f1=0.824
    202006:  n=2,570  acc=0.801  f1=0.797
    202002:  n=1,588  acc=0.844  f1=0.841
    201910:  n=2,400  acc=0.844  f1=0.837
    201906:  n=2,400  acc=0.833  f1=0.827
    201902:  n=2,400  acc=0.863  f1=0.860
    201810:  n=1,081  acc=0.884  f1=0.882
    201806:  n=2,440  acc=0.818  f1=0.806
    201802:  n=2,441  acc=0.897  f1=0.898
    201710:  n=2,577  acc=0.882  f1=0.882
    201706:  n=2,573  acc=0.780  f1=0.772
    201702:  n=2,449  acc=0.901  f1=0.896
    201610:  n=2,748  acc=0.920  f1=0.928
    201606:  n=2,635  acc=1.000  f1=1.000

Logistic Regression — IPC + GDELT + temporal:
------------------------------------------------------------------------
  Average Metrics (across 25 assessments, 58,058 total test samples):
    Accuracy:  0.8344
    Precision: 0.8394
    Recall:    0.8344
    F1 Score:  0.8328

  Per assessment:
    202406:  n=1,919  acc=0.712  f1=0.711
    202402:  n=2,376  acc=0.787  f1=0.789
    202310:  n=1,919  acc=0.765  f1=0.773
    202306:  n=2,613  acc=0.786  f1=0.788
    202302:  n=1,376  acc=0.818  f1=0.819
    202210:  n=2,399  acc=0.785  f1=0.784
    202206:  n=2,568  acc=0.807  f1=0.801
    202202:  n=2,535  acc=0.806  f1=0.803
```

```
    202110:  n=2,535  acc=0.866  f1=0.864
    202106:  n=2,536  acc=0.799  f1=0.798
    202102:  n=2,489  acc=0.855  f1=0.854
    202010:  n=2,491  acc=0.818  f1=0.821
    202006:  n=2,570  acc=0.807  f1=0.803
    202002:  n=1,588  acc=0.838  f1=0.834
    201910:  n=2,400  acc=0.848  f1=0.841
    201906:  n=2,400  acc=0.836  f1=0.830
    201902:  n=2,400  acc=0.863  f1=0.862
    201810:  n=1,081  acc=0.876  f1=0.875
    201806:  n=2,440  acc=0.814  f1=0.803
    201802:  n=2,441  acc=0.896  f1=0.897
    201710:  n=2,577  acc=0.877  f1=0.877
    201706:  n=2,573  acc=0.784  f1=0.778
    201702:  n=2,449  acc=0.902  f1=0.899
    201610:  n=2,748  acc=0.915  f1=0.915
    201606:  n=2,635  acc=1.000  f1=1.000

Logistic Regression — GDELT only:
----------------------------------------------------------------------
  Average Metrics (across 25 assessments, 58,058 total test samples):
    Accuracy:  0.8370
    Precision: 0.8403
    Recall:    0.8370
    F1 Score:  0.8345

  Per assessment:
    202406:  n=1,919  acc=0.688  f1=0.687
    202402:  n=2,376  acc=0.830  f1=0.831
    202310:  n=1,919  acc=0.776  f1=0.786
    202306:  n=2,613  acc=0.809  f1=0.808
    202302:  n=1,376  acc=0.849  f1=0.850
    202210:  n=2,399  acc=0.806  f1=0.806
    202206:  n=2,568  acc=0.820  f1=0.813
    202202:  n=2,535  acc=0.817  f1=0.815
    202110:  n=2,535  acc=0.879  f1=0.878
    202106:  n=2,536  acc=0.791  f1=0.791
    202102:  n=2,489  acc=0.858  f1=0.856
    202010:  n=2,491  acc=0.820  f1=0.823
    202006:  n=2,570  acc=0.791  f1=0.783
    202002:  n=1,588  acc=0.846  f1=0.845
    201910:  n=2,400  acc=0.853  f1=0.847
    201906:  n=2,400  acc=0.816  f1=0.804
    201902:  n=2,400  acc=0.869  f1=0.864
    201810:  n=1,081  acc=0.861  f1=0.860
    201806:  n=2,440  acc=0.800  f1=0.789
    201802:  n=2,441  acc=0.897  f1=0.898
    201710:  n=2,577  acc=0.883  f1=0.884
    201706:  n=2,573  acc=0.764  f1=0.753
    201702:  n=2,449  acc=0.905  f1=0.902
    201610:  n=2,748  acc=0.895  f1=0.891
    201606:  n=2,635  acc=1.000  f1=1.000

Logistic Regression — IPC only (no geo):
----------------------------------------------------------------------
  Average Metrics (across 25 assessments, 58,058 total test samples):
```

```
    Accuracy:  0.7448
    Precision: 0.7513
    Recall:    0.7448
    F1 Score:  0.7330

  Per assessment:
    202406:  n=1,919  acc=0.651  f1=0.648
    202402:  n=2,376  acc=0.677  f1=0.645
    202310:  n=1,919  acc=0.629  f1=0.632
    202306:  n=2,613  acc=0.465  f1=0.434
    202302:  n=1,376  acc=0.737  f1=0.737
    202210:  n=2,399  acc=0.735  f1=0.721
    202206:  n=2,568  acc=0.737  f1=0.719
    202202:  n=2,535  acc=0.774  f1=0.767
    202110:  n=2,535  acc=0.781  f1=0.783
    202106:  n=2,536  acc=0.717  f1=0.705
    202102:  n=2,489  acc=0.779  f1=0.776
    202010:  n=2,491  acc=0.735  f1=0.740
    202006:  n=2,570  acc=0.705  f1=0.687
    202002:  n=1,588  acc=0.775  f1=0.774
    201910:  n=2,400  acc=0.802  f1=0.802
    201906:  n=2,400  acc=0.770  f1=0.745
    201902:  n=2,400  acc=0.675  f1=0.625
    201810:  n=1,081  acc=0.851  f1=0.847
    201806:  n=2,440  acc=0.815  f1=0.799
    201802:  n=2,441  acc=0.899  f1=0.896
    201710:  n=2,577  acc=0.780  f1=0.784
    201706:  n=2,573  acc=0.625  f1=0.588
    201702:  n=2,449  acc=0.840  f1=0.828
    201610:  n=2,748  acc=0.813  f1=0.801
    201606:  n=2,635  acc=0.852  f1=0.841

Logistic Regression — IPC + GDELT base (no geo):
--------------------------------------------------------------------------
  Average Metrics (across 25 assessments, 58,058 total test samples):
    Accuracy:  0.7440
    Precision: 0.7583
    Recall:    0.7440
    F1 Score:  0.7332

  Per assessment:
    202406:  n=1,919  acc=0.650  f1=0.647
    202402:  n=2,376  acc=0.678  f1=0.648
    202310:  n=1,919  acc=0.629  f1=0.634
    202306:  n=2,613  acc=0.466  f1=0.435
    202302:  n=1,376  acc=0.736  f1=0.736
    202210:  n=2,399  acc=0.735  f1=0.724
    202206:  n=2,568  acc=0.738  f1=0.721
    202202:  n=2,535  acc=0.771  f1=0.764
    202110:  n=2,535  acc=0.779  f1=0.781
    202106:  n=2,536  acc=0.717  f1=0.706
    202102:  n=2,489  acc=0.777  f1=0.774
    202010:  n=2,491  acc=0.735  f1=0.740
    202006:  n=2,570  acc=0.705  f1=0.686
    202002:  n=1,588  acc=0.774  f1=0.773
    201910:  n=2,400  acc=0.802  f1=0.803
```

```
    201906:  n=2,400  acc=0.786  f1=0.773
    201902:  n=2,400  acc=0.676  f1=0.627
    201810:  n=1,081  acc=0.848  f1=0.845
    201806:  n=2,440  acc=0.810  f1=0.792
    201802:  n=2,441  acc=0.891  f1=0.888
    201710:  n=2,577  acc=0.779  f1=0.782
    201706:  n=2,573  acc=0.628  f1=0.592
    201702:  n=2,449  acc=0.836  f1=0.825
    201610:  n=2,748  acc=0.808  f1=0.799
    201606:  n=2,635  acc=0.843  f1=0.834

Logistic Regression — IPC + GDELT + temporal (no geo):
----------------------------------------------------------------------
  Average Metrics (across 25 assessments, 58,058 total test samples):
    Accuracy:  0.7410
    Precision: 0.7524
    Recall:    0.7410
    F1 Score:  0.7279

  Per assessment:
    202406:  n=1,919  acc=0.648  f1=0.645
    202402:  n=2,376  acc=0.677  f1=0.648
    202310:  n=1,919  acc=0.625  f1=0.631
    202306:  n=2,613  acc=0.467  f1=0.436
    202302:  n=1,376  acc=0.736  f1=0.736
    202210:  n=2,399  acc=0.732  f1=0.724
    202206:  n=2,568  acc=0.737  f1=0.722
    202202:  n=2,535  acc=0.770  f1=0.763
    202110:  n=2,535  acc=0.779  f1=0.781
    202106:  n=2,536  acc=0.715  f1=0.706
    202102:  n=2,489  acc=0.777  f1=0.775
    202010:  n=2,491  acc=0.735  f1=0.740
    202006:  n=2,570  acc=0.703  f1=0.683
    202002:  n=1,588  acc=0.771  f1=0.770
    201910:  n=2,400  acc=0.802  f1=0.802
    201906:  n=2,400  acc=0.785  f1=0.771
    201902:  n=2,400  acc=0.677  f1=0.628
    201810:  n=1,081  acc=0.844  f1=0.841
    201806:  n=2,440  acc=0.810  f1=0.793
    201802:  n=2,441  acc=0.831  f1=0.793
    201710:  n=2,577  acc=0.813  f1=0.776
    201706:  n=2,573  acc=0.629  f1=0.593
    201702:  n=2,449  acc=0.832  f1=0.823
    201610:  n=2,748  acc=0.802  f1=0.796
    201606:  n=2,635  acc=0.827  f1=0.821

Logistic Regression — GDELT only (no geo):
----------------------------------------------------------------------
  Average Metrics (across 25 assessments, 58,058 total test samples):
    Accuracy:  0.4977
    Precision: 0.4647
    Recall:    0.4977
    F1 Score:  0.3868

  Per assessment:
    202406:  n=1,919  acc=0.315  f1=0.272
```

```
    202402:  n=2,376  acc=0.423  f1=0.351
    202310:  n=1,919  acc=0.505  f1=0.443
    202306:  n=2,613  acc=0.412  f1=0.307
    202302:  n=1,376  acc=0.446  f1=0.347
    202210:  n=2,399  acc=0.372  f1=0.280
    202206:  n=2,568  acc=0.327  f1=0.225
    202202:  n=2,535  acc=0.374  f1=0.262
    202110:  n=2,535  acc=0.465  f1=0.356
    202106:  n=2,536  acc=0.364  f1=0.251
    202102:  n=2,489  acc=0.496  f1=0.371
    202010:  n=2,491  acc=0.463  f1=0.334
    202006:  n=2,570  acc=0.448  f1=0.311
    202002:  n=1,588  acc=0.431  f1=0.294
    201910:  n=2,400  acc=0.501  f1=0.364
    201906:  n=2,400  acc=0.500  f1=0.359
    201902:  n=2,400  acc=0.603  f1=0.477
    201810:  n=1,081  acc=0.597  f1=0.466
    201806:  n=2,440  acc=0.553  f1=0.418
    201802:  n=2,441  acc=0.631  f1=0.510
    201710:  n=2,577  acc=0.645  f1=0.531
    201706:  n=2,573  acc=0.574  f1=0.445
    201702:  n=2,449  acc=0.701  f1=0.608
    201610:  n=2,748  acc=0.666  f1=0.562
    201606:  n=2,635  acc=0.631  f1=0.526

Random Forest — IPC only:
----------------------------------------------------------------------
  Average Metrics (across 25 assessments, 58,058 total test samples):
    Accuracy:  0.7346
    Precision: 0.7738
    Recall:    0.7346
    F1 Score:  0.7399

  Per assessment:
    202406:  n=1,919  acc=0.688  f1=0.666
    202402:  n=2,376  acc=0.638  f1=0.631
    202310:  n=1,919  acc=0.546  f1=0.576
    202306:  n=2,613  acc=0.692  f1=0.692
    202302:  n=1,376  acc=0.664  f1=0.657
    202210:  n=2,399  acc=0.669  f1=0.680
    202206:  n=2,568  acc=0.736  f1=0.739
    202202:  n=2,535  acc=0.750  f1=0.753
    202110:  n=2,535  acc=0.770  f1=0.780
    202106:  n=2,536  acc=0.705  f1=0.705
    202102:  n=2,489  acc=0.789  f1=0.790
    202010:  n=2,491  acc=0.697  f1=0.702
    202006:  n=2,570  acc=0.732  f1=0.727
    202002:  n=1,588  acc=0.733  f1=0.738
    201910:  n=2,400  acc=0.793  f1=0.794
    201906:  n=2,400  acc=0.752  f1=0.749
    201902:  n=2,400  acc=0.792  f1=0.800
    201810:  n=1,081  acc=0.749  f1=0.769
    201806:  n=2,440  acc=0.729  f1=0.745
    201802:  n=2,441  acc=0.699  f1=0.743
    201710:  n=2,577  acc=0.790  f1=0.815
    201706:  n=2,573  acc=0.717  f1=0.715
```

```
    201702:  n=2,449  acc=0.822  f1=0.816
    201610:  n=2,748  acc=0.830  f1=0.832
    201606:  n=2,635  acc=0.883  f1=0.883

Random Forest — IPC + GDELT base:
----------------------------------------------------------------------
  Average Metrics (across 25 assessments, 58,058 total test samples):
    Accuracy:  0.7406
    Precision: 0.7648
    Recall:    0.7406
    F1 Score:  0.7386

  Per assessment:
    202406:  n=1,919  acc=0.675  f1=0.658
    202402:  n=2,376  acc=0.672  f1=0.658
    202310:  n=1,919  acc=0.618  f1=0.639
    202306:  n=2,613  acc=0.698  f1=0.705
    202302:  n=1,376  acc=0.725  f1=0.722
    202210:  n=2,399  acc=0.688  f1=0.697
    202206:  n=2,568  acc=0.718  f1=0.713
    202202:  n=2,535  acc=0.761  f1=0.757
    202110:  n=2,535  acc=0.749  f1=0.744
    202106:  n=2,536  acc=0.684  f1=0.681
    202102:  n=2,489  acc=0.783  f1=0.774
    202010:  n=2,491  acc=0.690  f1=0.686
    202006:  n=2,570  acc=0.702  f1=0.687
    202002:  n=1,588  acc=0.721  f1=0.722
    201910:  n=2,400  acc=0.747  f1=0.739
    201906:  n=2,400  acc=0.745  f1=0.736
    201902:  n=2,400  acc=0.806  f1=0.802
    201810:  n=1,081  acc=0.765  f1=0.780
    201806:  n=2,440  acc=0.743  f1=0.749
    201802:  n=2,441  acc=0.833  f1=0.848
    201710:  n=2,577  acc=0.811  f1=0.821
    201706:  n=2,573  acc=0.699  f1=0.688
    201702:  n=2,449  acc=0.812  f1=0.808
    201610:  n=2,748  acc=0.825  f1=0.819
    201606:  n=2,635  acc=0.843  f1=0.829

Random Forest — IPC + GDELT + temporal:
----------------------------------------------------------------------
  Average Metrics (across 25 assessments, 58,058 total test samples):
    Accuracy:  0.7319
    Precision: 0.7582
    Recall:    0.7319
    F1 Score:  0.7315

  Per assessment:
    202406:  n=1,919  acc=0.669  f1=0.649
    202402:  n=2,376  acc=0.648  f1=0.637
    202310:  n=1,919  acc=0.573  f1=0.599
    202306:  n=2,613  acc=0.633  f1=0.639
    202302:  n=1,376  acc=0.781  f1=0.783
    202210:  n=2,399  acc=0.680  f1=0.691
    202206:  n=2,568  acc=0.743  f1=0.742
    202202:  n=2,535  acc=0.735  f1=0.731
```

```
    202110:  n=2,535  acc=0.763  f1=0.763
    202106:  n=2,536  acc=0.668  f1=0.665
    202102:  n=2,489  acc=0.764  f1=0.751
    202010:  n=2,491  acc=0.713  f1=0.717
    202006:  n=2,570  acc=0.707  f1=0.693
    202002:  n=1,588  acc=0.727  f1=0.728
    201910:  n=2,400  acc=0.722  f1=0.715
    201906:  n=2,400  acc=0.740  f1=0.728
    201902:  n=2,400  acc=0.770  f1=0.776
    201810:  n=1,081  acc=0.749  f1=0.764
    201806:  n=2,440  acc=0.750  f1=0.750
    201802:  n=2,441  acc=0.814  f1=0.830
    201710:  n=2,577  acc=0.799  f1=0.810
    201706:  n=2,573  acc=0.684  f1=0.680
    201702:  n=2,449  acc=0.814  f1=0.812
    201610:  n=2,748  acc=0.806  f1=0.802
    201606:  n=2,635  acc=0.845  f1=0.834

Random Forest — GDELT only:
----------------------------------------------------------------------
  Average Metrics (across 25 assessments, 58,058 total test samples):
    Accuracy:  0.6773
    Precision: 0.7086
    Recall:    0.6773
    F1 Score:  0.6742

  Per assessment:
    202406:  n=1,919  acc=0.641  f1=0.610
    202402:  n=2,376  acc=0.599  f1=0.591
    202310:  n=1,919  acc=0.524  f1=0.540
    202306:  n=2,613  acc=0.667  f1=0.674
    202302:  n=1,376  acc=0.576  f1=0.546
    202210:  n=2,399  acc=0.632  f1=0.638
    202206:  n=2,568  acc=0.673  f1=0.666
    202202:  n=2,535  acc=0.668  f1=0.665
    202110:  n=2,535  acc=0.723  f1=0.721
    202106:  n=2,536  acc=0.632  f1=0.634
    202102:  n=2,489  acc=0.711  f1=0.703
    202010:  n=2,491  acc=0.621  f1=0.614
    202006:  n=2,570  acc=0.678  f1=0.673
    202002:  n=1,588  acc=0.674  f1=0.684
    201910:  n=2,400  acc=0.685  f1=0.681
    201906:  n=2,400  acc=0.685  f1=0.678
    201902:  n=2,400  acc=0.754  f1=0.761
    201810:  n=1,081  acc=0.665  f1=0.673
    201806:  n=2,440  acc=0.670  f1=0.680
    201802:  n=2,441  acc=0.746  f1=0.764
    201710:  n=2,577  acc=0.765  f1=0.775
    201706:  n=2,573  acc=0.645  f1=0.626
    201702:  n=2,449  acc=0.726  f1=0.706
    201610:  n=2,748  acc=0.783  f1=0.777
    201606:  n=2,635  acc=0.786  f1=0.775

Random Forest — IPC only (no geo):
----------------------------------------------------------------------
  Average Metrics (across 25 assessments, 58,058 total test samples):
```

```
    Accuracy:  0.7390
    Precision: 0.7564
    Recall:    0.7390
    F1 Score:  0.7313

  Per assessment:
    202406:  n=1,919  acc=0.651  f1=0.648
    202402:  n=2,376  acc=0.694  f1=0.677
    202310:  n=1,919  acc=0.616  f1=0.635
    202306:  n=2,613  acc=0.465  f1=0.434
    202302:  n=1,376  acc=0.737  f1=0.737
    202210:  n=2,399  acc=0.725  f1=0.728
    202206:  n=2,568  acc=0.749  f1=0.743
    202202:  n=2,535  acc=0.779  f1=0.776
    202110:  n=2,535  acc=0.781  f1=0.785
    202106:  n=2,536  acc=0.701  f1=0.697
    202102:  n=2,489  acc=0.779  f1=0.776
    202010:  n=2,491  acc=0.735  f1=0.740
    202006:  n=2,570  acc=0.705  f1=0.687
    202002:  n=1,588  acc=0.775  f1=0.774
    201910:  n=2,400  acc=0.802  f1=0.802
    201906:  n=2,400  acc=0.789  f1=0.776
    201902:  n=2,400  acc=0.676  f1=0.629
    201810:  n=1,081  acc=0.848  f1=0.851
    201806:  n=2,440  acc=0.812  f1=0.799
    201802:  n=2,441  acc=0.884  f1=0.889
    201710:  n=2,577  acc=0.757  f1=0.775
    201706:  n=2,573  acc=0.621  f1=0.586
    201702:  n=2,449  acc=0.838  f1=0.830
    201610:  n=2,748  acc=0.817  f1=0.812
    201606:  n=2,635  acc=0.735  f1=0.697

Random Forest — IPC + GDELT base (no geo):
--------------------------------------------------------------------------
  Average Metrics (across 25 assessments, 58,058 total test samples):
    Accuracy:  0.7394
    Precision: 0.7596
    Recall:    0.7394
    F1 Score:  0.7339

  Per assessment:
    202406:  n=1,919  acc=0.645  f1=0.642
    202402:  n=2,376  acc=0.688  f1=0.673
    202310:  n=1,919  acc=0.604  f1=0.622
    202306:  n=2,613  acc=0.468  f1=0.441
    202302:  n=1,376  acc=0.733  f1=0.734
    202210:  n=2,399  acc=0.721  f1=0.725
    202206:  n=2,568  acc=0.745  f1=0.739
    202202:  n=2,535  acc=0.775  f1=0.773
    202110:  n=2,535  acc=0.779  f1=0.783
    202106:  n=2,536  acc=0.698  f1=0.694
    202102:  n=2,489  acc=0.773  f1=0.771
    202010:  n=2,491  acc=0.730  f1=0.735
    202006:  n=2,570  acc=0.702  f1=0.683
    202002:  n=1,588  acc=0.772  f1=0.771
    201910:  n=2,400  acc=0.797  f1=0.797
```

```
    201906:  n=2,400  acc=0.782  f1=0.770
    201902:  n=2,400  acc=0.675  f1=0.628
    201810:  n=1,081  acc=0.841  f1=0.844
    201806:  n=2,440  acc=0.812  f1=0.800
    201802:  n=2,441  acc=0.888  f1=0.890
    201710:  n=2,577  acc=0.760  f1=0.775
    201706:  n=2,573  acc=0.630  f1=0.600
    201702:  n=2,449  acc=0.839  f1=0.834
    201610:  n=2,748  acc=0.813  f1=0.810
    201606:  n=2,635  acc=0.816  f1=0.816

Random Forest — IPC + GDELT + temporal (no geo):
----------------------------------------------------------------------
  Average Metrics (across 25 assessments, 58,058 total test samples):
    Accuracy:  0.7290
    Precision: 0.7488
    Recall:    0.7290
    F1 Score:  0.7233

  Per assessment:
    202406:  n=1,919  acc=0.647  f1=0.645
    202402:  n=2,376  acc=0.676  f1=0.662
    202310:  n=1,919  acc=0.605  f1=0.623
    202306:  n=2,613  acc=0.478  f1=0.451
    202302:  n=1,376  acc=0.729  f1=0.731
    202210:  n=2,399  acc=0.718  f1=0.721
    202206:  n=2,568  acc=0.735  f1=0.729
    202202:  n=2,535  acc=0.766  f1=0.764
    202110:  n=2,535  acc=0.772  f1=0.775
    202106:  n=2,536  acc=0.699  f1=0.697
    202102:  n=2,489  acc=0.766  f1=0.767
    202010:  n=2,491  acc=0.721  f1=0.729
    202006:  n=2,570  acc=0.694  f1=0.677
    202002:  n=1,588  acc=0.754  f1=0.756
    201910:  n=2,400  acc=0.782  f1=0.784
    201906:  n=2,400  acc=0.776  f1=0.764
    201902:  n=2,400  acc=0.676  f1=0.632
    201810:  n=1,081  acc=0.836  f1=0.841
    201806:  n=2,440  acc=0.802  f1=0.793
    201802:  n=2,441  acc=0.875  f1=0.881
    201710:  n=2,577  acc=0.717  f1=0.715
    201706:  n=2,573  acc=0.611  f1=0.577
    201702:  n=2,449  acc=0.826  f1=0.821
    201610:  n=2,748  acc=0.751  f1=0.738
    201606:  n=2,635  acc=0.811  f1=0.811

Random Forest — GDELT only (no geo):
----------------------------------------------------------------------
  Average Metrics (across 25 assessments, 58,058 total test samples):
    Accuracy:  0.4733
    Precision: 0.4619
    Recall:    0.4733
    F1 Score:  0.3969

  Per assessment:
    202406:  n=1,919  acc=0.301  f1=0.253
```

```
    202402:  n=2,376  acc=0.415  f1=0.348
    202310:  n=1,919  acc=0.497  f1=0.435
    202306:  n=2,613  acc=0.304  f1=0.250
    202302:  n=1,376  acc=0.439  f1=0.347
    202210:  n=2,399  acc=0.375  f1=0.279
    202206:  n=2,568  acc=0.332  f1=0.237
    202202:  n=2,535  acc=0.384  f1=0.291
    202110:  n=2,535  acc=0.448  f1=0.355
    202106:  n=2,536  acc=0.348  f1=0.253
    202102:  n=2,489  acc=0.462  f1=0.383
    202010:  n=2,491  acc=0.440  f1=0.342
    202006:  n=2,570  acc=0.422  f1=0.326
    202002:  n=1,588  acc=0.423  f1=0.334
    201910:  n=2,400  acc=0.485  f1=0.390
    201906:  n=2,400  acc=0.478  f1=0.383
    201902:  n=2,400  acc=0.578  f1=0.505
    201810:  n=1,081  acc=0.576  f1=0.497
    201806:  n=2,440  acc=0.523  f1=0.449
    201802:  n=2,441  acc=0.576  f1=0.522
    201710:  n=2,577  acc=0.591  f1=0.539
    201706:  n=2,573  acc=0.531  f1=0.466
    201702:  n=2,449  acc=0.653  f1=0.620
    201610:  n=2,748  acc=0.632  f1=0.575
    201606:  n=2,635  acc=0.621  f1=0.541

CatBoost — IPC only:
----------------------------------------------------------------------
  Average Metrics (across 25 assessments, 58,058 total test samples):
    Accuracy:  0.7398
    Precision: 0.7762
    Recall:    0.7398
    F1 Score:  0.7385

  Per assessment:
    202406:  n=1,919  acc=0.637  f1=0.628
    202402:  n=2,376  acc=0.686  f1=0.669
    202310:  n=1,919  acc=0.588  f1=0.620
    202306:  n=2,613  acc=0.465  f1=0.434
    202302:  n=1,376  acc=0.737  f1=0.737
    202210:  n=2,399  acc=0.717  f1=0.723
    202206:  n=2,568  acc=0.742  f1=0.743
    202202:  n=2,535  acc=0.759  f1=0.759
    202110:  n=2,535  acc=0.766  f1=0.775
    202106:  n=2,536  acc=0.737  f1=0.733
    202102:  n=2,489  acc=0.799  f1=0.802
    202010:  n=2,491  acc=0.737  f1=0.741
    202006:  n=2,570  acc=0.702  f1=0.682
    202002:  n=1,588  acc=0.755  f1=0.754
    201910:  n=2,400  acc=0.787  f1=0.790
    201906:  n=2,400  acc=0.777  f1=0.764
    201902:  n=2,400  acc=0.670  f1=0.632
    201810:  n=1,081  acc=0.804  f1=0.824
    201806:  n=2,440  acc=0.800  f1=0.806
    201802:  n=2,441  acc=0.853  f1=0.870
    201710:  n=2,577  acc=0.760  f1=0.780
    201706:  n=2,573  acc=0.651  f1=0.637
```

```
    201702:  n=2,449  acc=0.847  f1=0.840
    201610:  n=2,748  acc=0.818  f1=0.815
    201606:  n=2,635  acc=0.902  f1=0.903

CatBoost — IPC + GDELT base:
------------------------------------------------------------------------
  Average Metrics (across 25 assessments, 58,058 total test samples):
    Accuracy:  0.7398
    Precision: 0.7748
    Recall:    0.7398
    F1 Score:  0.7375

  Per assessment:
    202406:  n=1,919  acc=0.638  f1=0.629
    202402:  n=2,376  acc=0.672  f1=0.654
    202310:  n=1,919  acc=0.576  f1=0.604
    202306:  n=2,613  acc=0.470  f1=0.439
    202302:  n=1,376  acc=0.738  f1=0.738
    202210:  n=2,399  acc=0.719  f1=0.725
    202206:  n=2,568  acc=0.742  f1=0.743
    202202:  n=2,535  acc=0.760  f1=0.761
    202110:  n=2,535  acc=0.765  f1=0.774
    202106:  n=2,536  acc=0.737  f1=0.732
    202102:  n=2,489  acc=0.799  f1=0.802
    202010:  n=2,491  acc=0.736  f1=0.741
    202006:  n=2,570  acc=0.701  f1=0.680
    202002:  n=1,588  acc=0.755  f1=0.754
    201910:  n=2,400  acc=0.787  f1=0.790
    201906:  n=2,400  acc=0.778  f1=0.765
    201902:  n=2,400  acc=0.674  f1=0.624
    201810:  n=1,081  acc=0.805  f1=0.824
    201806:  n=2,440  acc=0.798  f1=0.803
    201802:  n=2,441  acc=0.861  f1=0.877
    201710:  n=2,577  acc=0.766  f1=0.784
    201706:  n=2,573  acc=0.648  f1=0.633
    201702:  n=2,449  acc=0.845  f1=0.838
    201610:  n=2,748  acc=0.819  f1=0.815
    201606:  n=2,635  acc=0.905  f1=0.906

CatBoost — IPC + GDELT + temporal:
------------------------------------------------------------------------
  Average Metrics (across 25 assessments, 58,058 total test samples):
    Accuracy:  0.7401
    Precision: 0.7746
    Recall:    0.7401
    F1 Score:  0.7377

  Per assessment:
    202406:  n=1,919  acc=0.641  f1=0.633
    202402:  n=2,376  acc=0.673  f1=0.656
    202310:  n=1,919  acc=0.569  f1=0.598
    202306:  n=2,613  acc=0.469  f1=0.438
    202302:  n=1,376  acc=0.737  f1=0.737
    202210:  n=2,399  acc=0.722  f1=0.728
    202206:  n=2,568  acc=0.748  f1=0.748
    202202:  n=2,535  acc=0.760  f1=0.760
```

```
    202110:  n=2,535  acc=0.765  f1=0.774
    202106:  n=2,536  acc=0.735  f1=0.731
    202102:  n=2,489  acc=0.799  f1=0.803
    202010:  n=2,491  acc=0.737  f1=0.741
    202006:  n=2,570  acc=0.701  f1=0.681
    202002:  n=1,588  acc=0.763  f1=0.760
    201910:  n=2,400  acc=0.792  f1=0.794
    201906:  n=2,400  acc=0.778  f1=0.765
    201902:  n=2,400  acc=0.680  f1=0.635
    201810:  n=1,081  acc=0.804  f1=0.822
    201806:  n=2,440  acc=0.789  f1=0.790
    201802:  n=2,441  acc=0.863  f1=0.878
    201710:  n=2,577  acc=0.761  f1=0.780
    201706:  n=2,573  acc=0.649  f1=0.633
    201702:  n=2,449  acc=0.845  f1=0.838
    201610:  n=2,748  acc=0.820  f1=0.816
    201606:  n=2,635  acc=0.902  f1=0.903

CatBoost — GDELT only:
----------------------------------------------------------------------
  Average Metrics (across 25 assessments, 58,058 total test samples):
    Accuracy:  0.6860
    Precision: 0.7385
    Recall:    0.6860
    F1 Score:  0.6875

  Per assessment:
    202406:  n=1,919  acc=0.639  f1=0.614
    202402:  n=2,376  acc=0.607  f1=0.601
    202310:  n=1,919  acc=0.543  f1=0.559
    202306:  n=2,613  acc=0.673  f1=0.681
    202302:  n=1,376  acc=0.597  f1=0.561
    202210:  n=2,399  acc=0.641  f1=0.646
    202206:  n=2,568  acc=0.656  f1=0.649
    202202:  n=2,535  acc=0.665  f1=0.660
    202110:  n=2,535  acc=0.715  f1=0.709
    202106:  n=2,536  acc=0.651  f1=0.648
    202102:  n=2,489  acc=0.755  f1=0.746
    202010:  n=2,491  acc=0.669  f1=0.662
    202006:  n=2,570  acc=0.701  f1=0.695
    202002:  n=1,588  acc=0.642  f1=0.650
    201910:  n=2,400  acc=0.715  f1=0.709
    201906:  n=2,400  acc=0.705  f1=0.711
    201902:  n=2,400  acc=0.759  f1=0.773
    201810:  n=1,081  acc=0.671  f1=0.691
    201806:  n=2,440  acc=0.683  f1=0.699
    201802:  n=2,441  acc=0.714  f1=0.750
    201710:  n=2,577  acc=0.750  f1=0.773
    201706:  n=2,573  acc=0.676  f1=0.661
    201702:  n=2,449  acc=0.690  f1=0.699
    201610:  n=2,748  acc=0.791  f1=0.797
    201606:  n=2,635  acc=0.841  f1=0.842

CatBoost — IPC only (no geo):
----------------------------------------------------------------------
  Average Metrics (across 25 assessments, 58,058 total test samples):
```

```
    Accuracy:  0.7390
    Precision: 0.7564
    Recall:    0.7390
    F1 Score:  0.7314

  Per assessment:
    202406:  n=1,919  acc=0.651  f1=0.648
    202402:  n=2,376  acc=0.694  f1=0.677
    202310:  n=1,919  acc=0.616  f1=0.635
    202306:  n=2,613  acc=0.465  f1=0.434
    202302:  n=1,376  acc=0.737  f1=0.737
    202210:  n=2,399  acc=0.725  f1=0.728
    202206:  n=2,568  acc=0.749  f1=0.743
    202202:  n=2,535  acc=0.779  f1=0.776
    202110:  n=2,535  acc=0.781  f1=0.785
    202106:  n=2,536  acc=0.701  f1=0.697
    202102:  n=2,489  acc=0.779  f1=0.776
    202010:  n=2,491  acc=0.735  f1=0.740
    202006:  n=2,570  acc=0.705  f1=0.687
    202002:  n=1,588  acc=0.775  f1=0.774
    201910:  n=2,400  acc=0.802  f1=0.802
    201906:  n=2,400  acc=0.789  f1=0.776
    201902:  n=2,400  acc=0.676  f1=0.629
    201810:  n=1,081  acc=0.848  f1=0.851
    201806:  n=2,440  acc=0.812  f1=0.799
    201802:  n=2,441  acc=0.884  f1=0.889
    201710:  n=2,577  acc=0.757  f1=0.775
    201706:  n=2,573  acc=0.621  f1=0.587
    201702:  n=2,449  acc=0.838  f1=0.830
    201610:  n=2,748  acc=0.817  f1=0.812
    201606:  n=2,635  acc=0.735  f1=0.697

CatBoost — IPC + GDELT base (no geo):
--------------------------------------------------------------------------
  Average Metrics (across 25 assessments, 58,058 total test samples):
    Accuracy:  0.7392
    Precision: 0.7621
    Recall:    0.7392
    F1 Score:  0.7322

  Per assessment:
    202406:  n=1,919  acc=0.648  f1=0.644
    202402:  n=2,376  acc=0.692  f1=0.675
    202310:  n=1,919  acc=0.615  f1=0.634
    202306:  n=2,613  acc=0.464  f1=0.433
    202302:  n=1,376  acc=0.736  f1=0.737
    202210:  n=2,399  acc=0.722  f1=0.726
    202206:  n=2,568  acc=0.746  f1=0.741
    202202:  n=2,535  acc=0.778  f1=0.776
    202110:  n=2,535  acc=0.782  f1=0.786
    202106:  n=2,536  acc=0.703  f1=0.698
    202102:  n=2,489  acc=0.779  f1=0.776
    202010:  n=2,491  acc=0.735  f1=0.740
    202006:  n=2,570  acc=0.705  f1=0.687
    202002:  n=1,588  acc=0.774  f1=0.773
    201910:  n=2,400  acc=0.800  f1=0.800
```

```
    201906:  n=2,400  acc=0.788  f1=0.776
    201902:  n=2,400  acc=0.675  f1=0.628
    201810:  n=1,081  acc=0.847  f1=0.851
    201806:  n=2,440  acc=0.813  f1=0.801
    201802:  n=2,441  acc=0.884  f1=0.888
    201710:  n=2,577  acc=0.763  f1=0.779
    201706:  n=2,573  acc=0.630  f1=0.598
    201702:  n=2,449  acc=0.844  f1=0.837
    201610:  n=2,748  acc=0.814  f1=0.810
    201606:  n=2,635  acc=0.740  f1=0.711

CatBoost — IPC + GDELT + temporal (no geo):
----------------------------------------------------------------------
  Average Metrics (across 25 assessments, 58,058 total test samples):
    Accuracy:  0.7405
    Precision: 0.7633
    Recall:    0.7405
    F1 Score:  0.7352

  Per assessment:
    202406:  n=1,919  acc=0.648  f1=0.644
    202402:  n=2,376  acc=0.692  f1=0.675
    202310:  n=1,919  acc=0.608  f1=0.629
    202306:  n=2,613  acc=0.465  f1=0.434
    202302:  n=1,376  acc=0.734  f1=0.736
    202210:  n=2,399  acc=0.722  f1=0.726
    202206:  n=2,568  acc=0.744  f1=0.739
    202202:  n=2,535  acc=0.774  f1=0.772
    202110:  n=2,535  acc=0.781  f1=0.785
    202106:  n=2,536  acc=0.702  f1=0.697
    202102:  n=2,489  acc=0.779  f1=0.777
    202010:  n=2,491  acc=0.735  f1=0.740
    202006:  n=2,570  acc=0.706  f1=0.687
    202002:  n=1,588  acc=0.770  f1=0.769
    201910:  n=2,400  acc=0.797  f1=0.798
    201906:  n=2,400  acc=0.788  f1=0.775
    201902:  n=2,400  acc=0.675  f1=0.627
    201810:  n=1,081  acc=0.841  f1=0.846
    201806:  n=2,440  acc=0.811  f1=0.800
    201802:  n=2,441  acc=0.882  f1=0.887
    201710:  n=2,577  acc=0.758  f1=0.776
    201706:  n=2,573  acc=0.632  f1=0.599
    201702:  n=2,449  acc=0.842  f1=0.836
    201610:  n=2,748  acc=0.814  f1=0.812
    201606:  n=2,635  acc=0.815  f1=0.815

CatBoost — GDELT only (no geo):
----------------------------------------------------------------------
  Average Metrics (across 25 assessments, 58,058 total test samples):
    Accuracy:  0.4626
    Precision: 0.4608
    Recall:    0.4626
    F1 Score:  0.3861

  Per assessment:
    202406:  n=1,919  acc=0.305  f1=0.256
```

```
    202402:  n=2,376  acc=0.411  f1=0.335
    202310:  n=1,919  acc=0.494  f1=0.426
    202306:  n=2,613  acc=0.281  f1=0.213
    202302:  n=1,376  acc=0.438  f1=0.341
    202210:  n=2,399  acc=0.368  f1=0.266
    202206:  n=2,568  acc=0.328  f1=0.233
    202202:  n=2,535  acc=0.376  f1=0.282
    202110:  n=2,535  acc=0.442  f1=0.349
    202106:  n=2,536  acc=0.348  f1=0.252
    202102:  n=2,489  acc=0.460  f1=0.377
    202010:  n=2,491  acc=0.438  f1=0.345
    202006:  n=2,570  acc=0.428  f1=0.332
    202002:  n=1,588  acc=0.408  f1=0.314
    201910:  n=2,400  acc=0.467  f1=0.369
    201906:  n=2,400  acc=0.464  f1=0.375
    201902:  n=2,400  acc=0.547  f1=0.487
    201810:  n=1,081  acc=0.548  f1=0.475
    201806:  n=2,440  acc=0.506  f1=0.439
    201802:  n=2,441  acc=0.547  f1=0.500
    201710:  n=2,577  acc=0.572  f1=0.524
    201706:  n=2,573  acc=0.508  f1=0.443
    201702:  n=2,449  acc=0.651  f1=0.618
    201610:  n=2,748  acc=0.618  f1=0.565
    201606:  n=2,635  acc=0.612  f1=0.536

======================================================================
SUMMARY TABLE
======================================================================
              Model                      Feature Set  N Features  IPC  GDELT
Base  GDELT Temporal  Geographic  Accuracy  Precision   Recall  F1 Score  To
tal Test  N Evals
Logistic Regression                        IPC only        3332    1
0               0        3331  0.837650   0.843496 0.837650  0.835777
58058       25
Logistic Regression                      GDELT only        3485    0
119              35        3331  0.836975   0.840327 0.836975  0.834485
58058       25
Logistic Regression                IPC + GDELT base        3451    1
119               0        3331  0.834990   0.840817 0.834990  0.833342
58058       25
Logistic Regression          IPC + GDELT + temporal        3486    1
119              35        3331  0.834383   0.839443 0.834383  0.832770
58058       25
      Random Forest                        IPC only        3332    1
0               0        3331  0.734557   0.773789 0.734557  0.739921
58058       25
      Random Forest                IPC + GDELT base        3451    1
119               0        3331  0.740582   0.764767 0.740582  0.738552
58058       25
           CatBoost                        IPC only        3332    1
0               0        3331  0.739843   0.776193 0.739843  0.738462
58058       25
           CatBoost          IPC + GDELT + temporal        3486    1
119              35        3331  0.740081   0.774619 0.740081  0.737743
58058       25
           CatBoost                IPC + GDELT base        3451    1
```

```
119               0        3331  0.739786   0.774793 0.739786  0.737467
58058       25
          CatBoost IPC + GDELT + temporal (no geo)         155    1
119              35           0  0.740505   0.763336 0.740505  0.735217
58058       25
      Random Forest       IPC + GDELT base (no geo)         120    1
119               0           0  0.739448   0.759563 0.739448  0.733875
58058       25
Logistic Regression       IPC + GDELT base (no geo)         120    1
119               0           0  0.743953   0.758250 0.743953  0.733167
58058       25
Logistic Regression               IPC only (no geo)           1    1
0                 0           0  0.744840   0.751333 0.744840  0.732967
58058       25
          CatBoost       IPC + GDELT base (no geo)         120    1
119               0           0  0.739158   0.762085 0.739158  0.732221
58058       25
      Random Forest          IPC + GDELT + temporal        3486    1
119              35        3331  0.731875   0.758174 0.731875  0.731485
58058       25
          CatBoost               IPC only (no geo)           1    1
0                 0           0  0.739011   0.756428 0.739011  0.731352
58058       25
      Random Forest               IPC only (no geo)           1    1
0                 0           0  0.738995   0.756411 0.738995  0.731334
58058       25
Logistic Regression IPC + GDELT + temporal (no geo)         155    1
119              35           0  0.740967   0.752363 0.740967  0.727868
58058       25
      Random Forest IPC + GDELT + temporal (no geo)         155    1
119              35           0  0.728951   0.748761 0.728951  0.723251
58058       25
          CatBoost                       GDELT only        3485    0
119              35        3331  0.686005   0.738486 0.686005  0.687458
58058       25
      Random Forest                      GDELT only        3485    0
119              35        3331  0.677264   0.708609 0.677264  0.674160
58058       25
      Random Forest             GDELT only (no geo)         154    0
119              35           0  0.473324   0.461901 0.473324  0.396861
58058       25
Logistic Regression             GDELT only (no geo)         154    0
119              35           0  0.497700   0.464699 0.497700  0.386839
58058       25
          CatBoost             GDELT only (no geo)         154    0
119              35           0  0.462602   0.460773 0.462602  0.386147
58058       25

======================================================================
RANDOM FOREST: AVERAGE FEATURE IMPORTANCES
======================================================================

── IPC only (averaged over 25 splits) ──
Feature                                                 Importance
----------------------------------------------------------------------
  IPC_last                                                  0.1277
```

ADMIN0_yemen	0.0730
ADMIN0_south sudan	0.0564
ADMIN0_nigeria	0.0520
ADMIN1_hajjah	0.0337
ADMIN1_borno	0.0320
ADMIN1_unity	0.0290
ADMIN0_somalia	0.0258
ADMIN0_kenya	0.0222
ADMIN2_leer	0.0205
ADMIN0_sudan	0.0190
ADMIN0_uganda	0.0175
ADMIN0_ethiopia	0.0175
ADMIN1_sadah	0.0162
ADMIN1_taizz	0.0152
ADMIN0_mozambique	0.0133
ADMIN1_tigray	0.0132
ADMIN1_jonglei	0.0128
ADMIN0_zimbabwe	0.0123
ADMIN1_hadramaut	0.0118
ADMIN1_al hudaydah	0.0116
ADMIN0_burundi	0.0103
ADMIN1_upper nile	0.0091
ADMIN1_amran	0.0087
ADMIN1_lahij	0.0081
... (3307 more features omitted)	

── IPC + GDELT base (averaged over 25 splits) ──

Feature	Importance
IPC_last	0.1115
ADMIN0_yemen	0.0565
ADMIN0_south sudan	0.0405
ADMIN0_nigeria	0.0320
ADMIN1_hajjah	0.0270
ADMIN1_borno	0.0186
ADMIN0_somalia	0.0171
ADMIN0_sudan	0.0165
ADMIN1_sadah	0.0146
ADMIN0_kenya	0.0134
ADMIN1_taizz	0.0102
ADMIN0_uganda	0.0099
ADMIN0_ethiopia	0.0097
ADMIN1_upper nile	0.0081
ADMIN1_tigray	0.0071
ADMIN1_jonglei	0.0070
ADMIN0_burundi	0.0068
ADMIN0_mozambique	0.0067
ADMIN0_zimbabwe	0.0066
evt_sentiment.neu_sum	0.0066
evt_compound_score_sum	0.0065
ADMIN1_hadramaut	0.0063
ADMIN1_al hudaydah	0.0062
evt_displaced_count_mean	0.0061
evt_pos_score_sum	0.0061
... (3426 more features omitted)	

── IPC + GDELT + temporal (averaged over 25 splits) ──

Feature	Importance
IPC_last	0.1165
ADMIN0_yemen	0.0508
ADMIN0_nigeria	0.0360
ADMIN0_south sudan	0.0302
ADMIN1_hajjah	0.0257
ADMIN1_borno	0.0158
ADMIN0_somalia	0.0127
ADMIN0_sudan	0.0124
ADMIN0_kenya	0.0120
evt_sentiment.compound_mean_rolling_6	0.0117
evt_compound_score_mean_rolling_6	0.0114
evt_displaced_freq_mean_rolling_6	0.0109
evt_food_insecurity_freq_mean_rolling_6	0.0100
ADMIN0_uganda	0.0097
ADMIN1_sadah	0.0094
ADMIN1_taizz	0.0087
evt_compound_score_mean_rolling_3	0.0085
evt_sentiment.compound_mean_rolling_3	0.0079
evt_displaced_freq_mean_rolling_3	0.0078
ADMIN1_upper nile	0.0072
ADMIN0_ethiopia	0.0072
evt_food_insecurity_freq_mean_rolling_3	0.0071
ADMIN1_unity	0.0070
evt_fatalities_freq_mean_rolling_6	0.0069
evt_sentiment.pos_sum	0.0063
... (3461 more features omitted)	

── GDELT only (averaged over 25 splits) ──

Feature	Importance
ADMIN0_yemen	0.0541
ADMIN0_south sudan	0.0348
ADMIN0_nigeria	0.0345
ADMIN1_hajjah	0.0280
ADMIN0_somalia	0.0174
ADMIN1_borno	0.0162
evt_sentiment.compound_mean_rolling_6	0.0142
ADMIN0_sudan	0.0139
ADMIN0_kenya	0.0138
evt_displaced_freq_mean_rolling_6	0.0135
evt_compound_score_mean_rolling_6	0.0133
evt_food_insecurity_freq_mean_rolling_6	0.0125
ADMIN1_sadah	0.0121
ADMIN1_taizz	0.0107
ADMIN0_uganda	0.0105
evt_compound_score_mean_rolling_3	0.0093
ADMIN0_ethiopia	0.0090
evt_displaced_freq_mean_rolling_3	0.0088
evt_sentiment.compound_mean_rolling_3	0.0087
evt_food_insecurity_freq_mean_rolling_3	0.0079
evt_fatalities_freq_mean_rolling_6	0.0078
ADMIN1_upper nile	0.0077
ADMIN1_unity	0.0076

```
  ADMIN1_jonglei                                                0.0075
  ADMIN0_burundi                                                0.0072
  ... (3460 more features omitted)

── IPC only (no geo) (averaged over 25 splits) ──
Feature                                                      Importance
------------------------------------------------------------------------
  IPC_last                                                      1.0000

── IPC + GDELT base (no geo) (averaged over 25 splits) ──
Feature                                                      Importance
------------------------------------------------------------------------
  IPC_last                                                      0.5774
  evt_sentiment.neu_sum                                         0.0116
  evt_sentiment.pos_sum                                         0.0112
  evt_neu_score_sum                                             0.0103
  evt_compound_score_sum                                        0.0093
  evt_sentiment.compound_sum                                    0.0091
  evt_displaced_freq_mean                                       0.0091
  evt_neg_score_mean                                            0.0090
  evt_neu_score_mean                                            0.0090
  evt_sentiment.compound_mean                                   0.0090
  evt_compound_score_mean                                       0.0089
  evt_sentiment.neg_mean                                        0.0088
  evt_sentiment.neu_mean                                        0.0088
  evt_neg_score_sum                                             0.0085
  evt_sentiment.neg_sum                                         0.0084
  evt_pos_score_mean                                            0.0083
  evt_pos_score_sum                                             0.0081
  evt_sentiment.pos_mean                                        0.0081
  evt_pred_resource_cash_aid_mean                               0.0076
  evt_food_insecurity_freq_mean                                 0.0074
  evt_displaced_count_sum                                       0.0071
  evt_pred_resource_shelter_sum                                 0.0065
  evt_displaced_count_mean                                      0.0064
  evt_fatalities_count_mean                                     0.0060
  evt_fatalities_freq_sum                                       0.0059
  ... (95 more features omitted)

── IPC + GDELT + temporal (no geo) (averaged over 25 splits) ──
Feature                                                      Importance
------------------------------------------------------------------------
  IPC_last                                                      0.4024
  evt_sentiment.compound_mean_rolling_6                         0.0244
  evt_food_insecurity_freq_mean_rolling_6                       0.0234
  evt_displaced_freq_mean_rolling_6                             0.0229
  evt_compound_score_mean_rolling_6                             0.0220
  evt_sentiment.compound_mean_rolling_3                         0.0164
  evt_fatalities_freq_mean_rolling_6                            0.0163
  evt_displaced_freq_mean_rolling_3                             0.0163
  evt_compound_score_mean_rolling_3                             0.0153
  evt_food_insecurity_freq_mean_rolling_3                       0.0126
  evt_sentiment.pos_sum                                         0.0122
  evt_fatalities_freq_mean_rolling_3                            0.0105
  evt_agriculture_freq_mean_rolling_6                           0.0097
  evt_pos_score_sum                                             0.0090
```

```
  evt_neu_score_sum                                          0.0084
  evt_sentiment.neu_sum                                      0.0084
  evt_sentiment.compound_mean_lag1                           0.0077
  evt_compound_score_mean_lag1                               0.0075
  evt_compound_score_sum                                     0.0073
  evt_compound_score_mean_lag2                               0.0070
  evt_sentiment.neu_mean                                     0.0067
  evt_sentiment.compound_mean_lag2                           0.0067
  evt_pos_score_mean                                         0.0064
  evt_neu_score_mean                                         0.0064
  evt_sentiment.neg_mean                                     0.0064
  ... (130 more features omitted)

── GDELT only (no geo) (averaged over 25 splits) ──
Feature                                                  Importance
──────────────────────────────────────────────────────────────────
  evt_displaced_freq_mean_rolling_6                          0.0439
  evt_compound_score_mean_rolling_6                          0.0434
  evt_sentiment.compound_mean_rolling_6                      0.0428
  evt_food_insecurity_freq_mean_rolling_6                    0.0367
  evt_fatalities_freq_mean_rolling_6                         0.0290
  evt_sentiment.compound_mean_rolling_3                      0.0281
  evt_compound_score_mean_rolling_3                          0.0276
  evt_displaced_freq_mean_rolling_3                          0.0264
  evt_food_insecurity_freq_mean_rolling_3                    0.0220
  evt_fatalities_freq_mean_rolling_3                         0.0186
  evt_sentiment.pos_sum                                      0.0181
  evt_agriculture_freq_mean_rolling_6                        0.0173
  evt_compound_score_mean_lag1                               0.0139
  evt_pos_score_sum                                          0.0139
  evt_neu_score_sum                                          0.0139
  evt_sentiment.compound_mean_lag1                           0.0130
  evt_compound_score_mean_lag2                               0.0123
  evt_sentiment.compound_mean_lag2                           0.0120
  evt_neu_score_mean                                         0.0120
  evt_sentiment.neu_mean                                     0.0120
  evt_sentiment.neu_sum                                      0.0119
  evt_displaced_freq_mean_lag2                               0.0118
  evt_compound_score_sum                                     0.0112
  evt_sentiment.pos_mean                                     0.0111
  evt_sentiment.compound_sum                                 0.0110
  ... (129 more features omitted)
```

Why Literature Shows Better Results: Key Differences

Potential Reasons for Better Performance in Literature:

1. **Different Prediction Tasks**:

- **Transitions/Changes**: Predicting when CS_score will worsen (crisis onset) rather than exact level
- **Early Warning**: Predicting 2-4 periods ahead (forecasting) rather than next period (nowcasting)
- **Binary Classification**: Predicting crisis (CS≥3) vs non-crisis (CS<3) rather than 5-class classification

2. **Feature Engineering**:
   - **Temporal Patterns**: Using lagged and rolling features (as we just added) rather than only current-period values
   - **Anomaly Detection**: Focusing on deviations from historical patterns
   - **Interaction Features**: Combining GDELT with other data sources (weather, prices, etc.)
3. **Data Coverage**:
   - **Spatial Aggregation**: Some studies aggregate to ADMIN0 or ADMIN1 level where coverage is better
   - **Temporal Aggregation**: Using quarterly rather than monthly data
   - **Filtering**: Focusing on regions/periods with good GDELT coverage
4. **Evaluation Metrics**:
   - **Recall for Rare Events**: Focusing on detecting crises (high recall for CS≥3) rather than overall accuracy
   - **Early Detection**: Measuring how early they detect transitions, not just accuracy
5. **Baseline Comparison**:
   - **Weaker Baselines**: Some studies don't include `previous_CS` in baseline, making improvement easier to show
   - **Different Baselines**: Comparing against simpler models or using different evaluation windows

Our Current Approach:

- ✅ Now includes temporal GDELT features (lags, rolling windows, anomalies, escalation)
- ✅ Uses same framework as previous work (allowing fair comparison)
- ⚠️ Still predicting exact CS_score level (highly autocorrelated)
- ⚠️ Using monthly data with CS_score available every 4 months

Potential Improvements:

1. **Predict Transitions**: Predict if CS_score will worsen (CS_t+1 > CS_t) rather than exact value

2. **Forecasting Horizon**: Predict 2-4 periods ahead instead of next period
3. **Crisis Detection**: Binary classification (crisis vs non-crisis) with focus on recall
4. **Feature Selection**: Use only temporal GDELT features, filter out low-importance current-period features

Feature Engineering Summary

Features Created:

1. **List Aggregations** (from events):
   - Mean, Max, Sum, Count for all list features
   - Prefixed with `evt_`
2. **Derived Features**:
   - Crisis severity indicators
   - Coverage intensity metrics
3. **Temporal Features**:
   - Lag features (CS_score_lag1, lag2, lag3)
   - Moving averages (ma2, ma3)
   - Change and percentage change features
   - Cyclical period encoding (sin/cos)
4. **Geographic Features**:
   - Aggregated CS_score by ADMIN0 and ADMIN1 levels
   - Standard deviations by geographic level
   - Count of regions with same score
5. **Interaction Features**:
   - Ratios (casualty_rate, aid_per_casualty)
   - Normalized features (tone_abs_normalized)
   - Coverage metrics (mentions_per_source, articles_per_source)

Next Steps for ML:

1. **Feature Selection**: Consider using:
   - Correlation-based selection
   - Mutual information
   - Recursive feature elimination
   - L1 regularization (Lasso)
2. **Categorical Encoding**: If you have categorical features:
   - One-hot encoding for low cardinality

- Target encoding for high cardinality
- Embedding for very high cardinality

3. **Scaling**: Consider:
    - StandardScaler or MinMaxScaler for numeric features
    - Especially important for distance-based algorithms
4. **Model Suggestions**:
    - Random Forest (handles non-linear relationships well)
    - Gradient Boosting (XGBoost, LightGBM, CatBoost)
    - Neural Networks (if you have enough data)
    - Consider class weights if classes are imbalanced